COURSE DESCRIPTIONS

Graduate Programs - 2025

Board of Studies – June 11, 2025
Board of Faculty – June 13, 2025
Academic Council – June 16, 2025

List of Courses

1. Master of Science (MS) Computer Science

Year 1 – Semester II		
Code	Title	Credit Hours
CS615	Theory of Computations	3(3+0)
CS617	Advanced Operating Systems	3(3+0)
XX***	CS Elective – I	3(3+0)
XX***	CS Elective – II	3(3+0)
Year 1 – Semester II		
Code	Title	Credit Hours
CS616	Advanced Analysis of Algorithms	3(3+0)
CS631	Advanced Computer Architecture	3(3+0)
XX***	CS Elective – III	3(3+0)
XX***	CS Elective – IV	3(3+0)
Year 2 – Semester I ––––– Year 4 – Semester VIII (max)		
Code	Title	Credit Hours
CS699	MS Research Report	6(0+6)

2. Master of Science (SE) Software Engineering

Year 1 – Semester II		
Code	Title	Credit Hours
SE601	Advanced Requirements Engineering	3(3+0)
SE611	Advanced Software System Architecture	3(3+0)
SE***	SE Domain Elective – I	3(3+0)
XX***	General Elective – I	3(3+0)
Year 1 – Semester II		
Code	Title	Credit Hours
SE641	Software Testing and Quality Assurance	3(3+0)
SE***	SE Domain Elective – II	3(3+0)
XX***	General Elective – II	3(3+0)
XX***	General Elective – III	3(3+0)
Year 2 – Semester I ––––– Year 4 – Semester VIII (max)		
Code	Title	Credit Hours
SE699	MS Research Report	6(0+6)

3. Master of Science (DS) Data Science

Year 1 – Semester II		
Code	Title	Credit Hours
DS601	Tools and Techniques for Data Science	3(3+0)
DS602	Statistical and Math. Methods for Data Analysis	3(3+0)
DS***	DS Domain Elective – I	3(3+0)
XX***	General Elective – I	3(3+0)
Year 1 – Semester II		
Code	Title	Credit Hours
CS661	Advance Machine Learning	3(3+0)
DS***	DS Domain Elective – II	3(3+0)

XX***	General Elective – II	3(3+0)
XX***	General Elective – III	3(3+0)
Year 2 – Semester I –----- Year 4 – Semester VIII (max)		
Code	Title	Credit Hours
DS699	MS Research Report	6(0+6)

4. PhD Computer Science

Year 1 – Semester I		
Code	Title	Credit Hours
CS***	CS Elective – I	3(3+0)
CS***	CS Elective – II	3(3+0)
CS***	CS Elective – III	3(3+0)
XX***	Non-Credit CS Course – I	0
Year 1 – Semester II		
Code	Title	Credit Hours
CS***	CS Elective – I	3(3+0)
CS***	CS Elective – II	3(3+0)
CS***	CS Elective – III	3(3+0)
XX***	Non-Credit CS Course – I	0
Year 2 – Semester I –----- Year 8 – Semester XVI (max)*		
Code	Title	Credit Hours
CS999	Thesis	6(0+6)

5. Course Description for MS(CS/SE/DS) and PhD (CS)

⇨ **Program -MS/PhD (CS)**	CS Courses begin from here

Program - MS/PhD (CS)	**Domain 0: Fundamental Courses**

Code	Course Title	CHs	Pre-Req
CS601	**Research Methods with an IT Perspective**	3(3+0)	None

Recommended Texts

- Research Design. Qualitative, Quantitative, and Mixed Methods Approaches. By John W. Creswell, J. David Creswell, SAGE Publications, Inc; 5th edition (January 2, 2018), ISBN-13 : 978-1506386706
- Research Methodology By Panneerselvam R, 2nd Edition, PHI, 2014
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course provides a rigorous foundation in research design, methodology, and scientific communication in computing and AI. It covers qualitative and quantitative methods, literature review strategies, experimental design, and reproducibility practices in AI research.

Course Objectives

- To comprehend the advantages and disadvantages of each of these approaches.
- How do you pick the best method(s) for your investigation?
- How should these methods be used to conduct investigations?

Week Wise Topics to be Covered

W1	Scientific Inquiry in Computing and AI
W2	Literature Review Techniques and Tools
W3	Formulating Hypotheses and Research Questions
W4	Quantitative Methods and Experimental Design
W5	Qualitative Research in HCI and AI Ethics
W6	Surveys, Case Studies, and User Studies
W7	Statistical Analysis and Validity
W8	Data Collection and Annotation Methodologies
MID EXAM/ MID OF SEMESTER	
W9	Reproducibility and Benchmarking in AI Research
W10	Ethics, Bias, and Responsible Research Conduct
W11	Open Science and Research Transparency
W12	Writing and Reviewing Scientific Papers
W13	Visualization and Communication of Research Results
W14	Proposal Writing and Grant Applications
W15	Reproducibility and Benchmarking in AI Research
W16	Ethics, Bias, and Responsible Research Conduct

Program - MS/PhD (CS)	**Domain - 1: Programming and Computing**

Code	Course Title	CHs	Pre-Req
CS611	**Advanced Parallel Programming**	3(3+0)	Nil

Recommended Texts

- Parallel Programming: Concepts and Practice, Bertil Schmidt, Jorge Gonzalez-Dominguez, Christian Hundt , Moritz Schlarb, Morgan Kaufmann; 1st edition (November 27, 2017) , ISBN-13 : 978-0128498903
- Parallel Programming: for Multicore and Cluster Systems 2nd ed. 2013 Edition , Thomas Rauber , Gudula Rünger , Springer publisher (June 8, 2013), ISBN: 978-3642378003
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

The course focuses on core concepts of parallel programming and bringing its importance in front of students while considering the modern applications and easy access to multi-core and multi-processor architecture now a days. The course will discuss real world problems and their solution by parallelizing it and observe the speed that could be gained by doing so.

Course Objectives

- Define parallel computing, explain why parallel computing is becoming mainstream
- Identify opportunities for parallelism in code segments and applications
- Describe three methods of dividing independent work
- Decide whether a variable in a multithreaded program should be shared or private.

Week Wise Topics to be Covered

W1	Introduction and overview of Parallel Computing.	W9	Deadlock
W2	Introduction to OpenMP. Problem Decomposition	W10	Task Decomposition
W3	Finding Parallelism	W11	Implementing Task Decomposition
W4	Shared Memory Consideration	W12	Predicting Parallel Performance
W5	Domain Decomposition.	W13	OpenMPI, MPI program structure Communicators.Point2Point Communication
W6	Confronting Race Condition	W14	Collective Communication. Virtual Topology
W7	Multi-core machines	W15	Case Studies / Projects / Demo
W8	Models of Parallel Computers and Computation	W16	Presentations / Peer Review / Q&A Sessions / Discussions
MID EXAM/ MID OF SEMESTER		Final EXAM	

Code	Course Title	CHs	Pre-Req
CS612	**Advanced Distributed Systems**	3(3+0)	Nil

Recommended Texts

- Distributed Systems: Design Concepts, Sunil Kumar, Narosa Publishing House (January 1, 2017), ISBN-13 : 978-8184874099
- The Practice of Cloud System Administration: Designing and Operating Large Distributed Systems, Volume 2 1st Edition by Thomas A. Limoncelli , Strata R. Chalup , Christina J. Hogan , Publisher: Addison-Wesley Professional (September 13, 2014), ISBN: 978-0321943187
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

In general this course will cover design and implementations of distributed systems. Different categories of distributed systems such as Cluster Computing, Grid Computing, Cloud Computing and Supercomputing will be covered.

Course Objectives

- Understand existing distributed systems
- Explain suitability of architecture for specific category of real world problems
- Have hands on experience on Cluster, Grid and Cloud computing systems

Week Wise Topics to be Covered

W1	Distributed System Models	W9	Workflow Systems
W2	System Architectures & Client-Server Models	W10, W11	Grid Computing
W3	Programming Systems and Models	W12, W13	Cloud Computing
W4	Processes and threads	W14	Virtualization . IaaS Clouds
W5	Remote Procedure Call	W15	File Systems. Distributed File systems
W6	High-Performance Computing	W16	Data-Intensive Computing. Many-core Computing
W7	MapReduce		
W8	Many-Task Computing		
MID EXAM/ MID OF SEMESTER			

Code	Course Title	CHs	Pre-Req
CS613	**Analysis and Design of Parallel Algorithms**	3(3+0)	Nil

Recommended Texts

- Programming Massively Parallel Processors, Third Edition: A Hands-on Approach 3rd Edition , David B. Kirk , Wen-mei W. Hwu, Morgan Kaufmann Publisher (December 14, 2016), ISBN: 978-0128119860
- Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, The MIT Press (March 22, 2022), ISBN-13: 978-0262046305
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course covers the design and analysis of parallel algorithms for already existing sequential problems. It will be explained what factors to consider while doing a problem transformation. In the light of available parallel programming paradigm existing problems will be categorized and their corresponding parallel algorithm will be designed and analyzed.

Course Objectives

- Design and analysis of parallel algorithms for
- Different Models of Computation,
- Graph Traversal
- Speed up gained by transforming sequential algorithm into parallel algorithm

Week Wise Topics to be Covered

W1	Analysis of Parallel Algorithms: Time Complexity, Asymptotic Notations, Number of Processors, overall Cost	W8	Message Passing Programming: Shared Memory
		MID EXAM/ MID OF SEMESTER	
W2	Different Models of Computation: Combinational Circuits	W9	Message Passing Libraries
		W10	Data Parallel Programming
W3	Parallel RandomAccess Machines (PRAM) Interconnection Networks	W11	Data Structures for Parallel Algorithms: Linked List, Arrays Pointers, and Hypercube Network
W4	Sorting: Combinational Circuit for Sorting the String. Merge Sort Circuit	W12	Introduction to OpenMP
W5	Sorting Using Interconnection Networks	W13, W14	Design and analysis Flloyd Warshal Algorithm. Design and analysis Bellman-Ford Algorithm. Design and analysis dijkstra's algorithm
W6	Matrix Computation		
W7	Concurrently Read Concurrently Write (CRCW). Concurrently Read Exclusively Write (CREW)	W15, W16	Design and analysis of different real world problems

Code	Course Title	CHs	Pre-Req
CS614	**Algorithms for Bio-Informatics**	3(3+0)	Nil

Recommended Texts

- Algorithms in Bioinformatics: A Practical Introduction, Wing-Kin Sung, Chapman and Hall/CRC; 1st edition (30 Sept. 2020), ISBN-13 : 978-0367659318
- Bioinformatics Algorithms: An Active Learning Approach, 2nd edition ,Phillip Compeau , Pavel Pevzner, Active Learning Publishers (2015), ISBN: 978-0990374626
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

Processing biological data requires complex computations on large volumes of data. To ensure that these computations complete within a reasonable amount of time, one must design the algorithms (computer procedures) after a careful study of the characteristics of underlying data and making use of existing algorithm design principles. This module aims to introduce students to the algorithmic solution of computational problems in bioinformatics.

Course Objectives

- Know how dynamic programming (DP) can be used to solve string comparison problems such as "longest common subsequence", "edit distance", "local alignment", and "global alignment".
- Know how the DP solution for aligning two strings can be extended to aligning multiple strings.
- Know how k-mer indexes can be used for exact and approximate string search.
- Know what a keyword tree is, and how this index structure is built and used for string search.

Week Wise Topics to be Covered

W1	Algorithms and Complexity	W9	Sorting by Reversals
W2	Biological algorithms vs Computer algorithms	W10	Dynamic Programming Algorithms
W3	Analyze DNA: Copying DNA, Cutting and Pasting DNA, Measuring DNA Length, Probing DNA	W11	The Power of DNA Sequence Comparison
W4	Exhaustive Search: Impractical Restriction Mapping Algorithms, A Practical Restriction Mapping Algorithm	W12	The Change Problem Revisited. The Manhattan Tourist Problem
W5	Regulatory Motifs in DNA Sequences	W13	Edit Distance and Alignments, Longest Common Subsequences, Global Sequence Alignment
W6, W7	The Motif Finding Problem, Search Trees, Finding Motifs, Finding a Median String	W14	Gene Prediction
W8	Greedy Algorithms. Genome Rearrangements	W15	Graph Algorithms: Graphs and Genetics, DNA Sequencing, Shortest Superstring Problem
MID EXAM/ MID OF SEMESTER		W16	Hidden Markov Models

Code	Course Title	CHs	Pre-Req
CS615	**Theory of Computation**	3(3+0)	None

Recommended Texts

- Formal Languages and Automata Theory: Introduction to Abstract and Theories of Computation, Joseph Bole, Independently published (June 23, 2021), ISBN-13 : 979-8525270657
- Introduction to the Theory of Computation 3rd edition, Michael Sipser, Cengage Learning Publisher (2014), ISBN: 978-8131525296
- Introduction to languages and the theory of computation, by John Martin, ISBN: 0073191469, 2010
- Introducing The Theory Of Computation, Wayne Goddard, 2008, ISBN-13: 978-0763741259
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

Introduces the foundations of formal language theory, computability, and complexity. Shows relationship between automata and various classes of languages. Addresses the issue of which problems can be solved by computational means, and studies complexity of solutions.

Course Objectives

- To introduce the students to the mathematical foundations of computation including automata theory, the notations of algorithm, decidability, complexity and computability.
- To enhance student's ability to understand and conduct mathematical proofs for computation and algorithms

Week Wise Topics to be Covered

W1	Introduction, Terminologies of Languages, Descriptive definition	W9	Context Free Grammar-CFG, Examples of CFG
W2	Examples using Descriptive definition	W10	Tree , Ambiguity, Parsing
W3	Recursive Definition Examples using Recursive definition	W11	Languages that are not context-free: pumping lemma for CFLs, Pushdown Automata
W4	Regular Expressions (RE), Examples of Regular Expressions (RE)	W12	Recursively Enumerable Languages Turing machines, Examples
W5	Finite Automaton (FA), Examples of Finite Automaton (FA)	W13	Computability Theory, Transformations, Decidability
W6	Martin Technique, Non- Deterministic Finite Automata- NFA, Conversion of NFA to DFA	W14	Complexity Theory, Time and space complexity
W7	Union of Two FAs, Concatenation of Two FAs	W15	Case Studies / Projects / Demo
W8	Transition Graph-TG, Generalize Transition Graph- GTG	W16	Presentations / Peer Review / Q&A Sessions / Discussions
MID EXAM/ MID OF SEMESTER		Final EXAM	

Code	Course Title	CHs	Pre-Req
CS616	**Advanced Analysis of Algorithms**	3(3+0)	Nil

Recommended Texts

- Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022). Introduction to Algorithms (4th ed.). MIT Press. https://doi.org/10.7551/mitpress/2040.001.0001
- Ailon, N., & Chazelle, B. (2021). Advanced Algorithmics: Parallel & Approximation. Cambridge University Press. https://doi.org/10.1017/9781108889250
- At least 5 research papers from Scopus-indexed international conferences
- At least 5 research papers from Q1/Q2 WoS-indexed journals

Course Description

This course explores advanced algorithmic strategies including graph algorithms, approximation techniques, randomized algorithms, and complexity theory, emphasizing both theoretical rigor and practical application.

Course Objectives

- To understand the design and analysis of advanced algorithms.
- To implement approximation and randomized algorithms.
- To understand computational complexity.
- To implement parallel and distributed algorithmic solutions.

Week Wise Topics to be Covered

W1	Review of algorithm analysis and asymptotic notation	**W9**	Number theoretic algorithms
W2	Divide and conquer strategies	**W10**	Geometric algorithms
W3	Dynamic programming and greedy algorithms	**W11**	String matching and text processing
W4	Graph algorithms: shortest paths and MST	**W12**	Parallel algorithms and models
W5	Network flows and matching	**W13**	External memory and cache-efficient algorithms
W6	NP-completeness and reductions	**W14**	Number theoretic algorithms
W7	Approximation algorithms	**W15**	**Case Studies / Projects / Demo**
W8	Randomized algorithms	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS617	**Advanced Operating Systems**	3(3+0)	None

Recommended Texts

- Distributed Systems: Principles and Paradigms, Andrew S. Tanenbaum, Maarten van Steen , CreateSpace Independent Publishing Platform; 2nd edition (February 26, 2016) , ISBN-13 : 978-1530281756
- Modern Operating Systems, Tanenbaum, Andrew S. Tanenbaum, Pearson India; 4th edition (March 25, 2016), ISBN-13 : 978-9332575776
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

The course will focus on the engineering and performance trade-offs in the design of operating systems. The purpose will be to teach not only what operating systems are and how they work today, but also why they are designed the way they are and how they are likely to evolve in the future. The emphasize will be on the practical aspects of the topics through the case study of Linux kernel as an example of a commercial operating system.

Course Objectives

- Understand in great detail how and why different parts of an operating system work.
- Understand the engineering tradeoffs involved in the design of various sub-modules of an operating system.
- Understand how operating systems are structured, what are alternative OS architectures and how different modules interact together to form a cohesive and complex system.

Week Wise Topics to be Covered

W1	Review of Operating Systems concepts	W9	Synchronization in distributed systems
W2	Hardware concepts of distributed systems	W10	Clock synchronization
W3	Software concepts and design issues	W11	Mutual exclusion; Election algorithms
W4	Communication in distributed systems	W12	Transaction and concurrent control
W5	Threads and thread usage Multithreading operating system	W13	Deadlock in distributed systems
W6	Client – server model Implementation of Client-server model	W14	Processor Allocation
W7	Remote procedure call	W15	Case Studies / Projects / Demo
W8	Implementation of remote procedure call	W16	Presentations / Peer Review / Q&A Sessions / Discussions
MID EXAM/ MID OF SEMESTER		Final EXAM	

Code	**Course Title**	**CHs**	**Pre-Req**
CS811	**Cluster and Grid Computing**	3(3+0)	Nil

Recommended Texts

- Grid, Cloud, and Cluster Computing and Applications (The 2017 WorldComp International Conference Proceedings) Hamid R. Arabnia (Editor), Fernando G. Tinetti (Editor), CSREA (February 1, 2018), ISBN-13 : 978-1601324580
- Scheduling Strategies in Grid Computing , T. Kokilavani , D. I. George Amalarethinam , Scholars' Press Publisher (September 22, 2016), ISBN: 978-3659842610
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course will cover two parallel computing architectures: Cluster Computing and Grid Computing. MPI parallel programming will give the student the opportunity to practically understand the coordination of multiple processes in a cluster. Furthermore, load balancing and performance evaluation on real cluster should also be an acquired skill. For Grid Computing, Globus toolkit, a standard Grid middleware, is going to give a deeper knowledge of working of Grid computing.

Course Objectives

- Understand the parallel computing environment.
- Understand the architecture and working mechanism of cluster computing.
- Understand the notion of Grid Computing, its architecture and working mechanism.
- Get the knowledge of today's real world Grid computing applications, their challenges.

Week Wise Topics to be Covered

W1	Introduction to Cluster and Grid Computing	**W9**	GridSim and SimGrid for Simulation
W2	Architectures for High-Performance Distributed Systems	**W10**	Hands-on Lab: Deploying a Local Computing Cluster
W3	Resource Management and Scheduling Techniques	**W11**	Grid-based Problem Solving Environments
W4	Parallel Programming Models for Grid	**W12**	Cloud vs Grid Paradigms
W5	Middleware Systems and Services	**W13**	Interoperability and Standards in Grid Computing
W6	Load Balancing and Fault Tolerance	**W14**	Project Development and Testing
W7	Data Management and Grid Security	**W15**	**Case Studies / Projects / Demo**
W8	Real-world Case Studies in Scientific Grids	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS812	**High Performance Computing**	3(3+0)	Nil

Recommended Texts

- Introduction to High Performance Scientific Computing , lulu.com publisher (December 28, 2015), ISBN: 978-1257992546
- Parallel and High Performance Computing, Robert Robey, Yuliana Zamora, Manning Publications (June 22, 2021), ISBN-13 : 978-1617296468
- Problem-solving in High Performance Computing: A Situational Awareness Approach with Linux1st Edition , Igor Ljubuncic , Morgan Kaufmann publisher (October 1, 2015), ISBN: 978-0128010198
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course is a tour through various research topics in High Performance Computing (HPC) systems, covering topics in cluster computing, grid computing, supercomputing, and cloud computing. This course will help in learning the design principles for building large network-based computational systems to support data intensive computing. Readings and discussions will help identify research problems and understand methods and general approaches to design, implement, and evaluate HPC systems to support data intensive computing.

Course Objectives

- Explain all existing High Performance Computing Systems
- Explain the middle ware used by HPC systems
- Explain HPC system's scheduling mechanism
- Explain the storage mechanism of these systems

Week Wise Topics to be Covered

W1	Paradigms: Supercomputing (e.g. IBM BlueGene/P/Q, Cray XT6)	W9	Light-weight Task Scheduling (e.g. Falkon, Sparrow, MATRIX)
W2	Grid Computing (e.g. XSEDE, OSG)	W10	Storage Systems: File Systems (e.g. EXT3)
W3	Cloud Computing (e.g. Amazon AWS, Google App Engine, Windows Azure)	W11	Shared File Systems (e.g. NFS)
W4	Many-core Computing (e.g. NVIDIA GPUs, Xeon Phi)	W12	Distributed File Systems (e.g. HDFS, FusionFS)
W5	Parallel Programming Systems: MapReduce (e.g. Hadoop)	W13	Parallel File Systems (e.g. GPFS, PVFS, Lustre)
W6	Workflows (e.g. Swift), MPI (e.g. MPICH)	W14	Distributed NoSQL Key/Value Stores (e.g. Casandra, MongoDB, ZHT)
W7	OpenMP, Multi-Threading (e.g. PThreads)	W15	Relational Databases (e.g. MySQL)
W8	Job Management Systems: Batch scheduling (e.g. Condor, Slurm, SGE, PBS)		
MID EXAM/ MID OF SEMESTER			

Code	Course Title	CHs	Pre-Req
CS813	**Federated Learning**	3(3+0)	Nil

Recommended Texts

- M. Irfan Uddin, and Wali Khan Mashwani. Federated Learning: Unlocking the Power of Collaborative Intelligence. CRC Press Taylor & Francis Group, USA, September 2024. ISSN 9781032724324. doi:10.1201/9781003466581.
- Kairouz, P., McMahan, H. B., et al. (2021). Advances and Open Problems in Federated Learning. Foundations and Trends in Machine Learning, 14(1–2), 1–210. https://doi.org/10.1561/2200000083
- Li, T., Sahu, A. K., Zaheer, M., Sanjabi, M., Talwalkar, A., & Smith, V. (2020). Federated Optimization in Heterogeneous Networks. In Proceedings of Machine Learning and Systems. https://doi.org/10.48550/arXiv.1812.06127
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course introduces federated learning and privacy-aware machine learning methods. It covers distributed training, differential privacy, secure aggregation, and real-world use cases of decentralized AI in healthcare, IoT, and finance.

Course Objectives

- To understand the fundamentals of federated learning and decentralized AI.
- To implement privacy-preserving machine learning models.
- To understand system-level and communication constraints in federated settings.
- To implement collaborative learning architectures using real-world data.

Week Wise Topics to be Covered

W1	Introduction to Federated Learning and Privacy Concepts	**W9**	FL Applications in Education
W2	Federated Averaging and Training Paradigms	**W10**	FL for Healthcare and Edge Devices
W3	Differential Privacy Techniques	**W11**	Fairness and Bias in Federated Systems
W4	Secure Multiparty Computation and Homomorphic Encryption	**W12**	Real-World Deployment Challenges
W5	Communication Efficiency and System Design	**W13**	Evaluation Frameworks and Benchmarks
W6	Personalization in Federated Learning	**W14**	Federated Learning Research Frontiers
W7	Robustness and Adversarial Attacks	**W15**	**Case Studies / Projects / Demo**
W8	FL Applications in Industry	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Program - MS/PhD (CS)	**Domain - 2: Data and Information Management**

Code	Course Title	CHs	Pre-Req
CS622	**Advanced Data Mining**	3(3+0)	Nil

Recommended Texts

- Advanced Data Mining Techniques, David L. Olson,DursunDelen, Springer publisher, 2016, ISBN-13: 978-3540769163
- Data Mining: Introductory and Advanced Topics Margaret H. Dunham, Pearson publisher, 2016, ISBN-13: 9780130888921
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

Data Mining studies algorithms and computational paradigms that allow computers to find patterns and regularities in databases, perform prediction and forecasting, and generally improve their performance through interaction with data. It is currently regarded as the key element of a more general process called Knowledge Discovery that deals with extracting useful knowledge from raw data. The knowledge discovery process includes data selection, cleaning, coding, using different statistical and machine learning techniques, and visualization of the generated structures.

Course Objectives

- To introduce students to the basic concepts and techniques of Data Mining.
- To develop skills of using recent data mining software for solving practical problems.
- To gain experience of doing independent study and research.

Week Wise Topics to be Covered

W1	Introduction to Data Mining and Knowledge Discovery	**W9**	Dimensionality Reduction Techniques
W2	Frequent Pattern Mining and Association Rule Learning	**W10**	Graph Mining and Social Network Analysis
W3	Classification Techniques: Decision Trees, SVM	**W11**	Anomaly and Outlier Detection
W4	Clustering Algorithms: k-means, DBSCAN	**W12**	Mining Data from Non-traditional Sources
W5	Advanced Classification: Ensemble Methods	**W13**	Ethical Considerations in Data Mining
W6	Text and Web Mining Techniques	**W14**	Project Implementation and Evaluation
W7	Mining Time Series and Sequential Data	**W15**	**Case Studies / Projects / Demo**
W8	Scalability and Data Stream Mining	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS623	**Big Data**	3(3+0)	Nil

Recommended Texts

- White, T. (2020). Hadoop: The Definitive Guide (4th ed.). O'Reilly Media. https://doi.org/10.1007/978-1-4842-0076-6
- Karau, H., & Warren, R. (2020). High Performance Spark: Best Practices for Scaling and Optimizing Apache Spark. O'Reilly Media. https://doi.org/10.1007/978-1-4842-4392-3
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course introduces basic technology (algorithms, architectures, systems) and advanced research topics in connection with large-scale data management and information extraction techniques for big data. The course will start by introducing Big data models, databases and query languages, and cover modern distributed database systems and algorithms and Big data systems adopted in industry and science applications. Implementation of a distributed database on a standalone machine will be covered.

Course Objectives

- To understand the architecture and ecosystem of big data technologies.
- To implement distributed data processing with Hadoop and Spark.
- To understand real-time streaming and batch processing.
- To implement scalable storage and NoSQL database solutions.

Week Wise Topics to be Covered

W1	Introduction to Big Data Ecosystems	**W9**	Graph Processing using GraphX and Pregel
W2	Hadoop and MapReduce Programming	**W10**	Big Data Tools: Hive, Pig, and Flink
W3	HDFS and Data Storage Architecture	**W11**	Big Data in Cloud Environments
W4	NoSQL Databases: HBase, Cassandra	**W12**	Privacy and Security in Big Data
W5	Data Ingestion and ETL Pipelines	**W13**	Big Data Use Cases in Industry
W6	Apache Spark and Distributed Computing	**W14**	Project Development and Testing
W7	Big Data Analytics with Spark SQL	**W15**	**Case Studies / Projects / Demo**
W8	Streaming Data and Real-time Processing	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Program - MS/PhD (CS)	**Domain - 3 : Computer Systems and Electronics**

Code	Course Title	CHs	Pre-Req
CS631	**Advanced Computer Architecture**	3(3+0)	None

Recommended Texts

- Computer Organization and Architecture – Designing or Performance, Willian Stallings, Pearson, 10th Edition(January 2015), ISBN-13: 978-013410613
- Computer Architecture – A Quantitative Approach, 6th Edition, by John L. Hennessy and David A. Patterson,Elsevier (November 2017), ISBN: 9780128119051
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

The course covers the topics of organization and architecture of computer systems hardware; instruction set architectures; addressing modes; register transfer notation; processor design and memory systems.

Course Objectives

- To enable students to understand the basic components of a computer system
- To enable students to understand the issues that are limiting the performance of processor, caches and memories
- To enable students to highlight architecture and organization parameters for processor performance
- To enable students to understand the cycle of instruction execution in the pipeline of a processor

Week Wise Topics to be Covered

W1	Introduction to Advanced Processor Architectures	**W9**	Power-aware and Energy-efficient Design
W2	Instruction-Level Parallelism and Superscalar Architectures	**W10**	Graphics and Vector Processing Architectures
W3	VLIW and EPIC Architectures	**W11**	Benchmarking and Performance Evaluation
W4	Multithreading and Simultaneous Multithreading	**W12**	RISC-V and Open Architectures
W5	Memory Hierarchy and Cache Optimization	**W13**	Modern Trends in Processor Design
W6	Multiprocessors and Shared Memory Systems	**W14**	Project Design and Simulation
W7	Interconnection Networks and Communication	**W15**	**Case Studies / Projects / Demo**
W8	Scalable and Distributed Architectures	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS731	**Embedded Systems**	3(3+0)	Nil

Recommended Texts

- Marwedel, P. (2021). Embedded System Design: Embedded Systems Foundations of Cyber-Physical Systems and the Internet of Things (4th ed.). Springer. https://doi.org/10.1007/978-3-030-60910-8
- Iniewski, K. (2020). Embedded Systems: Hardware, Design, and Implementation (2nd ed.). Wiley. https://doi.org/10.1002/9781119702414
- At least 5 research papers from Scopus-indexed international conferences
- At least 5 research papers from Q1/Q2 WoS-indexed journals

Course Description

This course covers embedded systems design including microcontrollers, real-time operating systems, and low-level interfacing. Emphasis on real-time constraints and safety-critical design.

Course Objectives

- To understand real-time embedded systems.
- To implement microcontroller-based systems.
- To understand timing and resource constraints.
- To implement secure and reliable firmware.

Week Wise Topics to be Covered

W1	Introduction to embedded systems and applications	**W9**	Wireless and IoT integration
W2	Microcontroller architectures and development tools	**W10**	Security in embedded systems
W3	Programming in embedded C and I/O interfacing	**W11**	Debugging and testing techniques
W4	Interrupts, timers, and counters	**W12**	RTOS case studies and projects
W5	Real-time operating systems concepts	**W13**	Embedded system performance tuning
W6	Task scheduling and context switching	**W14**	Wireless and IoT integration
W7	Power-aware embedded design	**W15**	**Case Studies / Projects / Demo**
W8	Communication protocols (SPI, I2C, CAN)	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS831	**AI Accelerator Architectures and Edge AI**	3(3+0)	Nil

Recommended Texts

- Sze, V. (2020). Efficient Processing of Deep Neural Networks: A Tutorial and Survey. Foundations and Trends® in Electronic Design Automation, 12(3), 139–372. https://doi.org/10.1561/1000000052
- Mittal, S. (2022). AI Accelerator Architectures for Edge Computing. Springer. https://doi.org/10.1007/978-3-030-95591-2
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course investigates the design and deployment of AI models on specialized hardware accelerators and resource-constrained edge devices. It covers AI chips, low-power optimizations, model compression, and deployment strategies for real-time AI systems in embedded and mobile environments.

Course Objectives

- To understand the architectural design of AI accelerators.
- To implement edge AI models on hardware-constrained platforms.
- To understand system-level design and integration of AI chips.
- To implement performance optimization strategies on AI accelerators.

Week Wise Topics to be Covered

W1	Introduction to AI Hardware and Edge Computing	**W9**	Edge Deployment Frameworks: TensorFlow Lite, ONNX, TVM
W2	Overview of AI Accelerators: GPUs, TPUs, NPUs, FPGAs	**W10**	TinyML and Ultra-Low-Power Inference
W3	Hardware-Software Co-Design for AI Workloads	**W11**	Distributed and Federated Learning on Edge
W4	Edge AI Applications and Constraints	**W12**	Security and Privacy in Edge AI Systems
W5	Model Compression and Quantization	**W13**	Neuromorphic and Emerging Hardware Paradigms
W6	Pruning, Knowledge Distillation, and Efficient Architectures	**W14**	AI on Mobile, IoT, and Wearables
W7	Memory and Energy Optimization in AI Inference	**W15**	**Case Studies / Projects / Demo**
W8	Benchmarking Edge Devices for AI	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS932	**Embodied AI and Cognitive Robotics**	3(3+0)	Nil

Recommended Texts

- Pfeifer, R., & Bongard, J. (2007). How the Body Shapes the Way We Think: A New View of Intelligence. MIT Press. https://doi.org/10.7551/mitpress/9780262162395.001.0001
- Arkin, R. C. (2022). Behavior-Based Robotics (Updated Ed.). MIT Press. https://doi.org/10.7551/mitpress/9780262014410.001.0001
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS
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Course Description

This course focuses on AI systems embedded in physical bodies or simulators, capable of interacting with and learning from the real world. It covers sensorimotor learning, perception-action loops, embodied cognition, and cognitive architectures for robots.

Course Objectives

- To understand the foundations of embodied intelligence.
- To implement robotic agents with perception and motor skills.
- To understand learning-driven control for cognitive robotics.
- To implement multi-modal fusion for agent behavior modeling.

Week Wise Topics to be Covered

W1	Foundations of Embodied AI: Physical grounding of intelligence	**W9**	Multi-modal Integration in Cognitive Agents
W2	Sensors, Actuators, and Control	**W10**	Language and Action Grounding
W3	Perception in Robots: Vision, depth, proprioception	**W11**	Human-Robot Interaction (HRI)
W4	Motor Control and Kinematics	**W12**	Developmental Robotics and Learning from Demonstration
W5	Reinforcement Learning for Embodied Agents	**W13**	Robotics Simulators: MuJoCo, Isaac Gym, CoppeliaSim
W6	Sim2Real Transfer and Domain Adaptation	**W14**	Ethics and Safety in Embodied AI
W7	Embodied Cognitive Architectures	**W15**	**Case Studies / Projects / Demo**
W8	Memory and Planning in Robots	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Program - MS/PhD (CS)	**Domain – 4 : Multimedia and Web**

Code	Course Title	CHs	Pre-Req
CS641	**Multimedia Computing and Applications**	3(3+0)	Nil

Recommended Texts

- Multimedia Computing 1st Edition, Gerald Friedland, Ramesh Jain, Cambridge University Press, 2014, ISBN-13: 978-0521764513
- Multimedia Foundations: Core Concepts for Digital Design 2nd Edition, Vic Costello, Focal Press, 2016, ISBN-13: 978-0415740036
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

The course covers the important multimedia computing methods by presenting a comprehensive coverage of the underlying content processing, content transformation and resource optimization techniques

Course Objectives

- It lays the foundation for graduate students to build advanced multimedia computing applications comprising of images, videos, and audio.
- To enable students to understand content processing, content transformation and resource optimization techniques
- By considering the research issues in the multimedia systems areas, it will also prepare the student in formulating novel approaches for future multimedia computing applications

Week Wise Topics to be Covered

W1	Introduction to Multimedia Systems	**W9**	Applications in Entertainment and Education
W2	Audio, Image, and Video Data Representation	**W10**	Computer Vision in Multimedia Systems
W3	Multimedia Compression Standards	**W11**	AI in Multimedia Content Generation
W4	Content-Based Multimedia Retrieval	**W12**	Security and Copyright in Multimedia
W5	Multimedia Networking and Streaming	**W13**	Emerging Trends in Multimedia Tech
W6	Human-centered Multimedia Interaction	**W14**	Project Implementation and Testing
W7	Multimedia Databases and Indexing	**W15**	**Case Studies / Projects / Demo**
W8	Multimodal Fusion and Analysis	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS642	**Computer Vision**	3(3+0)	Nil

Recommended Texts

- Computer Vision: Algorithms and Applications, Richard Szeliski, Springer; 2nd ed. 2022 edition (March 7, 2022), ISBN-13 : 978-3030343712
- Compute Vision: A modern Approach, David A. Forsyth, Jean Ponce, Pearson; 2nd edition (October 26, 2011), ISBN-13 : 978-0136085928
- Computer Vision: Models, Learning, and Inference, Simon J. D. Prince, Cambridge University Press, ISBN-13: 978-1107011793, 2012
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course provides an introduction to computer vision, including fundamentals of image formation, feature detection and matching, image classification, scene understanding, and deep learning with neural networks. We will develop basic methods for applications that include finding known models in images, image stabilization, automated alignment, tracking, boundary detection, and recognition. We will develop the intuitions and mathematics of the methods in class, and then learn about the difference between theory and practice in projects.

Course Objectives

- Be familiar with both the theoretical and practical aspects of computing with images;
- Have described the foundation of image formation, measurement, and analysis;
- Have implemented common methods for robust image matching and alignment;
- Understand the geometric relationships between 2D images and the 3D world.

Week Wise Topics to be Covered

W1	Introduction to Image Formation and Cameras	**W9**	Pose Estimation and Action Recognition
W2	Image Filtering, Edge Detection, and Histograms	**W10**	Vision for Autonomous Systems
W3	Feature Detection and Matching	**W11**	Visual SLAM and Scene Understanding
W4	Geometric Vision and 3D Reconstruction	**W12**	Multimodal Perception and Sensor Fusion
W5	Motion and Optical Flow Estimation	**W13**	Ethics in Vision-based Surveillance
W6	Object Detection and Recognition	**W14**	Vision System Project Implementation
W7	Deep Learning in Vision: CNNs and Vision Transformers	**W15**	**Case Studies / Projects / Demo**
W8	Semantic and Instance Segmentation	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS643	**Semantic Web**	3(3+0)	Nil

Recommended Texts

- Semantic Web for the Working Ontologist: Effective Modeling for Linked Data, RDFS, and OWL, James Hendler, Fabien Gandon, Dean Allemang, ACM Books; 3rd edition (August 3, 2020), ISBN-13 : 978-1450376143
- The Semantic Web Explained: The Technology and Mathematics Behind Web 3.0, Peter Szeredi, Gergely Lukacsy, Tamas Benko, Gergely LukCsy, Tam S Benk, Cambridge University Press,2014, ISBN13: 9780521700368
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course enables students to understand basics of Semantic Web, its importance and various techniques.

Course Objectives

- Providing graduate and advanced level undergraduate students with the knowledge and understanding of Semantic Web and it importance, the Semantic Web technologies
- Teach them a set of tools and techniques to equip them with the required skills in area.

Week Wise Topics to be Covered

W1	RDF and Linked Data Principles	**W9**	Ontology Engineering Practices
W2	Ontologies and OWL	**W10**	Semantic Web in Knowledge Graphs
W3	SPARQL and Query Processing	**W11**	Trust and Provenance in the Semantic Web
W4	Semantic Annotation and Metadata	**W12**	Applications in Bioinformatics and Smart Cities
W5	Reasoning over Semantic Data	**W13**	Ethical and Legal Issues in the Semantic Web
W6	Publishing and Consuming Linked Open Data	**W14**	Project Development and Testing
W7	Semantic Web Standards and Tools	**W15**	**Case Studies / Projects / Demo**
W8	RDF and Linked Data Principles	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	**Course Title**	**CHs**	**Pre-Req**
CS644	**Data Compression**	3(3+0)	Nil

Recommended Texts

- Understanding Compression: Data Compression for Modern Developers 1st Edition, Colt McAnlis, AleksHaecky, O'Reilly Media, 2016, ISBN-13: 978-1491961537
- Data Compression: The Complete Reference, David Salomon, G. Motta, D. Bryant, Springer, 2006, ISBN- 978-1846286025.
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course enables students to understand basics of data compression, its importance and various useful techniques.

Course Objectives

- To introduce a wonderful treasure chest of information
- To introduce students to wide range of data compression methods, from simple text compression methods to the use of wavelets in image compression

Week Wise Topics to be Covered

W1	Introduction to Information Theory and Entropy	**W9**	Context Modeling and Prediction
W2	Lossless Compression Techniques: Huffman, Arithmetic Coding	**W10**	Recent Advances in Deep Compression
W3	Dictionary-Based Methods: LZW and Variants	**W11**	Compression for Resource-Constrained Devices
W4	Lossy Compression: Transform Coding	**W12**	Applications in Web, Cloud, and IoT
W5	Image Compression: JPEG and JPEG2000	**W13**	Security and Compression Interactions
W6	Audio Compression: MP3, AAC	**W14**	Project Design and Experimentation
W7	Video Compression: MPEG Standards	**W15**	**Case Studies / Projects / Demo**
W8	Performance Metrics and Evaluation	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS645	**Web Metrics**	3(3+0)	Nil

Recommended Texts

- Google Analytics Uncovered: How to Set Up and Maximize Ecommerce Data in Google Analytics, Eric Leuenberger, Independently published (November 28, 2018), ISBN-13 : 978-1790477920
- Advanced Web Metrics with Google Analytics 3rd Edition, Brian Clifton, Sybex Publisher, 2012, ISBN-13: 978-1118168448
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

The course covers the basic web usability techniques by introducing Google Analytic and its various applications and techniques.

Course Objectives

- Enabling students to Gain a conversational fluency of Google Analytics terms including key metrics and dimensions
- Discuss when a metrics can be used as a Key Performance Indicator (KPI)
- Set up properly Google Analytics tracking code

Week Wise Topics to be Covered

W1	Web Traffic; Importance and Monitoring. Web Analytics; Information, Decisions. ROI Methodologies
W2	Page Tags and Log files. Cookies in Web Analytics. Data and Its Accuracy
W3	Privacy Considerations. GA Reports. GA Interface
W4	Discoverability. Navigating Your Way Around. Google Analytics Accounts
W5	Backup: Keeping a Local Copy of Your Data. When and How to Use Accounts and Profiles
W6	Agencies and Hosting Providers. Setting Up Client Accounts
W7	Getting AdWords Data. Linking to Your AdWords Account. Advanced Implementation. The GA Workhorse
W8	E-Commerce Tracking. Online Campaign Tracking. Event Tracking.
MID EXAM/ MID OF SEMESTER	
W9	Customizing the GATC. Best Practices Configuration Guide
W10	Initial Configuration. Goals and Funnels,
W11	Why Segmentation Is Important. Filtering: Segmenting Visitors Using Filters
W12	Best-Practices Configuration Guide. Setting the Default Page. Excluding Unnecessary Parameters. Enabling E-commerce Reporting
W13	Enabling Site Search. Configuring Data-Sharing Settings. Goal Conversions and Funnels. The Importance of Defining Goals. What Funnel Shapes Can Tell You
W14	Using Visitor Data to Drive Website Improvement. Selecting and Preparing KPIs. What Is a KPI? Preparing KPIs
W15	Presenting Hierarchical KPIs via Segmentation. E-commerce Manager KPI Examples. Marketer KPI Examples
W16	Content Creator KPI Examples. Webmaster KPI Examples. Presentations. Using KPIs for Web. Why the Fuss about Web

Code	Course Title	CHs	Pre-Req
CS646	**Web Usability**	3(3+0)	Nil

Recommended Texts

- Design, usability - Wiley The Essential Guide To User Interface Design 3Rd Edition 2007, ISBN : 978-0-470-05342-3
- Handbook of Usability Testing ,How to Plan, Design and Conduct Effective Tests .May.2008 (Wiley) , ISBN : 978-0470185483
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course gives you the overview about, What Web Usability actually is? What is its importance in Web development? What are Rules/principles to make web sites more usable?

Course Objectives

- To help design better web sites by showing how the various user experience tools and techniques fit into real-world design and development processes.
- Giving hands-on practice in all the key areas of usability, from identifying your customers through to usability testing your web site with them.

Week Wise Topics to be Covered

W1	Introduction to Usability and Human-Centered Design	**W9**	Web Usability for Mobile and Responsive Interfaces
W2	Usability Principles and Heuristics	**W10**	Cognitive Load and Visual Design
W3	Information Architecture and Navigation Design	**W11**	User-Centered Design in Agile Environments
W4	Interaction Design Patterns	**W12**	Ethics and User Trust in UX
W5	Accessibility and Inclusive Design	**W13**	Trends in Web Usability Research
W6	Usability Testing Methods and Metrics	**W14**	Project Usability Testing and Iteration
W7	Prototyping Tools and Techniques	**W15**	**Case Studies / Projects / Demo**
W8	Evaluating Real-World Websites	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS647	**Medical Image Analysis**	3(3+0)	Nil

Recommended Texts

- Medical Image Analysis by Atam Dhawan
- Deep Learning for Medical Image Analysis by Zhou, Greenspan, and Shen
- Relevant journal articles (e.g., IEEE TMI, MICCAI proceedings)
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course introduces fundamental concepts, techniques, and tools used in the analysis of medical images. It covers medical imaging modalities, image pre-processing, segmentation, registration, feature extraction, classification, deep learning-based methods, and real-world applications in diagnostics and therapy.

Course Objectives

- Understand various medical imaging modalities and their characteristics.
- Apply classical and deep learning methods for image analysis.
- Evaluate image segmentation, classification, and registration techniques.
- Design and implement solutions for medical image interpretation and diagnosis.

Week Wise Topics to be Covered

W1	Introduction to Medical Imaging Modalities (MRI, CT, PET, Ultrasound)	**W9**	3D Visualization and Surface/Volume Rendering
W2	Medical Image File Formats and Standards (DICOM, NIfTI)	**W10**	Disease Detection and Classification using Machine Learning
W3	Image Preprocessing: Noise Reduction, Normalization, Histogram Equalization	**W11**	Deep Learning in Medical Image Analysis (CNNs, Transfer Learning)
W4	Image Enhancement and Contrast Adjustment, Deblurring blurred images	**W12**	Evaluation Metrics: Dice Coefficient, ROC, Precision-Recall
W5	Image Segmentation: Thresholding, Region Growing, Watershed	**W13**	Case Studies in Oncology, Neurology, and Cardiology
W6	Advanced Segmentation: Graph Cuts, Active Contours, U-Net	**W14**	Explainability and Interpretability, Explainable AI in medical imaging
W7	Feature Extraction: Texture, Shape, and Intensity Descriptors	**W15**	**Case Studies / Projects / Demo**
W8	Image Registration: Rigid, Non-Rigid, and Affine Transformations	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS648	**Vision Learning Models**	3(3+0)	Nil

Recommended Texts

- Deep Learning for Vision Systems by Mohamed Elgendy – https://www.oreilly.com/library/view/deep-learning-for/9781492030664/
- Computer Vision: Algorithms and Applications by Richard Szeliski – http://szeliski.org/Book/
- Deep Learning by Ian Goodfellow, Yoshua Bengio, and Aaron Courville – https://www.deeplearningbook.org/
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course explores the foundational and advanced concepts behind deep learning models used in computer vision tasks. It emphasizes learning-based approaches for image recognition, object detection, segmentation, and generative modeling.

Course Objectives

- Understand and apply key vision tasks such as image classification, object detection, and segmentation
- Explore CNN architectures and training paradigms
- Implement and experiment with state-of-the-art models such as ResNet, YOLO, and ViT
- Evaluate model performance using appropriate metrics and datasets

Week Wise Topics to be Covered

W1	Introduction to Vision Learning Models and Classical Computer Vision	**W9**	Vision Transformers (ViT, DeiT, and Hybrid Models)
W2	Neural Networks Fundamentals and CNN Basics	**W10**	Self-supervised Learning in Vision (SimCLR, MoCo)
W3	Deep Convolutional Architectures	**W11**	3D Vision and Depth Estimation
W4	Image Classification Case Studies and Transfer Learning	**W12**	Multi-view and Stereo Vision
W5	Object Detection	**W13**	Generative Models for Vision (GANs, Diffusion Models)
W6	YOLO, SSD and Real-Time Detection Architectures	**W14**	Vision in Adverse Conditions
W7	Semantic and Instance Segmentation (FCN, U-Net, Mask R-CNN)	**W15**	**Case Studies / Projects / Demo**
W8	Evaluation Metrics and Dataset Pipelines (COCO, Pascal VOC)	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	**Course Title**	**CHs**	**Pre-Req**
CS649	**Advance digital image processing**	3(3+0)	Nil

Recommended Texts

- Digital Image Processing by Rafael C. Gonzalez and Richard E. Woods – https://www.pearson.com/store/p/digital-image-processing/P100002980645
- Handbook of Image and Video Processing by Al Bovik – https://www.sciencedirect.com/book/9780121197926/handbook-of-image-and-video-processing
- IEEE Transactions on Image Processing – https://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=83
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course delves into advanced techniques in digital image processing, emphasizing algorithm design, practical implementations, and the mathematical principles underlying image analysis and enhancement.

Course Objectives

- Understand spatial and frequency domain image processing techniques
- Apply morphological, restoration, and color image processing methods
- Analyze and implement multi-resolution and wavelet-based techniques
- Explore applications including remote sensing, medical imaging, and image compression

Week Wise Topics to be Covered

W1	Review of Basic Image Processing and Linear Filtering	**W9**	Multiresolution Processing and Wavelet Transforms
W2	Advanced Image Enhancement Techniques	**W10**	Image Compression (JPEG, JPEG2000, MPEG)
W3	Image Restoration and Degradation Models	**W11**	Pattern Recognition and Image Classification
W4	Morphological Image Processing and Shape Analysis	**W12**	Image Registration and Motion Analysis
W5	Color Image Processing and Color Models	**W13**	Medical Image Processing and Biomedical Applications
W6	Image Segmentation Techniques (Thresholding, Clustering, Region-Based)	**W14**	Remote Sensing Image Processing
W7	Feature Extraction and Description (SIFT, SURF, ORB)	**W15**	**Case Studies / Projects / Demo**
W8	Frequency Domain Processing and FFT	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS841	**Advanced Topics in Semantic Web**	3(3+0)	Nil

Recommended Texts

- Advances in Web Semantics I, Chang, Elizabeth J.; Sycara, Katia, Springer 2009, ISBN 978-3-540-89784-2
- Foundations of Semantic Web Technologies, Pascal Hitzler, Markus Krotzsch, Sebastian Rudolph, CRC Press, 2009, ISBN 9781420090505
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course enable students to understand core concepts and ideas of Semantic Web and Ontology engineering

Course Objectives

- To understand the architecture and standards of the Semantic Web, including RDF, OWL, and SPARQL.
- To implement ontology modeling and reasoning techniques for semantic data integration.
- To analyze and apply linked data principles in real-world semantic web applications.
- To evaluate current research and technologies in semantic web services and intelligent agents.

Week Wise Topics to be Covered

W1	OWL-S - SOA requirements. Web semantics - Web services - Web-based services. Algorithms - artificial intelligence	W9	Scope, coverage, compliance. The Architecture of Proton
W2	Dynamic information retrieval. Failure level - failure scale -intelligent information retrieval. Mental health	W10	Proton knowledge management module Semantic information access Knowledge Access and the Semantic WEB
W3	Multi-agent systems - Multi-site software development	W11	Limitations of Current Search Technology. Role of Semantic Technology
W4	Ontology - ontology engineering. Ontology extraction - ontology-based multi-agent system	W12	Exploiting Domain-specific Knowledge Natural Language Generation from Ontologies. Generation from Taxonomies
W5	Representation languages - reputation Risk assessing agent - Semantic Web	W13	Generation of Interactive Information Sheets. Generation of Interactive Information Sheets. Ontosum and Miakt Summary Generators
W6	Semantic networks - software engineering ontology. Ontologies for Knowledge Management	W14	Device independence: information anywhere. Device Independence Architectures and Technologies. Ontology engineering methodologies
W7	Ontology usage Scenario, Ontology usage Scenario	W15	Diligent methodology. Argumentation support. Semantic Web Services? Approaches and Perspectives
W8	Topic-ontologies versus Schema-ontologies, Proton ontology	W16	The wsmo approach. The conceptual model?the web services modeling
MID EXAM/ MID OF SEMESTER			

Code	Course Title	CHs	Pre-Req
CS842	**Advanced Computer Vision**	3(3+0)	Nil

Recommended Texts

- Szeliski, R. (2022). Computer Vision: Algorithms and Applications (2nd ed.). Springer. https://doi.org/10.1007/978-3-030-59717-1
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press. (Chapters on vision) https://www.deeplearningbook.org
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course provides a deep dive into advanced methods and systems in computer vision. Topics include 3D vision, visual tracking, scene understanding, multimodal vision, and generative vision models. Emphasis is placed on recent research and large-scale vision architectures.

Course Objectives

- To understand deep learning techniques for vision tasks
- To implement advanced models for detection, segmentation, and recognition
- To understand recent architectures like Transformers in vision
- To implement state-of-the-art models in practical applications

Week Wise Topics to be Covered

W1	Introduction to Advanced Vision Systems: Overview and challenges	**W9**	Vision Transformers (ViT, Swin)
W2	Convolutional Neural Networks Revisited: Architectures and innovations	**W10**	Self-Supervised and Contrastive Learning for Vision
W3	Object Detection and Localization: R-CNN, YOLO, DETR	**W11**	Multimodal Vision and V+L Models
W4	Instance and Semantic Segmentation: Mask R-CNN, DeepLab	**W12**	Generative Models in Vision: GANs, diffusion models
W5	3D Computer Vision: Point clouds, depth estimation, SLAM	**W13**	Vision Applications in Healthcare, Security, and AR/VR
W6	Video Analysis and Action Recognition	**W14**	Ethical Considerations in Vision AI: Bias, surveillance, and privacy
W7	Visual Tracking and Motion Analysis	**W15**	**Case Studies / Projects / Demo**
W8	Multiview and Stereo Vision	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	**Course Title**	**CHs**	**Pre-Req**
CS843	**Advanced Social Network Analysis**	3(3+0)	Nil

Recommended Texts

- Newman, M. E. J. (2021). Networks: An Introduction (2nd ed.). Oxford University Press. https://doi.org/10.1093/oso/9780198805090.001.0001
- Wasserman, S., & Faust, K. (1994). Social Network Analysis: Methods and Applications. Cambridge University Press. https://doi.org/10.1017/CBO9780511815478
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course delves into the theoretical foundations and computational techniques for analyzing complex social networks. Topics include graph theory, network dynamics, influence modeling, community detection, and applications in online platforms, epidemiology, and misinformation.

Course Objectives

- To understand graph-theoretic and statistical models of social networks
- To implement community detection and influence propagation analyses
- To understand dynamic and multilayer network structures
- To implement metrics and visualization for network insights

Week Wise Topics to be Covered

W1	Introduction to Social Networks: Structure, metrics, and typologies	**W9**	Signed and Directed Networks
W2	Graph Theory Basics: Degree, centrality, paths, cliques	**W10**	Social Network Embeddings: Node2vec, GraphSAGE
W3	Network Models: Random graphs, small-world, scale-free	**W11**	Graph Neural Networks for Social Data
W4	Community Detection: Modularity, Louvain, spectral clustering	**W12**	Network Robustness and Resilience
W5	Dynamic Networks: Temporal evolution, link prediction	**W13**	Privacy and Ethics in Social Network Analysis
W6	Influence and Diffusion: Threshold and cascade models	**W14**	Applications: Marketing, Epidemiology, and Recommender Systems
W7	Information Spread and Misinformation	**W15**	**Case Studies / Projects / Demo**
W8	Multiplex and Heterogeneous Networks	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS844	**Advanced Complex Networks**	3(3+0)	Nil

Recommended Texts

- Barabási, A.-L. (2016). Network Science (1st ed.). Cambridge University Press. https://doi.org/10.1017/CBO9781316276235
- Boccaletti, S., et al. (2020). The Structure and Dynamics of Complex Networks: A Review. Physics Reports. https://doi.org/10.1016/j.physrep.2020.01.005
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course delves into the modeling, analysis, and dynamics of complex networks found in biology, social systems, infrastructure, and technology. It emphasizes statistical mechanics, multilayer networks, critical phenomena, and emergent behaviors.

Course Objectives

- To understand theoretical foundations of complex network systems
- To implement modeling of scale-free, small-world, and temporal networks
- To understand network robustness and resilience analysis
- To implement algorithms for multiplex and multilayer networks

Week Wise Topics to be Covered

W1	Introduction to Complex Systems and Networks	**W9**	Dynamical Processes on Networks: Spreading, synchronization
W2	Structural Properties of Networks	**W10**	Percolation and Phase Transitions in Networks
W3	Random Graphs and Network Models: ER, WS, BA	**W11**	Network Resilience and Robustness
W4	Community Structure and Modularity	**W12**	Applications in Epidemiology and Infrastructure
W5	Spectral Properties and Network Laplacians	**W13**	Biological and Ecological Networks
W6	Temporal and Evolving Networks	**W14**	AI and GNNs in Complex Network Analysis
W7	Multilayer and Interdependent Networks	**W15**	**Case Studies / Projects / Demo**
W8	Centrality, Controllability, and Influence	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS845	**Explainable Artificial Intelligence**	3(3+0)	Nil

Recommended Texts

- Molnar, C. (2020). Interpretable Machine Learning. Lulu.com (available open-access). https://christophm.github.io/interpretable-ml-book/
- Doshi-Velez, F., & Kim, B. (2017). Towards a Rigorous Science of Interpretable Machine Learning. arXiv. https://doi.org/10.48550/arXiv.1702.08608
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course explores the emerging field of Explainable AI (XAI), focusing on interpretability techniques, transparency models, and human-centered AI design. It includes model-agnostic and model-specific methods, post-hoc explanations, and regulatory implications across critical application domains.

Course Objectives

- To understand principles of transparent and interpretable AI
- To implement explainability methods for deep and structured models
- To understand human-centric evaluation of model explanations
- To implement XAI pipelines in real-world systems

Week Wise Topics to be Covered

W1	Introduction to Explainable AI	**W9**	Architectures for Explainable AI
W2	Interpretability vs. Explainability	**W10**	XAI in Healthcare, Finance, and Legal Domains
W3	Model-Agnostic Explanation Techniques (e.g., LIME, SHAP)	**W11**	Adversarial Attacks and Robust Explanations
W4	Intrinsically Interpretable Models (e.g., Decision Trees, Rule-Based Models)	**W12**	Causal Inference and Counterfactual Explanations
W5	Explanation Techniques for Neural Networks (e.g., Saliency Maps, Grad-CAM)	**W13**	Fairness, Accountability, and Transparency (FAT) in AI
W6	Post-Hoc Explanation Methods for Black-Box Models	**W14**	Legal and Ethical Considerations in XAI
W7	Human-Centered Explanations and Cognitive Perspectives	**W15**	**Case Studies / Projects / Demo**
W8	Evaluation Metrics for Explainability	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS941	**Cognitive Modeling and Human-Like AI**	3(3+0)	Nil

Recommended Texts

- Sun, R. (2016). Cognitive Architecture: Designing for How We Respond to the Future (2nd ed.). Oxford University Press. https://doi.org/10.1093/acprof:oso/9780190268771.001.0001
- Thórisson, K. R. (2019). Artificial General Intelligence. Springer. https://doi.org/10.1007/978-3-030-07004-0
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course investigates the design and implementation of computational models that simulate human cognition. It combines insights from psychology, neuroscience, linguistics, and AI to create models that mimic reasoning, learning, problem-solving, and decision-making. Students will engage with both symbolic and sub-symbolic modeling techniques, emphasizing interpretability and alignment with human behavior.

Course Objectives

- To understand cognitive architectures and modeling frameworks
- To implement agents that mimic human decision-making processes
- To understand psychological constraints in AI systems
- To implement human-like learning models in computational settings

Week Wise Topics to be Covered

W1	Introduction to Cognitive Modeling: Overview, goals, and interdisciplinary context	**W9**	Cognitive Models in Perception and Attention
W2	Human Cognitive Architecture: Working memory, long-term memory, and executive functions	**W10**	Language and Thought: Models of natural language understanding and generation
W3	Symbolic Cognitive Models: Production systems, logic-based reasoning	**W11**	Theory of Mind and Social Cognition
W4	Connectionist Models: Neural networks and associative learning	**W12**	Emotion and Affect in Cognitive Architectures
W5	Bayesian Models of Cognition: Probabilistic reasoning and inference	**W13**	Validation and Evaluation of Cognitive Models
W6	Hybrid Cognitive Architectures: Combining symbolic and neural models	**W14**	Applications in HCI, Robotics, and Education
W7	Learning Mechanisms in Cognitive Models: Imitation, reinforcement, and supervised learning	**W15**	**Case Studies / Projects / Demo**
W8	Modeling Human Problem-Solving and Decision-Making	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS942	**Human-Centered and Ethical AI**	3(3+0)	Nil

Recommended Texts

- Artificial Unintelligence by Meredith Broussard – https://mitpress.mit.edu/9780262537018/artificial-unintelligence/
- Ethics of Artificial Intelligence and Robotics, Stanford Encyclopedia of Philosophy – https://plato.stanford.edu/entries/ethics-ai/
- Human-Centered AI by Ben Shneiderman – https://global.oup.com/academic/product/human-centered-ai-9780192845290
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course examines the social, ethical, and human-centered dimensions of AI technologies, highlighting fairness, accountability, transparency, and inclusiveness in system design and deployment.

Course Objectives

- Analyze ethical challenges and societal impacts of AI
- Understand frameworks for fairness, accountability, and transparency in AI
- Design and evaluate human-centered AI systems
- Explore interdisciplinary perspectives on responsible AI development

Week Wise Topics to be Covered

W1	Introduction to Human-Centered AI and Ethical Frameworks	**W9**	Designing Ethical AI Systems (HCD Principles, UX Design)
W2	Bias in AI, Sources, Types, and Case Studies	**W10**	Participatory Design and Stakeholder Engagement
W3	Fairness in Machine Learning and Algorithmic Audits	**W11**	AI and Accessibility for Marginalized Communities
W4	Privacy, Surveillance, and Data Governance	**W12**	Ethical Challenges in Autonomous Systems
W5	Transparency and Explainability in AI Systems	**W13**	AI in Healthcare, Education, and Law
W6	Human-AI Interaction and Trustworthy AI	**W14**	AI Misuse, Deepfakes, and Information Integrity
W7	Legal and Regulatory Aspects of AI Ethics	**W15**	**Case Studies / Projects / Demo**
W8	Social Implications, Labor, Access, and Global Impact	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Program - MS/PhD (CS)	**Domain – 5 : Software Engineering**

Code	Course Title	CHs	Pre-Req
CS651	**Requirement Engineering**	(3 0 3)	Nil

Recommended Texts

- Sommerville, I., & Sawyer, P. (2022). Requirements Engineering: Processes and Techniques (4th ed.). Wiley. https://doi.org/10.1002/9781119484542
- Nuseibeh, B., & Easterbrook, S. (2000). Requirements Engineering: A Roadmap. In ICSE 2000. https://doi.org/10.1145/336512.336531
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course enables students to understand basics of requirement engineering, its importance and various models used for gathering it

Course Objectives

- To understand elicitation and specification techniques
- To implement requirements modeling using formal and semi-formal methods
- To understand requirements validation and verification processes
- To implement traceability and change management in requirements

Week Wise Topics to be Covered

W1	Software Development	**W9**	Requirements in Agile and DevOps Environments
W2	Types of Requirements: Functional, Non-functional, and Domain Requirements	**W10**	Requirements Engineering Tools and Automation
W3	Stakeholder Identification and Requirement Elicitation Techniques	**W11**	Legal, Ethical, and Contractual Aspects of Requirements
W4	Requirements Analysis and Prioritization	**W12**	Requirements for Safety-Critical and Regulated Systems
W5	Modeling Requirements using Use Cases and Scenarios	**W13**	Global Software Development and Cross-Cultural Requirements
W6	Prototyping and Requirements Validation Techniques	**W14**	Project Work: Elicitation and Specification from Case Scenario
W7	Requirements Specification: Formats and Standards (IEEE, etc.)	**W15**	**Case Studies / Projects / Demo**
W8	Managing Requirements Volatility and Traceability	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS652	**Software System Architecture**	3(3+0)	Nil

Recommended Texts

- Bass, L., Clements, P., & Kazman, R. (2021). Software Architecture in Practice (4th ed.). Addison-Wesley Professional. https://doi.org/10.1007/978-3-662-46816-2
- Shaw, M., & Garlan, D. (1996). Software Architecture: Perspectives on an Emerging Discipline. Prentice Hall. ISBN-13: 978-0131829570
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course enables students to understand the concepts, principles and standards underlying modern software architecting.

Course Objectives

- To understand architectural styles and quality attributes
- To implement architectural modeling and documentation techniques
- To understand architectural evaluation and analysis methods
- To implement architecture in scalable and maintainable systems

Week Wise Topics to be Covered

W1	Introduction to Software Architecture. Views and Viewpoints. Architecture Description Languages (ADL)	W9	Architectural design decisions. System organization
W2	Rational Unified Process. Design Planning. Architectural Views	W10	Modular decomposition styles. Control styles
W3	Use Case View. Logical View. Implementation, Process, and Deployment Views	W11	Distributed systems architectures. Multiprocessor architectures. Client-server architectures. Distributed object architectures
W4	Metrics to Manage Software Risks. Patterns,	W12	Inter-organizational distributed computing. Application architectures
W5	CORBA-based Architecture. CORBA-IDL. Designing CORBA Systems	W13, W14	Data processing systems. Transaction processing systems. Event processing systems. Language processing systems
W6	Implementing CORBA Applications. Component-based Development, CORBA Component Model (CCM),	W15	Real-time software design. Real-time operating system
W7, W8	Specialized Software Architectures. System models. Context models. Behavioral models. Data models. Object models. Structured methods.	W16	Monitoring and control system. Presentations. Data acquisition systems
MID EXAM/ MID OF SEMESTER			

Code	Course Title	CHs	Pre-Req
CS653	**Advanced Software Quality Assurance**	3(3+0)	Nil

Recommended Texts

- Software Quality Assurance, Ivan Mistrík Richard Mark Soley Nour Ali John Grundy, PH. Bedir Tekinerdogan, Elsevier Science, 2015, ISBN: 9780128023013
- Software Quality Assurance: Principles and Practice, Nina S. Godbole, Alpha Science International Limited, 2004, ISBN-13: 978-1842651766
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course enable introduces students to processes, methods and techniques for developing quality software.

Course Objectives

- To introduce students to high quality software that.
- It includes the processes, methods and techniques for developing quality software, for assessing software quality, and for maintaining the quality of software.

Week Wise Topics to be Covered

W1	Introduction to Software Quality. Software Process, Software Process Models. eXtreme Programming	W9	Classification: QA as Dealing with Defects. Defect Prevention. Education and training. Formal method
W2	Software Testing, Systematic Testing, Black Box Testing. functional, input, output, partitioning and gray box testing,White Box Testing - coverage, path	W10	Defect Reduction. Inspection: Direct fault detection and removal. Testing: Failure observation and fault removal
W3	Decision and mutation testing. Continuous Testing - regression, defect testing, Test Automation - test maintenance and analysis	W11	Other techniques and risk identification. Defect Containment. Software fault tolerance
W4	Harnesses, tracking, tools. Software Inspection, refactoring	W12	Safety assurance and failure containment. Quality Engineering: Activities and Process
W5	Systematic Inspection. inspection process, formal reviews. Code Inspection, continuous inspection	W13	Quality Planning: Goal Setting and Strategy Formation. Quality Assessment and Improvement
W6	Alternative Verification and Validation Techniques. Dynamic analysis, static analysis,	W14	Defect Prevention and Process Improvement. Root Cause Analysis for Defect Prevention
W7	Formal methods, Software Metrics. Product Quality Metrics, Process Metrics	W15	Education and Training for Defect Prevention. Other Techniques for Defect Prevention
W8	Quality: Perspectives and	W16	Analysis and modeling for defect prevention. Presentations. Technologies, standards, and methodologies for defect prevention
MID EXAM/ MID OF SEMESTER			

Code	Course Title	CHs	Pre-Req
CS657	**Advanced Software Project Management**	3(3+0)	Nil

Recommended Texts

- Hughes, B., Cotterell, M., & Mall, R. (2020). Software Project Management (6th ed.). McGraw-Hill Education. https://www.mheducation.com/highered/product/software-project-management-hughes-cotterell/M9781260570570.html
- Pressman, R. S., & Maxim, B. R. (2020). Software Engineering: A Practitioner's Approach (9th ed.). McGraw-Hill Education. https://doi.org/10.1036/0078022126
- Sommerville, I. (2015). Software Engineering (10th ed.). Pearson. https://www.pearson.com/store/p/software-engineering/P100000623618
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course explores advanced practices and methodologies for managing large and complex software projects. It covers cost estimation, risk management, quality assurance, scheduling, resource allocation, and project governance. Emphasis is placed on both traditional and agile approaches to managing software projects at scale.

Course Objectives

- To understand the principles, frameworks, and life cycles of software project management.
- To implement estimation techniques, scheduling models, and risk management strategies.
- To analyze and manage quality assurance, configuration, and human factors in software projects.
- To evaluate agile, DevOps, and hybrid project management practices for scalable software delivery.

Week Wise Topics to be Covered

W1	Introduction to Software Project Management and Lifecycle Models	**W9**	Agile Project Management and Scrum Frameworks
W2	Project Initiation, Planning, and Stakeholder Analysis	**W10**	Scaling Agile with SAFe, LeSS, and Hybrid Approaches
W3	Effort and Cost Estimation Techniques (COCOMO, Function Points)	**W11**	Project Configuration and Change Management
W4	Scheduling Techniques and Critical Path Analysis	**W12**	Software Procurement and Outsourcing Management
W5	Risk Identification, Assessment, and Mitigation Strategies	**W13**	Ethical, Legal, and Cultural Considerations in Projects
W6	Project Quality Planning and Assurance Techniques	**W14**	Project Work: Planning and Documentation for a Software Project
W7	Human Resource and Communication Management	**W15**	**Case Studies / Projects / Demo**
W8	Project Monitoring, Control, and Earned Value Analysis	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS658	**Software Testing**	3(3+0)	Nil

Recommended Texts

- Foundations of Software Testing by by Cem Kaner, Rebecca L Fiedler, 2013 by Context-Driven Press, 2013, ISBN-13: 978-0989811927
- Software Testing: Concepts and Operations,Ali Mili, Fairouz Tchier,ISBN: 978-1-118-66287-8,July 2015, ISBN: 978-1-118-66287-8
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

The course covers the important software testing by studying various software testing techniques in details.

Course Objectives

- To find defects which may get created by the programmer while developing the software.
- To gain confidence in and providing information about the level of quality.
- To prevent defects, to make sure that the end result meets the business and user requirements
- To gain the confidence of the customers by providing them a quality product.

Week Wise Topics to be Covered

Week	Topics	Week	Topics
W1	Software Quality Attributes, Boolean Attributes, Statistical Attributes	W9	Myth 7: Testers are Responsible for Quality of Product, Myth 8: Test Automation should be used Wherever Possible to Reduce Time, Myth 9: Anyone can Test a Software Application, Myth 10: A Tester's Only Task is to Find Bugs, QA, QC, AND TESTING
W2	Operational Attributes, Usability Attributes, Business Attributes, Structural Attributes		
W3	Why is testing necessary? What is testing? Testing Principles, Testing Objectives		
W4	Fundamental test process, Fundamental test process, Psychology of testing Testers and code of ethics	W10	Testing, Quality Assurance, and Quality Control, Audit and Inspection, Testing and Debugging
W5	Paradoxes and main principles, Types of tests Test and maintenance. Test organization Test planning and estimation	W11	ISO STANDARDS: ISO/IEC 9126, ISO/IEC 9241, ISO/IEC 25000:2005
		W12	TYPES OF TESTING: Manual Testing, Automation Testing, What to Automate?
W6	Test progress monitoring and control, Who does Testing? When to Start Testing?	W13	When to Automate? How to Automate? Software Testing Tools, TESTING METHODS, Black Box Testing
W7	When to Stop Testing? Verification & Validation, MYTHS, Myth 1: Testing is Too Expensive, Myth 2: Testing is Time Consuming	W14	WhiteBox Testing, Grey Box Testing, White Box Testing, Grey Box Testing
W8	Only Fully Developed Products are Tested, Myth 4: Complete Testing is Possible, Myth 5: A Tested Software is Bug, Myth 6: Missed Defects are due to Testers	W15	Integration Testing, System Testing, Regression Testing, Acceptance Testing
		W16	Non Functional Testing, Usability Testing, Security Testing, Portability Testing
MID EXAM/ MID OF SEMESTER			

Code	Course Title	CHs	Pre-Req
CS659	**Foundations of Quantum Computing**	3(3+0)	Nil

Recommended Texts

- Nielsen, M. A., & Chuang, I. L. (2021). Quantum Computation and Quantum Information: 10th Anniversary Edition. Cambridge University Press. https://doi.org/10.1017/9781107295094
- Rieffel, E. G., & Polak, W. H. (2011). Quantum Computing: A Gentle Introduction. MIT Press. https://doi.org/10.7551/mitpress/9780262015066.001.0001
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals indexed in Q1 or Q2 WoS

Course Description

This course introduces the theoretical and computational foundations of quantum computing. It covers quantum mechanics principles relevant to computation, quantum logic gates, and key algorithms that outperform classical counterparts. Emphasis is placed on problem modeling and quantum algorithm design.

Course Objectives

- To understand the mathematical foundations of quantum computing
- To analyze and simulate quantum circuits and gates
- To apply quantum algorithms such as Grover's and Shor's
- To evaluate the computational advantages of quantum models

Week Wise Topics to be Covered

W1	Introduction to quantum computing and qubits	**W9**	Quantum complexity classes (BQP, QMA)
W2	Linear algebra and quantum mechanics essentials	**W10**	Quantum cryptography: BB84 and QKD
W3	Quantum states, bra-ket notation, and Bloch sphere	**W11**	Physical implementation of quantum systems
W4	Quantum gates and quantum circuits	**W12**	Simulating quantum systems with Qiskit
W5	Entanglement and quantum teleportation	**W13**	Recent advances in quantum hardware
W6	Deutsch-Jozsa and Grover's algorithm	**W14**	Project implementation and evaluation
W7	Shor's algorithm and quantum Fourier transform	**W15**	**Case Studies / Projects / Demo**
W8	Quantum error correction and noise models	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS751	**Green and Sustainable Software Engineering**	3(3+0)	Nil

Recommended Texts

- Naumann, S., & Kern, E. (2022). Sustainable Software Engineering: Concepts and Techniques. Springer. https://doi.org/10.1007/978-3-030-93962-3
- Lago, P., & Procaccianti, G. (2022). Software Sustainability: Research and Practice. Springer. https://doi.org/10.1007/978-3-031-05760-4
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals indexed in Q1 or Q2 WoS

Course Description

This course focuses on environmentally responsible and sustainable approaches in software engineering. Students will learn strategies to reduce software-induced energy consumption and promote long-term sustainability across the software lifecycle.

Course Objectives

- To understand principles of sustainable and green software design
- To evaluate energy consumption in software systems
- To implement energy-aware development practices
- To apply sustainability metrics to software projects

Week Wise Topics to be Covered

W1	Introduction to sustainability in software engineering	**W9**	Empirical studies and best practices
W2	Energy consumption in software systems	**W10**	Sustainability metrics and indicators
W3	Green design patterns and architectural principles	**W11**	Sustainability in DevOps and CI/CD
W4	Measurement and monitoring of software energy	**W12**	Policy and regulatory frameworks
W5	Green-aware software requirements and modeling	**W13**	AI for energy-efficient software
W6	Refactoring for energy efficiency	**W14**	Project implementation and validation
W7	Tools and frameworks for sustainable software	**W15**	**Case Studies / Projects / Demo**
W8	Case studies on green software projects	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS752	**Middle Ware**	3(3+0)	Nil

Recommended Texts

- Coulouris, G., Dollimore, J., Kindberg, T., & Blair, G. (2012). Distributed Systems: Concepts and Design (5th ed.). Pearson. https://doi.org/10.1016/C2010-0-64636-0
- Emmerich, W. (2000). Engineering Distributed Objects. Wiley. https://doi.org/10.1002/9781118799270
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals indexed in Q1 or Q2 WoS

Course Description

This course covers the design, development, and deployment of middleware technologies that connect distributed applications. Topics include service-oriented architecture, messaging, RPC, and modern frameworks used in cloud-native and IoT environments.

Course Objectives

- To understand middleware architecture and components
- To implement and evaluate different middleware paradigms
- To analyze the role of middleware in distributed applications
- To apply middleware solutions in real-time and cloud systems

Week Wise Topics to be Covered

W1	Introduction to middleware and distributed systems	**W9**	Middleware for cloud and microservices
W2	Types of middleware: message-oriented, object-oriented, etc.	**W10**	Middleware for mobile and IoT applications
W3	Remote procedure calls (RPC) and messaging systems	**W11**	Data integration and enterprise application middleware
W4	Middleware design patterns and architecture	**W12**	Modern trends: serverless and event-driven architectures
W5	Interoperability and communication protocols	**W13**	Case studies in middleware deployments
W6	Middleware for real-time and embedded systems	**W14**	Project development and evaluation
W7	Service-oriented architecture and web services	**W15**	**Case Studies / Projects / Demo**
W8	Security and fault tolerance in middleware	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS851	**AI for Code and Software Engineering**	3(3+0)	Nil

Recommended Texts

- Allamanis, M. (2021). Machine Learning for Source Code Analysis and Generation. Foundations and Trends® in Programming Languages, 7(1–2), 1–160. https://doi.org/10.1561/2500000031
- Raychev, V., Vechev, M., & Yahav, E. (2016). Code Completion with Statistical Language Models. Springer. https://doi.org/10.1007/978-3-319-39306-8_3
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course examines how AI, especially deep learning and NLP, can be applied to code understanding, generation, completion, and software maintenance. Topics include code representation learning, program synthesis, code summarization, and intelligent development tools.

Course Objectives

- To understand machine learning techniques applied to software engineering tasks
- To implement models for code completion, synthesis, and bug detection
- To understand AI-driven software lifecycle automation
- To implement intelligent systems that assist software developers

Week Wise Topics to be Covered

W1	Introduction to AI for Software Engineering	**W9**	Refactoring and Clone Detection
W2	Code as Data: Structure, Syntax, Semantics	**W10**	Test Case Generation and Fuzzing with AI
W3	Representing Code: ASTs, CFGs, embeddings	**W11**	Machine Learning for Version Control and CI/CD
W4	Code Language Models: GPT, Codex, CodeBERT	**W12**	Ethics and Fairness in AI-Generated Code
W5	Program Synthesis and Code Generation	**W13**	Intelligent IDEs and Human-in-the-Loop Tools
W6	Code Completion and Autocompletion Models	**W14**	Large Language Models and Software Engineering Tasks
W7	Bug Detection and Automated Repair	**W15**	**Case Studies / Projects / Demo**
W8	Code Summarization and Documentation Generation	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS852	**Engineering Large-Scale AI Systems**	3(3+0)	Nil

Recommended Texts

- Hellerstein, J. M., & Gonzalez, J. E. (2020). Data Systems for Machine Learning. Cambridge University Press. https://doi.org/10.48550/arXiv.1809.00343
- Zaharia, M., & Xin, R. S. (2022). Spark + AI Summit: Building Scalable AI Infrastructure. O'Reilly Media. https://databricks.com/resources
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course addresses the design, development, deployment, and maintenance of AI systems at scale. It covers distributed learning, model deployment, monitoring, optimization, and system reliability in cloud and edge environments, preparing students to engineer production-grade AI solutions.

Course Objectives

- To understand design principles for scalable AI systems
- To implement distributed and cloud-based model training pipelines
- To understand performance, reproducibility, and fault tolerance
- To implement continuous integration and deployment for AI

Week Wise Topics to be Covered

W1	Overview of AI Systems Engineering Lifecycle	**W9**	Monitoring and Observability (ML Ops and Metrics)
W2	Architectures for Scalable AI (Microservices, Pipelines)	**W10**	Edge and Federated AI Deployments
W3	Data Engineering for AI at Scale	**W11**	Cloud Platforms and Auto-Scaling
W4	Model Training Infrastructure: GPUs, TPUs, Clusters	**W12**	Security and Privacy in AI Systems
W5	Distributed and Parallel Training (Horovod, DDP)	**W13**	Cost-Aware and Energy-Efficient AI Engineering
W6	Model Optimization: Pruning, Quantization, Distillation	**W14**	Real-World AI System Failures and Successes
W7	Serving Models at Scale (ONNX, TensorRT, TFX)	**W15**	**Case Studies / Projects / Demo**
W8	Model Deployment: Docker, Kubernetes, CI/CD Pipelines	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS951	**Quantum Computing and Machine Learning**	3(3+0)	Nil

Recommended Texts

- Schuld, M., & Petruccione, F. (2021). Machine Learning with Quantum Computers (2nd ed.). Springer. https://doi.org/10.1007/978-3-030-17801-2
- Nielsen, M. A., & Chuang, I. L. (2021). Quantum Computation and Quantum Information (10th Anniversary ed.). Cambridge University Press. https://doi.org/10.1017/CBO9780511976667
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course introduces quantum computing concepts and explores their intersection with machine learning. Students will learn quantum algorithms, quantum-enhanced ML models, and how to implement hybrid quantum-classical systems for data-driven tasks.

Course Objectives

- To understand quantum algorithms applicable to machine learning
- To implement quantum circuits for optimization and classification
- To understand quantum speedups and their limitations
- To implement hybrid quantum-classical learning models

Week Wise Topics to be Covered

W1	Foundations of Quantum Computing	**W9**	Variational Quantum Circuits and VQAs
W2	Qubits, Superposition, and Entanglement	**W10**	Quantum Neural Networks and Quantum Boltzmann Machines
W3	Quantum Gates and Circuits	**W11**	Hybrid Quantum-Classical Models
W4	Quantum Algorithms: Deutsch-Jozsa, Grover's, Shor's	**W12**	Implementing QML in Qiskit and PennyLane
W5	Quantum Simulation and State Preparation	**W13**	Challenges in Quantum ML: Noise, Decoherence, and Scaling
W6	Introduction to Quantum Machine Learning (QML)	**W14**	Applications in Chemistry, Finance, Optimization
W7	Quantum Data Encoding and Feature Maps	**W15**	**Case Studies / Projects / Demo**
W8	Quantum Kernels and Quantum SVMs	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS661	**Advanced Machine Learning**	3(3+0)	NIL

Recommended Texts

- Machine Learning Paradigms, Artificial Immune Systems and their Applications in Software Personalization, Dionysios Sotiropoulos, George A. Tsihrintzis, Springer International Publishing, 2016, ISBN 978-3-319-47192-1, 2016
- E. Alpaydin, Introduction to Machine Learning, 2nd edition, The MIT Press., 2010, ISBN 978-0-262-01243-0, 2010
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course provides a broad introduction to machine learning, datamining, and statistical pattern recognition. Topics include: (i) Supervised learning (parametric/non-parametric algorithms, support vector machines, kernels, neural networks). (ii) Unsupervised learning (clustering, dimensionality reduction, recommender systems). (iii) Best practices in machine learning (bias/variance theory; innovation process in machine learning and AI). The course will also draw from numerous case studies and applications, so that you'll also learn how to apply learning algorithms to building smart robots (perception, control), text understanding (web search, anti-spam), computer vision, medical informatics, audio, database mining, and other areas.

Course Objectives

- Main objective of the course is providing graduate students with an understanding of practical and theoretical methods, mathematics, and algorithms required for applications and research in Machine Learning.
- Apply user centered design and usability engineering principles as they design a wide variety of software user interfaces.

Week Wise Topics to be Covered

W1	Introduction to Machine Learning. Types of Machine Learning, Supervised Learning. Classification, Classification based on Distance Measurement	W8	Example of Machine Learning: Photo OCR
		MID EXAM/ MID OF SEMESTER	
W2	Linear Regression with One variable. Linear Regression with multiple variable. Logistic Regression	W9	k-Nearest Neighbors Algorithm
		W10	Bayesian Learning, Decision Trees
		W11	Evolutionary Algorithms
W3	Regularization	W12	Clustering. Learning Vector Quantization. Back Propagation
W4	Neural Networks	W13	Bootstrap Aggregation
W5	Support Vector machine. Dimensionality Reduction	W14	Random Forest
W6	Anomaly Detection	W15	Stacked Generalization
W7	Large Scale Machine Learning	W16	Big Data and MapReduce

Code	Course Title	CHs	Pre-Req
CS662	**Advanced Artificial Neural Networks**	3(3+0)	Nil

Recommended Texts

- L. C. Jain, "Recent Advances in Artificial Neural Networks", Taylor & Francis, 13-Dec-2017
- Graupe Daniel, "Principles Of Artificial Neural Networks" (3rd Edition), World Scientific, 31-Jul-2013
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course will cover basic neural network architectures and learning algorithms, for applications in pattern recognition, image processing, and computer vision. Three forms of learning will be introduced (i.e., supervised, unsupervised and reinforcement learning) and applications of these will be discussed. The students will have a chance to try out several of these models on practical problems. It is primarily intended for students who are interested in doing research in the areas of Neural Networks and Computer Vision. There are many open problems in this area suitable for investigation by students leading to a professional paper or thesis.

Course Objectives

- To introduce the neural networks for classification and regression;
- To give design methodologies for artificial neural networks;
- To provide knowledge for network tuning and overfitting avoidance;
- To offer neural network implementations in Matlab/Octave/Python;

Week Wise Topics to be Covered

Week	Topic
W1	Introduction
W2	Fundamental concepts: neuron models and basic learning rules
W3	Learning of a single neuron and single layer neural networks: Part one: Learning of a single neuron, Part two: Learning of single layer neural networks.
W4	Multilayer neural networks and back-propagation
W5	Associative memory
W6	Self-organizing neural networks
W7	Self-organizing feature map
W8	Data visualization with self-organizing feature map, Data for Project V
MID EXAM/ MID OF SEMESTER	
W9	RBF neural networks and support vector machines
W10, W11	Neural Network Trees
W12, W13	Introduction to Convolutional Neural Network.
W14	Local Vs. Global Minima
W15	The action of well-trained nets
W16	Generalization and Overtraining

Code	Course Title	CHs	Pre-Req
CS663	**Soft Computing**	3(3+0)	Nil

Recommended Texts

- Principles of Soft Computing, 3ed Kindle Edition, S.N. Sivanandam, S.N. Deepa, Wiley (January 1, 2018), ISBN-13: 978-8126577132
- Soft Computing and Its Applications, Rafik A. Aliev, World Scientific Publishing Co, ISBN:9810247001, 2001
- An Introduction to Genetic Algorithm, Melanic Mitchell, MIT Press, ISBN:0-262-13316-4, 1998
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course provides a broad introduction to soft computing approaches like Neural networks, fuzzy logic, genetic algorithm. Soft computing is based on some biological inspired methodologies such as genetics, evolution, ant's behaviors, particles swarming, human nervous systems, etc. Soft computing provides solution to the problems when we don't have any mathematical modeling of problem solving (i.e., algorithm), need a solution to a complex problem in real time, easy to adapt with changed scenario and can be implemented with parallel computing. The course will also draw from numerous case studies and applications.

Course Objectives

- To give students knowledge of soft computing theories fundamentals, i.e. fundamentals of artificial neural networks, fuzzy sets and fuzzy logic and genetic algorithms.
- They will be able to design program systems using approaches of these theories for solving various real-world problems.

Week Wise Topics to be Covered

Week	Topics
W1	Introduction to Soft Computing, Introduction, Fuzzy Computing, Neural Computing, Genetic Algorithms
W2	Importance of tolerance of imprecision and uncertainty, Adaptive Resonance Theory, Applications. Biological and artificial neuron, neural networks
W3	Fundamentals of Neural Network: Introduction, Model of Artificial Neuron, Architectures, Taxonomy of NN Systems, Single-Layer NN System, Applications
W4	Back Propagation Network: Background, Back-Propagation Learning, Back-Propagation Algorithm.
W5	Adaptive feedforward multilayer networks. Neural networks as associative memories (Hopfield, BAM, SDM).
W6	Solving optimization problems using neural networks. Stochastic neural networks, Boltzmann machine. Rough sets, Chaos
W7	Fuzzy Set Theory: Introduction, Fuzzy set: Membership, Operations, Properties; Fuzzy Relations. Fuzzy Systems: Introduction, Fuzzy Logic, Fuzzification, Fuzzy Inference, Fuzzy Rule Based System
W8	Defuzzification, Solving optimization problems, Concept of GA, GA Operators: Encoding
MID EXAM/ MID OF SEMESTER	
W9	GA Operators: Crossover-I, GA Operators: Crossover-II, GA Operators: Mutation
W10	Hybrid Systems
W11	Integration of Neural Networks, Fuzzy Logic and Genetic Algorithms
W12	GA Based Back Propagation Networks. Fuzzy Back Propagation Networks
W13	Fuzzy Associative Memories
W14	Application of Neural Networks
W15	Application of Fuzzy Logic
W16	Application of Genetic Algorithm

Code	Course Title	CHs	Pre-Req
CS664	**Evolutionary Computing**	3(3+0)	Nil

Recommended Texts

- Introduction to Evolutionary Computing, A. E. Eiben, J. E. Smith, Springer, ISBN 978-3-662-05094-1, 2013
- Evolutionary Computation1: Basic Algorithms and Operators, Thomas Back, D.B Fogel, Z Michalewicz, IoP, ISBN 9780750306645, 2000
- Evolutionary Computation for Modeling and Optimization, Daniel Ashlock, Springer, ISBN: 978-0-387-31909-4, 2006
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course provides an introduction to Evolutionary computation. History of evolutionary computation; major areas: genetic algorithms, evolution strategies, evolution programming, genetic programming, classifier systems. This course will also cover constraint handling, coevolutionary systems parameter control and hybrid approaches

Course Objectives

- To be able to formulate and assess problems in evolutionary computation.
- To be able to assess the strengths and weaknesses of several approaches to evolutionary computation.
- To be able to assess and understand the key similarities and differences in various evolutionary computation models.
- To develop knowledge and experience in developing evolutionary algorithms for real world problems.

Week Wise Topics to be Covered

Week	Topic
W1	Basics of Evolutionary Computation, Natural evolution, Evolutionary algorithms basics
W2	Evolutionary strategies, Optimization
W3	Genetic algorithms, operators, selection and parameters
W4	Combinatorial optimization problems and GA representations
W5	GA convergence
W6,W7	Genetic programming and how it differs from GA
W8	Constraint Handling
MID EXAM/ MID OF SEMESTER	
W9,W10	Evolutionary Neural Networks
W11,W12	Co-Evolution
W13,W14	Swarm Intelligence
W15	Other Variants (DE/EDA/PSO/ACO)
W16	Evolutionary strategies application

Code	Course Title	CHs	Pre-Req
CS665	**Development of Natural Language Engineering Resources**	3(3+0)	Nil

Recommended Texts

- Speech and Language Processing, Daniel Jurafsky, James H Martin, Peter Norvig, Stuart Russell, Pearson Education, 2014, ISBN: 9780133252934
- Natural Language Processing with Python, Steven Bird, Ewan Klein and Edward Loper, 2009, ISBN13: 978-0-5965-5571-9
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

Natural Language Processing addresses fundamental questions at the intersection of human languages and computer science. How can computers acquire, comprehend and produce English? How can computational methods give us insight into observed human language phenomena?

Course Objectives

- Understand approaches to syntax and semantics in NLP.
- Understand approaches to discourse, generation, dialogue and summarization within NLP.
- Understand current methods for statistical approaches to machine translation.
- Understand machine learning techniques used in NLP

Week Wise Topics to be Covered

W1	Introduction to Computing with Python. Texts as Lists of Words	W7	Processing Raw Text. Accessing Text from the Web and from Disk
W2	Computing with Language. Simple Statistics	W8	Strings, Text Processing at the Lowest Level
W3	Making Decisions and Taking Control	**MID EXAM/ MID OF SEMESTER**	
		W9	Text Processing with Unicode Regular Expressions for Detecting Word Patterns
W4	Building Corpora, Accessing and analyzing Text Corpora,	W10	Useful Applications of Regular Expressions
W5	Lexical Resources, conditional Frequency Distributions, Reusing Code, Lexical Resources:WordNet	W11	Normalizing Text
		W12, W13	Regular Expressions for tokenizing Text
W6	Monolingual/bilingual/multilingual Corpora. FrameNet and ConceptNetetc	W14, W15	Formatting: From Lists to Strings
		W16	Segmentation

Code	Course Title	CHs	Pre-Req
CS666	**Optimized Input Methods**	3(3+0)	Nil

Recommended Texts

1. Optimization Methods: From Theory to Design Scientific and Technological Aspects in Mechanics, Marco Cavazzuti, Springer Science & Business Media, 2012, ISBN 978-3-642-31187-1
2. At least 5 research papers from Scopus indexed international conferences
3. At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

Overview of optimization and classification of optimization problems; Optimization using calculus, Kuhn-Tucker Conditions; Linear Programming - Graphical method, Simplex method, Revised simplex method, Sensitivity analysis, Examples of transportation, assignment, water resources and other applications; Dynamic Programming - Introduction, Sequential optimization, computational procedure, curse of dimensionality

Course Objectives

- A course that bridges NLE and HCI in order to Understand Input Methods and Optimization, Study and design Input Methods that are appropriate for different genre of computer users
- Evaluate input methods through automated and manual models
- Statistical Models used in input method design and optimization, Structure information effectively for onsite and offsite readers
- Create modularized input methods that are extendible, sharable, re-usable

Week Wise Topics to be Covered

Week	Topics	Week	Topics
W1	History of HCI. Memex, SketchPad, Augment/NLS, WIMP interfaces, Ergonomics	W8	overall text environment, Factors influencing legibility of text
		MID EXAM/ MID OF SEMESTER	
W2	Golden principles of ergonomics, one-to-many mapping for input, touch/blind typing, composing on small screens; Input Methods, Different modes of input	W9	Visual Designs, multi-tap, touch, multi-touch, with and without keyboard/keypad input methods, touch screen input, screen sizes; small, medium
W3	Models, Visual, Typographic, large, X-large screens and appropriate user interfaces, greedy approach in input design, , touch, single finger, two fingers	W10	hybrid input systems
W4	Hybrid Models, Optimization	W11	Menu driven, Iconic. Icon design principles
W5	Statistical Models, HMM models, Muscular memorization process, one-to-one	W12	visual associations, Silhouette, Recognition and Recall. Cultural and multi-national issues
W6	n-finger, single hand, both hands input, dasher, wearable and gaze systems	W13	Maru-batsu, tick-cross, Levels of realism
W7	Fonts, Lines, spacing, margins. Importance of single pixel/pt in user interface design	W14, W15	coherence, Legibility
		W16	viewing distances, hygiene in design, health hazards caused by computers

Code	Course Title	CHs	Pre-Req
CS667	**Machine Translation**	3(3+0)	Nil

Recommended Texts

- Koehn, P. (2020). Neural Machine Translation. Cambridge University Press. https://doi.org/10.1017/9781108918450
- Bahdanau, D., Cho, K., & Bengio, Y. (2016). Neural Machine Translation by Jointly Learning to Align and Translate. ICLR. https://doi.org/10.48550/arXiv.1409.0473
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS
-

Course Description

This is an introduction to the field of machine translation (systems that translate speech or text from one human language to another), with a focus on statistical approaches. Three major paradigms will be covered: word-based translation, phrase-based translation, and syntax-based translation. Students will gain hands-on experience with building translation systems and working with real-world data, and they will learn how to formulate and investigate research questions in machine translation.

Course Objectives

- To understand neural and statistical models for machine translation
- To implement encoder-decoder and attention-based translation systems
- To understand multilingual and domain-specific translation challenges
- To implement pre-trained and fine-tuned models for MT tasks

Week Wise Topics to be Covered

W1	History. Translation process	W9	The Interlingua Idea
W2	Approaches: Rule-based, Statistical, Example based	W10	Using Meaning
W3	Hybrid MT. Major issues	W11	Direct Translation. Using Statistical Techniques
W4	Disambiguation	W12	Quantifying Fluency. Statistical MT, Basis and Benefits of Word based translation, Phrase based translation
W5	Named entities		
W6	Applications, Evaluation	W13	Syntax based translation, Challenges with statistical machine translation
W7	Language Similarities and Differences The Transfer Metaphor	W14, W15	Compound words, Idioms, Morphology, Syntax, Out of vocabulary (OOV) words
W8	Syntactic Transformations Lexical Transfer	W16	Different word orders
MID EXAM/ MID OF SEMESTER			

CS868 -> Empty

Code	Course Title	CHs	Pre-Req
CS669	**Advanced Natural Language Engineering**	3(3+0)	Natural Language Engineering

Recommended Texts

- Learning spatial-semantic representations from natural language descriptions and scene classifications, Sachithra Hemachandra, Matthew R. Walter, Stefanie Tellex, Seth Teller, IEEE, 2014, ISBN: 978-1-4799-3685-4
- Natural Language Understanding and Cognitive Robotics, Masao Yokota, CRC Press; 1st edition (December 6, 2019), ISBN-13: 978-0367360313
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

Advanced Natural Language Engineering (ANLE) seeks to endow computers with the ability to intelligently process human language. ANLE components are used in conversational agents and other systems that engage in dialogue with humans, automatic translation between human languages, automatic answering of questions using large text collections, the extraction of structured information from text, tools that help human authors, and many, many more.

Course Objectives

- Understand approaches to syntax and semantics in NLE.
- Understand approaches to discourse, generation, dialogue and summarization within NLE.
- Understand current methods for statistical approaches to machine translation.

Week Wise Topics to be Covered

W1	Introduction to Natural Language Engineering	**W9**	Deep Learning in NLP: Transformers and BERT
W2	Text Preprocessing and Tokenization	**W10**	Language Models and Fine-Tuning
W3	Syntactic Parsing and POS Tagging	**W11**	Multilingual NLP and Transfer Learning
W4	Named Entity Recognition and Coreference	**W12**	Low-Resource and Domain-Specific NLP
W5	Semantic Analysis and Word Embeddings	**W13**	Bias, Fairness, and Evaluation in NLP Systems
W6	Machine Translation and Alignment Techniques	**W14**	Project Training and Testing
W7	Information Extraction and Text Summarization	**W15**	**Case Studies / Projects / Demo**
W8	Question Answering Systems	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS761	**Design and Development of Corpora**	3(3+0)	Nil

Recommended Texts

- Year book of Corpus Linguistics and Pragmatics 2016: Current Approaches to Discourse and Translation Studies, Jesús Romero-Trillo, Springer, 2016, ISBN: 978-3-319-41732-5
- Computational Linguistics and Intelligent Text Processing, Alexander Gelbukh, Springer, 2015, ISBN: 9783319181172
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

The goal of this course is to provide the student with the necessary tools and techniques for doing corpus-based studies and annotation projects. It will thus help prepare the student for doing original research using corpora. This course provides an in-depth introduction to the construction and use of corpora for linguistic analyses. Some specific goals of the course are to enable students to: make an informed choice of data for a corpus study, specify an appropriate annotation format for an annotation project, understand and conduct evaluations of annotator performance

Course Objectives

- To the end of this course, student will be able to: understand how the corpus can be developed and processed
- know what is in existing corpus
- know different type of corpora and their day to day applications

Week Wise Topics to be Covered

W1, W2	Corpus Linguistic basics	W9	POS Tagged Corpus
W3	Corpus Characteristics	W10	Monolingual, Bilingual
W4	Corpus Mark-up	W11	Multi-lingual Corpora. Parallel Corpora for Machine Translation
W5	Corpus Annotation	W12	Analyzing Corpus. Manipulating Corpus
W6	Multilingual Corpora. Existing available Corpora	W13	Stochastic techniques and their application to extract and analyze Knowledge from Corpus
W7	Corpora and Computational Linguistics How to develop a Corpus? Raw Corpus	W14, W15	Corpora applications
W8	Annotated Corpus	W16	Corpora future research
MID EXAM/ MID OF SEMESTER			

Code	Course Title	CHs	Pre-Req
CS762	**Information Architecture for the World Wide Web**	3(3+0)	Nil

Recommended Texts

- Information Architecture: For the Web and Beyond 4th Edition, Lou Rosenfeld, Peter Morville, Jorge Arango, O'Reilly Media, 2015, ISBN: 978-1491911686
- Reframing Information Architecture, Andrea Resmini, Springer International Publishing, 2014, ISBN: 978-3-319-06491-8
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

The World Wide Web is the latest advancement in information technology, and, as with the previous innovations, certain principles carry over and others must be completely reexamined and overhauled. Because the Web integrates so many technologies and content types into a single interface, it challenges designers of web sites and intranets greatly

Course Objectives

- Understand Information Architecture fundamentals, why it's important, and related concepts
- Understand at a high-level design an effective Information Architecture
- Understand the relationship between the practice of information architecture

Week Wise Topics to be Covered

W1	Introduction to Information Architecture (IA)	**W9**	IA in Content Management Systems
W2	User Research and Content Strategy	**W10**	IA for E-Commerce and Educational Platforms
W3	Navigation Systems and Labeling	**W11**	Search Systems and Faceted Browsing
W4	Taxonomies and Metadata	**W12**	AI-driven IA and Dynamic Personalization
W5	Site Mapping and Card Sorting	**W13**	Ethical Design and User Privacy
W6	Wireframing and Prototyping Tools	**W14**	Project Development in IA Tools
W7	IA in Responsive and Adaptive Design	**W15**	**Case Studies / Projects / Demo**
W8	Usability and Findability Testing	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	**Course Title**	**CHs**	**Pre-Req**
CS763	**Advance Human Computer Interaction**	3(3+0)	Nil

Recommended Texts

- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course explores advanced interaction models, user-centered design, and emerging technologies in human-computer interaction.

Course Objectives

• Design and evaluate advanced interactive systems
• Understand cognitive models in interaction design
• Explore emerging interaction paradigms (AR/VR, multimodal systems)
• Conduct usability testing and iterative design

Week Wise Topics to be Covered

W1	HCI Principles and Design Cycle	**W9**	AR/VR Interfaces and Interaction
W2	Cognitive Models and Theories	**W10**	Multimodal Interaction and Natural Interfaces
W3	Advanced Input/Output Devices	**W11**	Emotion-Aware and Affective Interfaces
W4	Interaction Styles and Interfaces	**W12**	Conversational Agents and Voice UIs
W5	User Modeling and Adaptive Interfaces	**W13**	HCI in Safety-Critical Systems
W6	Usability Testing and Evaluation	**W14**	Ethical and Social Issues in HCI
W7	Designing for Accessibility	**W15**	**Case Studies / Projects / Demo**
W8	Mobile and Ubiquitous Interfaces	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

CS764	Information Authoring

Code	Course Title	CHs	Pre-Req
CS765	**Knowledge Representation**	3(3+0)	Nil

Recommended Texts

- Knowledge Representation And Reasoning A Complete Guide - Gerardus Blokdyk, 5STARCooks (February 20, 2020), ISBN-13 : 978-1867335252
- Knowledge Representation, Reasoning, and the Design of Intelligent Agents: The Answer-Set Programming Approach 1st Edition, Michael Gelfond , Yulia Kahl, Cambridge University Press, 2014, ISBN-13: 978-1107029569.
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This introductory knowledge representation course will provide an overview of existing representational frameworks developed within AI, their key concepts and inference methods. The course will cover propositional and first-order logics, their object-oriented extensions (frames), temporal logic and reasoning, inheritance relations, probabilistic models for reasoning and decision making, as well as new topics related to Semantic web and knowledge-based ontologies.

Course Objectives

- Knowledge representation is the study of how knowledge about the world can be represented in a computer system and what kind of reasoning can be done with that knowledge.
- This knowledge representation course will provide the students an overview of existing representational frameworks developed within Artificial Intelligence, their key concepts and inference methods.

Week Wise Topics to be Covered

W1	Introduction, Object Oriented Representation	W8	Defaults and Negation
		MID EXAM/ MID OF SEMESTER	
W2	Structured Description	W9	Answer Set Programming, Adductive Reasoning
W3	Ontologies and Representation of Domain Knowledge	W10	Constraint Satisfaction
W4	Knowledge Representation in Social Context	W11	Representation of Actions
		W12	Reasoning with Actions
W5	Social Network Analysis	W13 W14	Practical Planning, Reformulation
W6	Logic Programming	W15 W16	Approximation and Abstraction
W7	Combining Rules and Objects		

Code	Course Title	CHs	Pre-Req
CS766	**Computational Linguistics**		Nil

Recommended Texts

- Puzzles in Logic, Languages and Computation: The Red Book (Recreational Linguistics), Dragomir Radev, James Pustejovsky, Springer, 2015, ISBN-13: 978-3642442438.
- How to do Linguistics with R: Data exploration and statistical analysis, Natalia Levshina, John Benjamins Publishing Company, 2015, ISBN-13: 978-9027212252
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

In this interdisciplinary introductory course, you will learn how computers can do useful things with human languages, such as translate from French into English, summarize a magazine article into a few sentences, and find the main topics in the day's news. You will also learn about how computational methods can help linguists explain language phenomena, including automatic discovery of different word senses and phrase structure

Course Objectives

- To understand Linguistics, analyze its sub-topics from computational viewpoint
- Apply stochastic and other computational models to detect, extract and solve linguistic knowledge in naturally occurring formal or informal text.

Week Wise Topics to be Covered

Week	Topics	Week	Topics
W1	Morphology, Words	W10	Feature Structures in the Grammar, Agreement, Head Features. Sub-categorization, Long Distance Dependencies
W2	Regular Expressions, Automata, Finite-State Transducers	W11	Implementing Unification, Unification Data Structures
W3	Computational Phonology	W12	The Unification Algorithm. Unification Parsing, Integrating Unification into an Earley Parser
W4	Probabilistic Models of Pronunciation and Spelling Syntax	W13	Types and Inheritance, Extensions to Typing Semantics. Representing Meaning, Semantic Analysis, Lexical Semantics
W5	N-gram Models of Syntax	W14	Multilingual Computing in Machine Translation. Word Sense Disambiguation and Information Retrieval Module,
W6	HMMs and Speech Recognition	W15	Pragmatics, Discourse, Dialog and Conversational Agents
W7	Word Classes, Part-of-Speech Tagging, Context-Free Grammars, Parsing with CFGs	W16	Natural Language Generation
W8	Probabilistic Context-Free Grammars, Learning PCFG Probabilities		
MID EXAM/ MID OF SEMESTER			
W9	Problems with PCFGs, Probabilistic Lexicalized CFGs. Dependency Grammars, Unification of Feature Structures		

Code	Course Title	CHs	Pre-Req
CS767	**User Interface Design in Global Perspective**	3(3+0)	Nil

Recommended Texts

- Designing the User Interface, Strategies for Effective Human-Computer Interaction, CTI Reviews, Cram101, 2016, ISBN-13: 978-1467294386.
- International User Interfaces ,By Elisa M. del Galdo , Jakob Nielsen, ISBN : 978-1-4503-1965-2
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

Understanding user interface design, or how to design with the user in mind, is not only a requirement for designers, but also skill product managers and developers should develop throughout their careers. This User Interface Design in Global Perspective course should help product developers, managers and marketers as well as designers develop unique critical, analytical perspectives in user experience and user interface. Theoretical knowledge of the human factor is at the heart of user-centered design and evaluation.

Course Objectives

- Understand basic guidelines for Globally/Locally acceptable user interfaces.
- Design and Analyze methods for multi-national user interfaces.
- Have practical experience in analysis methods for user interfaces.

Week Wise Topics to be Covered

W1	Global perspective to Design User Interface Legal issues. Ergonomics, Golden principles of ergonomics	W8	Process of interaction design. Interaction design and the user experience
		MID EXAM/ MID OF SEMESTER	
W2	health issues, hygienic designs. Developing a Cultural Model	W9	Understanding and conceptualizing interaction
W3	Culture and Design	W10	Understanding the problem space and conceptualizing design
W4	Cultural Learning Differences in Software User Training	W11	Conceptual models. Interface metaphors
W5	Arabization of Graphical User Interfaces.	W12	Interaction types. Studying Context across Cultures
W6	A Chinese Text Display Supported by an Algorithm for Chinese Segmentation. Design of Multilingual Documents, Icon and Symbol Design Issues for Graphical User Interfaces	W13	Case Study: Managing a Multiple-Language Document System
		W14 W15	Systems Supporting hybrid Human-Human Communication
W7	The user experience.	W16	Impact of Culture on User Interface Design

Code	Course Title	CHs	Pre-Req
CS861	**Knowledge based System Design**	3(3+0)	Nil

Recommended Texts

- Expert systems, Jean-Louis Ermine, 2001, ISBN13: 978-8-1203-0919-7
- Artificial Intelligence, Stuart Jonathan Russell, Peter Norvig and John F. Canny, 2003, ISBN13: 978-8-1297-0041-4
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

The basic purpose of the course is to discuss the application of artificial intelligence techniques, and more specifically knowledge based systems, in information processing and information systems design. The focus will be on knowledge based systems/expert systems as opposed to e.g. integrated with traditional data processing environments.

Course Objectives

- Main objective of the course is providing graduate and advance undergraduate students with an understanding of practical and theoretical methods, mathematics, and algorithms required for applications and research in Machine Learning.
- Apply user centered design and usability engineering principles as they design a wide variety of software user interfaces.

Week Wise Topics to be Covered

W1	Introduction To Knowledge Engineering	W9	Tools for Building Expert Systems, Case Based Reasoning
W2	The Human Expert and their Problem Solving Process		
W3	Knowledge Base	W10	Semantics of Expert Systems, Modeling of Uncertain Reasoning
W4	Inference Engine		
W5	Knowledge Acquisition	W11	Machine Learning
W6	Knowledge Representation	W12	Rule Generation and Refinement
W7	Rule Based Systems, Heuristic Classifications	W13 W14	Learning Evaluation
W8	Constructive Problem Solving	W15 W16	Testing and Tuning
MID EXAM/ MID OF SEMESTER			

Code	Course Title	CHs	Pre-Req
CS862	**Information Foraging**	3(3+0)	Nil

Recommended Texts

- New Directions in Human Information Behavior , By Amanda Spink, Charles Cole, ISBN 978-1-4020-3670-5
- Personalized Information Retrieval and Access: Concepts, Methods and Practices, edited by Rafael Andrés Gonzáles, Nong Chen, Ajantha Dahanayake, IGI Global 2008, ISBN : 978-1599045108
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

With the emergence of the "personal computing" paradigm in the 1970s, computers moved out of the machine room and became devices with which end-users would interact directly. The field of HCI grew up in response to the engineering and cognitive challenges of building systems that fit with user needs.

Course Objectives

- Information foraging theory as a new method for analyzing information-intensive work.
- Techniques of information foraging analysis for characterizing human information-seeking behavior.
- Models and empirical tools for analysis of adaptation to information environments cognitive mechanisms.
- Emphasis on applications to Web and information system design

Week Wise Topics to be Covered

Week	Topic	Week	Topic
W1	Information Foraging Theory, Elementary Foraging Models	W7 W8	Rational Analysis and Computational Cognitive Model of the Scatter/Gather Document Cluster Browser
			MID EXAM/ MID OF SEMESTER
W2 W3	Patch residence time and diet models, The Ecology of Information Foraging on the World Wide Web, Summary of the significance question coding categories and web protocol coding guide	W9	Stochastic Models of Information Foraging by Information Scent
		W10	Alternate methods based on the analysis of absorbing Markov chains
		W11 W12	Social Information Foraging
		W13	Design Heuristics
W4 W5	Rational Analyses of Information Scent and Web Foraging	W14	Engineering Models
		W15	Engineering Models Applications
W6	A Cognitive Model of Information Foraging on the Web	W16	Future Directions

Code	Course Title	CHs	Pre-Req
CS863	**Recommender Systems**	3(3+0)	Nil

Recommended Texts

- Recommender Systems: The Textbook 1st ed, Charu C. Aggarwal, Springer; 1st ed. 2016, ISBN-13: 978-3319296579.
- D. Jannach, M. Zanker, A. Felfernig, and G. Friedrich, Recommender Systems: AnIntroduction, Cambridge University Press., 2010, ISBN 978-0521493369.
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

Recommender systems guide people to interesting materials based on information from other people. A large design space of alternative ways to organize such systems exists. The information that other people provide may come from explicit ratings, tags, or reviews, or implicitly from how they spend their time or money. The information can be aggregated and used to select, filter, or sort items. The recommendations may be personalized to the preferences of different users.

Course Objectives

- To understand the theoretical foundations and classification of recommender systems, including collaborative, content-based, and hybrid approaches.
- To implement recommendation algorithms using techniques such as matrix factorization, neighborhood methods, and deep learning.
- To evaluate recommender systems using metrics like precision, recall, F1-score, and diversity.
- To explore current trends and challenges in scalable, context-aware, and explainable recommender systems..

Week Wise Topics to be Covered

W1	Introduction to Recommender Systems	**W9**	Deep Learning for Recommendations
W2	Content-Based Filtering Techniques	**W10**	Sequential and Session-Based Recommenders
W3	Collaborative Filtering: Memory-Based	**W11**	Explainable and Fair Recommendations
W4	Model-Based Collaborative Filtering	**W12**	Privacy in Recommender Systems
W5	Hybrid Recommendation Models	**W13**	Industrial Applications and Case Studies
W6	Evaluation Metrics and Challenges	**W14**	Project Development and Experimentation
W7	Context-Aware Recommenders	**W15**	**Case Studies / Projects / Demo**
W8	Cold Start and Sparsity Problems	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS864	**Parsing Technologies**	3(3+0)	Nil

Recommended Texts

- Trends in Parsing Technology: Dependency Parsing, Domain Adaptation, and Deep Parsing, Harry Bunt, Paola Merlo, Joakim Nivre, Springer Science & Business Media, 2010, ISBN-13: 978-9048193523.
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

The course introduces theories used in language technology. It attempts to cover the whole field from character encoding and statistical language models to semantics and conversational agents, going through syntax and parsing. It focuses on proven techniques as well as significant industrial or laboratory applications.

Course Objectives

- To introduce the advanced concepts of Parsing in computational linguistics and
- To introduce the advanced concepts of modern grammar formalisms.

Week Wise Topics to be Covered

W1	Tree Adjoining Grammars, Dependency Grammars, Discourse Processing	W8	Speech Understanding and Translation, Morphology and Finite State Transducers
W2	Statistical Parsing		**MID EXAM/ MID OF SEMESTER**
W3	Introduction to Semantic Processing, Semantic Knowledge Representation	W9	Inflectional Morphology, LFG Formalism
		W10	Derivational Morphology, Well-formedness
W4	Deep Structure and Logical Form, Overview of LFG	W11	Finite State Morphological Parsing, Computational Aspects
W5	Compositional Semantic Interpretation, Lexical Functional Grammar	W12	The Lexicon and Morpho-syntactic language units
W6	Semantic Grammars, Active-Passive and Dative Constructions, Wh-movement in Questions	W13	Morphological Parsing with Finite State Transducers
W7	Case Frames and Case Frame based Parsing, Natural Language Generation. Problems in NLG, Conditions-Handling, Pruning and beam search,	W14 W15	Orthographic Rules and Finite-State Transducers
		W16	Combining an FST Lexicon and Rules

Code	Course Title	CHs	Pre-Req
CS865	**Statistical Natural Language Engineering**	3(3+0)	Nil

Recommended Texts

- Foundations of Statistical Natural Language Processing, Christopher D. Manning and Hinrich Schütze, 1999, ISBN13: 978-0-2621-3360-9
- Statistical Language Learning, Eugene Charniak, 1996, ISBN13: 978-0-2625-3141-2
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

In this course we will explore statistical, model-based approaches to natural language processing. There will be a focus on corpus-driven methods that make use of supervised and unsupervised machine learning approaches and algorithms. We will explore generative and discriminative learning approaches, spanning from graphical models to neural networks.

Course Objectives

- To introduce the use of statistical techniques for processing human languages
- To introduce the use of statistical techniques for processing large text corpora

Week Wise Topics to be Covered

W1	Introduction, Mathematical Foundations	W9	Implementations, properties and variants
W2	Elementary Probability theory	W10	applications to Part-of-Speech Tagging, Probabilistic Context Free Grammars
W3	Information Theory	W11	Probabilistic Parsing, Applications
W5	Corpus-Based Work, Text Categorization, Collocations	W12	Statistical Alignment and Machine Translation
W6	Statistical Inference, n-gram Models over Sparse Data, Word Sense Disambiguation	W13 W14	Clustering
W7	Supervised and Dictionary based approaches, Lexical Acquisition	W15 W16	Linguistic Essentials
W8	Markov Models, HMM fundamentals		
MID EXAM/ MID OF SEMESTER			

Code	Course Title	CHs	Pre-Req
CS866	**Computational Intelligence**	3(3+0)	Nil

Recommended Texts

- Computational Intelligence: A Methodological Introduction, by Rudolf Kruse (Author), Christian Borgelt (Author), Christian Braune (Author), Sanaz Mostaghim (Author), Matthias Steinbrecher (Author), Frank Klawonn (Contributor), Christian Moewes (Contributor), Springer; 2nd ed. 2016 edition (September 17, 2016), ISBN-13 : 978-1447172949
- Computational Intelligence: Concepts to Implementations, R. Eberhart, and Y. Shi, Morgan Kaufmann 2007., ISBN 978-1558607590
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

Introducing concepts, models, algorithms, and tools for development of intelligent systems. Example topics include artificial neural networks, genetic algorithms, fuzzy systems, swarm intelligence, ant colony optimization, artificial life, and hybridizations of the above techniques. We will look at these techniques from a machine learning perspective. This domain is called Computational Intelligence, and is a numerical interpretation of biological intelligence.

Course Objectives

- The objective of this course is to introduce the techniques of computational intelligence, especially evolutionary computation, neural networks, fuzzy logic, genetic algorithms, and artificial immune systems.
- To provides the students with an understanding of how the techniques function in solving the problems in the real world how hybrid multiple techniques and the selection of the appropriate techniques can be applied for particular problems.

Week Wise Topics to be Covered

Week	Topic	Week	Topic
W1	Introduction to Computation Intelligence Concepts, Soft Computing,Introduction to Evolutionary Computing	W8	Neural Networks and Evolutionary Computation
			MID EXAM/ MID OF SEMESTER
W2	Evolutionary Programming and Evolution Strategies, Swarm Intelligence	W9	Explanation and Sensitivity Analysis
W3	Particle Swarm Optimization	W10	Evolutionary and swarm-based neural networks
W4	Supervised	W11	Fuzzy Logic
W5	Unsupervised	W12	Support Vector Machines, Probabilistic Reasoning
W6	Reinforcement Learning	W13	Bayesian Belief Networks
W7	Data partitioning and Cross Validation, Error metrics: Mean squared error, receiver operating characteristic curves	W14	Genetic Programming, Artificial Immune Systems
		W15	Classification
		W16	Learning and Adaptation

Code	Course Title	CHs	Pre-Req
CS867	**Speech Processing Techniques**	3(3+0)	Nil

Recommended Texts

- Speech & Language Processing, Dan Jurafsky and James H. Martin, 2000, ISBN13: 978-8-1317-1672-4
- Statistical Language Learning, Eugene Charniak, 1996, ISBN13: 978-0-2625-3141-2
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

Speech Processing offers a practical and theoretical understanding of how human speech can be processed by computers. It covers speech recognition, speech synthesis and spoken dialog systems. The course involves practical where the student will build working speech recognition systems, build their own synthetic voice and build a complete telephone spoken dialog system. This work will be based on existing toolkits.

Course Objectives

- To understand the fundamental principles of speech production and perception relevant to speech signal processing.
- To implement signal processing techniques for speech analysis, enhancement, and synthesis.
- To analyze models and algorithms for speech feature extraction, including MFCCs and LPC.
- To explore applications of speech processing in speech recognition, speaker identification, and speech synthesis.

Week Wise Topics to be Covered

Week	Topics
W1	The Role of Knowledge in Speech and Language, Engineering, Models and Algorithms, Language, Thought and Understanding
W2	The Vocal Organs, The Turing Test, Places of Articulation, A Short History of Speech and Language Processing,
W3	Consonants, Manner of Articulation, Formal Languages and recognizers,
W4	Vowels, The Phoneme and Phonological Rules
W5	Phonological Rules and Transducers
W6	Mapping Text to Phonemes for TTS, Pronunciation Dictionaries Beyond Dictionary Lookup, Text Analysis, An FSA based Pronunciation Lexicon English Pronunciation Variation
W7	Models of Pronunciation and Spelling, Introduction to Spell Checking-Spelling Error Patterns
W8	Detecting Non-word Errors, Computational Phonology and Pronunciation Modeling, Speech Sounds and Phonetic Transcription, Probabilistic Models, Applying the Bayesian Method to Spelling Correction
MID EXAM/ MID OF SEMESTER	
W9	Minimum Edit Distance, The Bayesian Method for Pronunciation
W10	Decision Tree Models of Pronunciation Variation, Weighted Automata
W11	Computing Likelihoods, The Forward Algorithm, Decoding, The Viterbi Algorithm
W12	Weighted Automata and Segmentation, Introduction to Speech Recognition. Overview of a Speech Recognition Architecture.
W13	Overview of Hidden Markov Models
W14	Advanced Methods of Decoding
W15	A* Decoding, Acoustic Processing of Speech
W16	Feature Extraction, Computing Acoustic Probabilities, Training a Recognizer

Code	Course Title	CHs	Pre-Req
CS868	**Advanced Information Retrieval**	3(3+0)	Nil

Recommended Texts

- C. Manning, P. Raghavan, and H. Schütze, Introduction to Information Retrieval, by Cambridge University Press, 2008, ISBN: 0521865719
- Mining the Web: Discovering Knowledge from Hypertext Data, Soumen Chakrabarti, 2003, ISBN-10: 1558607544
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

The course will cover: organization, representation, and access to information; categorization, indexing, and content analysis; data structures for unstructured data; design and maintenance of such databases, indexing and indexes, retrieval and classification schemes; use of codes, formats, and standards; analysis, construction and evaluation of search and navigation techniques; and search engines and how they relate to the above.

Course Objectives

- Understand the principles of information storage and retrieval systems and database
- Understand how effective information search and retrieval is interrelated with the organization and description information to be retrieved
- Use a set of tools and procedures for organizing information

Week Wise Topics to be Covered

W1	Introduction to Information Retrieval, Structured vs Unstructured Data, Information vs Data Retrieval, IR Basics	W7	MapReduce, Dynamic Indexing, MapReduce Architecture,
W2	Information Retrieval Models, Boolean Model, Vector Model, Inverted Index Construction, Query Processing, Dictionary and Postings	W8	Models of Natural Language, Index Size Estimation using Zif's law
		MID EXAM/ MID OF SEMESTER	
		W9	Vector Space Model: TF-IDF, Boolean vs Ranked Retrieval,
		W10	Vector Space Model: Ranking Revisited,
W3	Query Processing, Tokenization, Normalization, Stop words, Lemmatization, Stemming, Faster Postings Merges	W11	Evaluation of IR Systems, Evaluating Search Engines. Creating Test Collections for IR Evaluation
W4	skip pointers, Phrase Queries, Tolerant Information Retrieval, Dictionary data structures for inverted indexes, Hashtables	W12	Other Evaluation Measures, Query Operations. Relevance Feedback, Centroid, Rocchio Algorithm
W5	Tree, Wild card queries, Processing wild card queries, Spelling Correction, Document Correction, n-gram overlap	W13	Relevant Feedback in in vector spaces
		W14, W15	Text Classification
W6	Thesauri, Query Expansion, Soundex, Language Detection, Index Construction, Distributed Indexing	W16	Index Compression, Compression in inverted indexes

Code	Course Title	CHs	Pre-Req
CS869	**Machine Learning for Cybersecurity**	3(3+0)	Nil

Recommended Texts

- Machine Learning for Cybersecurity Cookbook by Emmanuel Tsukerman, (2019), ISBN: 978-1789614671
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course explores the intersection of machine learning and cybersecurity. Students will learn how to apply ML algorithms to solve real-world cybersecurity problems, such as malware detection, social engineering prevention, penetration testing, and intrusion detection. Emphasis is placed on both offensive and defensive applications of ML, with a focus on building secure and privacy-preserving AI systems. The course includes hands-on assignments and culminates in a project targeting an open research problem or an applied cybersecurity challenge.

Course Objectives

- Understand the principles, workflows, and prerequisites of applying machine learning techniques to cybersecurity domains.
- To implement ML models for malware detection, intrusion detection, and penetration testing using real-world datasets.
- To analyze the role of ML in social engineering defense and develop secure, privacy-preserving AI solutions.
- To evaluate the effectiveness of various ML methods in addressing advanced cybersecurity threats and propose novel applications or improvements.

Week Wise Topics to be Covered

W1	Introduction to ML for Cybersecurity: Concepts, Importance, Challenges	**W9**	Penetration Testing Enhanced by ML: Exploit Prediction, Access Path Identification
W2	Machine Learning Prerequisites for Security Tasks: Data Collection, Feature Engineering	**W10**	Intrusion Detection Systems (IDS) using Supervised and Unsupervised Learning
W3	Malware Detection with Classical ML Models (SVM, Random Forest, etc.)	**W11**	Autoencoders and Anomaly Detection in Network Traffic
W4	Deep Learning for Malware Detection: CNNs, RNNs, Autoencoders	**W12**	Secure & Private AI: Federated Learning, Differential Privacy, Homomorphic Encryption
W5	Adversarial ML and Evasion Attacks in Malware Classification	**W13**	Attacking and Poisoning ML Models in Cybersecurity
W6	ML in Social Engineering Detection: Email Phishing, Voice & Text Classification	**W14**	Explainability and Trust in AI-based Security Systems
W7	Graph-based Techniques in Phishing and Threat Intelligence	**W15**	Latest Research and Trends in ML for Cybersecurity (papers, tools, case studies)
W8	Midterm Review + Practical Lab on Malware Detection Project	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS961	**Nature Inspired Computation**	3(3+0)	Nil

Recommended Texts

- Nature-Inspired Computation in Data Mining and Machine Learning by Xin-She Yang, Xing-Shi He, (2020), ISBN: 978-3-030-28553-1
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

In this course the students will study the research papers exploring the transformative potential of nature-inspired algorithms. From optimizing functions to enhancing machine learning, each paper offers unique insights into the intersection of nature and technology.

Course Objectives

To comprehend nature-inspired algorithms. To analyze research papers on nature-inspired computation. To apply nature-inspired algorithms to diverse problems. To propose future research directions.

Week Wise Topics to be Covered

W1	Introduction to Nature-Inspired Computation and its Relevance in Data Mining and ML	**W9**	Case Study 1: Cybersecurity Applications Using Nature-Inspired ML
W2	Fundamentals of Bio-inspired Algorithms: Genetic Algorithms, Evolutionary Strategies	**W10**	Case Study 2: Healthcare Applications – Vital Signs, Kidney Disease Prediction
W3	Swarm Intelligence: Ant Colony Optimization and Particle Swarm Optimization	**W11**	Text Mining and NLP: Named Entity Recognition from Social Media in Regional Languages
W4	Flower Pollination Algorithm (FPA): Theory and Adaptive Improvements	**W12**	Classification & Clustering: Comparative Study using Nature-Inspired Methods
W5	Firefly and Bat Algorithms: Mechanisms and Applications	**W13**	Feature Selection using Hybrid Binary PSO + FPA with Rough Set Theory
W6	Hybrid Nature-Inspired Algorithms: Integration with Classical ML Models	**W14**	Deep Belief Networks and their Integration with Evolutionary Learning
W7	Optimization in Cloud using Nature-Inspired Algorithms	**W15**	Research Trends: Current Gaps, Open Problems, and Future Research Directions
W8	Midterm Review + Seminar on Nature-Inspired Models for Real-Time Systems	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS962	**Reinforcement Learning and Deep RL**	3(3+0)	Nil

Recommended Texts

- M. Irfan Uddin, and Wali Khan Mashwani. Deep Learning, Reinforcement Learning, and the Rise of Intelligent Systems. IGI Global, USA, February 2024. ISSN 9798369317389. doi:10.4018/979-8-3693-1738-9.
- Sutton, R. S., & Barto, A. G. (2018). Reinforcement Learning: An Introduction (2nd ed.). MIT Press. https://doi.org/10.1201/9780367332513
- Francois-Lavet, V., & Henderson, P. (2020). An Introduction to Deep Reinforcement Learning. Foundations and Trends® in Machine Learning. https://doi.org/10.1561/2200000071
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course provides an in-depth exploration of reinforcement learning (RL), including policy optimization, value-based methods, and deep reinforcement learning architectures. Emphasis is placed on theoretical foundations, algorithms, and practical implementations in simulated and real-world environments.

Course Objectives

- To understand value-based and policy-based RL algorithms
- To implement Q-learning, DQN, PPO, and A3C models
- To understand exploration-exploitation trade-offs and credit assignment
- To implement Deep RL solutions to real-world problems

Week Wise Topics to be Covered

W1	Introduction to Reinforcement Learning	**W9**	Multi-Agent RL
W2	Markov Decision Processes and Bellman Equations	**W10**	Exploration-Exploitation and Sample Efficiency
W3	Dynamic Programming and Temporal-Difference Learning	**W11**	Multi-Agent and Hierarchical RL
W4	Monte Carlo Methods and Policy Iteration	**W12**	RL in Robotics and Control
W5	Q-Learning and SARSA	**W13**	Challenges in Deep RL
W6	Deep Q-Networks (DQNs)	**W14**	Applications of RL in Game and Industry
W7	Policy Gradient Methods	**W15**	**Case Studies / Projects / Demo**
W8	Actor-Critic and A3C Algorithms	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS963	**Graph Representation Learning**	3(3+0)	Nil

Recommended Texts

- Hamilton, W. L. (2022). Graph Representation Learning. Synthesis Lectures on Artificial Intelligence and Machine Learning. https://doi.org/10.2200/S01108ED1V01Y202109AIM051
- Bronstein, M. M., Bruna, J., Cohen, T., & Velickovic, P. (2021). Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges. https://doi.org/10.48550/arXiv.2104.13478
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course provides a comprehensive exploration of machine learning techniques for graph-structured data, covering traditional methods, modern Graph Neural Networks (GNNs), and cutting-edge research topics. Students will learn to design, implement, and critically evaluate graph-based ML models, with applications in social networks, biology, recommender systems, and knowledge graphs.

Course Objectives

- To understand graph embedding and representation techniques
- To implement Graph Neural Networks (GNNs) for node and graph-level tasks
- To understand applications in social, biological, and knowledge networks
- To implement scalable GNN frameworks and models

Week Wise Topics to be Covered

W1	Introduction to Graphs and Network Data	**W9**	Graph Contrastive Learning
W2	Node, Edge, and Graph-Level Tasks	**W10**	Scalability and Efficiency in Large Graphs
W3	Graph Embedding Techniques: DeepWalk, node2vec, LINE	**W11**	GNNs in Knowledge Graphs and Recommender Systems
W4	Matrix Factorization and Random Walk Approaches	**W12**	GNNs for Molecular and Biological Data
W5	Spectral GNNs: Graph Convolutions, ChebNet	**W13**	Interpreting and Explaining GNNs
W6	Spatial GNNs: GCN, GAT, GraphSAGE	**W14**	GNN Benchmarks and Libraries (DGL, PyTorch Geometric)
W7	Pooling and Hierarchical GNNs	**W15**	**Case Studies / Projects / Demo**
W8	Dynamic, Temporal, and Heterogeneous Graphs	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS964	**Advanced NLP with Transformers**	3(3+0)	Nil

Recommended Texts

- Eisenstein, J. (2022). Introduction to Natural Language Processing (2nd ed.). MIT Press. https://doi.org/10.7551/mitpress/13645.001.0001
- Lin, J., & Wang, W. Y. (2021). Pretrained Transformers for Text Ranking: BERT and Beyond. Synthesis Lectures on Information Concepts, Retrieval, and Services. https://doi.org/10.2200/S01108ED1V01Y202108ICR078
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course offers an in-depth exploration of advanced Natural Language Processing using Transformer-based architectures. It covers foundational models like BERT and GPT, explores cutting-edge advancements, and delves into fine-tuning, prompt engineering, interpretability, and large-scale deployment. Emphasis is placed on recent research, ethical concerns, and real-world applications.

Course Objectives

- To understand the architecture and training of Transformer-based models
- To implement models like BERT, GPT, RoBERTa, and T5
- To understand fine-tuning and prompt-based learning
- To implement NLP applications using pre-trained models

Week Wise Topics to be Covered

W1	Overview of Transformer Architecture: Attention mechanism, self-attention, positional encoding	**W9**	Transformers for Multilingual NLP: XLM-R, mBERT, and translation tasks
W2	BERT and Encoder-Only Models: Pre-training objectives and downstream applications	**W10**	Transformers for Structured Data and Knowledge: TableBERT, TAPAS, and KG-BERT
W3	GPT and Decoder-Only Models: Language modeling and generative tasks	**W11**	Dialogue Systems and Transformers: BlenderBot, DialoGPT
W4	Encoder-Decoder Transformers: T5, BART, and sequence-to-sequence modeling	**W12**	Ethical and Societal Challenges: Bias, misinformation, fairness, and safety
W5	Transfer Learning in NLP: Fine-tuning, domain adaptation, and continual learning	**W13**	Recent Advances and Foundation Models: GPT-4, LLaMA, Claude, Gemini
W6	Prompt Engineering and In-Context Learning	**W14**	Large-Scale Pretraining Infrastructure: Data curation, model scaling, optimization
W7	Transformer Efficiency: Distillation, pruning, quantization, and efficient variants (e.g., Longformer, Performer)	**W15**	**Case Studies / Projects / Demo**
W8	Interpretability and Explainability in Transformer Models	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS965	**Machine Learning and AI for Scientific Discovery**	3(3+0)	Nil

Recommended Texts

- Karniadakis, G. E., Kevrekidis, I. G., Lu, L., Perdikaris, P., Wang, S., & Yang, L. (2021). Physics-Informed Machine Learning. Nature Reviews Physics, 3, 422–440. https://doi.org/10.1038/s42254-021-00314-5
- Brunton, S. L., & Kutz, J. N. (2022). Data-Driven Science and Engineering: Machine Learning, Dynamical Systems, and Control (2nd ed.). Cambridge University Press. https://doi.org/10.1017/9781009168465
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course investigates how ML and AI can accelerate hypothesis generation, experimentation, and analysis in scientific domains such as physics, biology, chemistry, and materials science. Emphasis is placed on symbolic regression, surrogate modeling, and generative modeling for discovery.

Course Objectives

- To understand ML models designed for hypothesis generation and testing
- To implement physics-informed and domain-aware learning models
- To understand reproducibility and scientific validity in AI research
- To implement ML workflows for scientific data and simulations

Week Wise Topics to be Covered

W1	AI and Scientific Method: From observation to theory	**W9**	Interpretable ML in Science
W2	Data-Driven Hypothesis Generation	**W10**	AI for Physics-Informed Systems
W3	Surrogate Modeling and Simulation Replacement	**W11**	ML in Genomics and Systems Biology
W4	Symbolic Regression and Equation Discovery	**W12**	AI in Drug Discovery and Healthcare Research
W5	Generative Models for Molecules and Materials	**W13**	Automation in Lab and Scientific Instruments
W6	Bayesian Optimization for Experimental Design	**W14**	Open Science and AI-Driven Collaborations
W7	Active Learning in Science	**W15**	**Case Studies / Projects / Demo**
W8	Graph-Based Models for Scientific Structures	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS966	**AutoML and Neural Architecture Search**	3(3+0)	Nil

Recommended Texts

- Hutter, F., Kotthoff, L., & Vanschoren, J. (2019). Automated Machine Learning: Methods, Systems, Challenges. Springer. https://doi.org/10.1007/978-3-030-05318-0
- Elsken, T., Metzen, J. H., & Hutter, F. (2019). Neural Architecture Search: A Survey. Journal of Machine Learning Research, 20(55), 1–21. https://www.jmlr.org/papers/volume20/18-598/18-598.pdf
- At least 5 research papers from Scopus indexed international conference
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course explores the automation of the machine learning pipeline, focusing on methods for hyperparameter optimization, neural architecture search (NAS), and meta-learning. Students will study cutting-edge techniques and apply them to design efficient and effective ML systems with minimal human intervention.

Course Objectives

- To understand the motivation and components of AutoML pipelines
- To implement hyperparameter optimization and meta-learning techniques
- To understand NAS algorithms like reinforcement-based, evolutionary, and gradient-based methods
- To implement scalable and efficient NAS in ML systems

Week Wise Topics to be Covered

W1	Introduction to AutoML: Motivation, scope, and components	**W9**	Multi-Objective and Resource-Aware NAS
W2	Hyperparameter Optimization: Grid search, random search	**W10**	AutoML for Data Preprocessing and Feature Engineering
W3	Bayesian Optimization and SMBO	**W11**	End-to-End AutoML Systems: AutoKeras, Auto-sklearn, TPOT
W4	Meta-Learning and Learning to Learn	**W12**	Benchmarking and Evaluation in AutoML
W5	Foundations of Neural Architecture Search (NAS)	**W13**	Scalability and Real-World Applications
W6	Search Strategies in NAS: Reinforcement learning, evolution	**W14**	Fairness, Transparency, and Ethical Challenges in AutoML
W7	Differentiable NAS and Gradient-Based Methods	**W15**	**Case Studies / Projects / Demo**
W8	One-Shot and Weight-Sharing NAS	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS967	**Lifelong Machine Learning**	3(3+0)	Nil

Recommended Texts

- Silver, D. L., Yang, Q., & Li, L. (2013). Lifelong Machine Learning Systems: Beyond Learning Algorithms. In AAAI Spring Symposium Series. https://www.aaai.org/ocs/index.php/SSS/SSS13/paper/view/5752
- Parisi, G. I., Kemker, R., Part, J. L., Kanan, C., & Wermter, S. (2019). Continual Lifelong Learning with Neural Networks: A Review. Neural Networks, 113, 54–71. https://doi.org/10.1016/j.neunet.2019.01.012
- Chen, Z., & Liu, B. (2018). Lifelong Machine Learning. Synthesis Lectures on Artificial Intelligence and Machine Learning, 12(3), 1–207. https://doi.org/10.2200/S00834ED1V01Y201810AIM041
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course explores advanced learning paradigms where data is scarce, dynamic, or evolving. It covers lifelong learning, few-shot learning, and zero-shot learning, focusing on theoretical foundations, algorithms, and applications in real-world AI systems.

Course Objectives

- To understand the principles and motivations behind lifelong and continual learning in machine learning systems.
- To implement algorithms that retain, transfer, and accumulate knowledge across tasks.
- To evaluate challenges such as catastrophic forgetting and knowledge consolidation.
- To explore applications of lifelong learning in robotics, natural language processing, and recommendation system

Week Wise Topics to be Covered

W1	Overview of Generalization in Low-Data Regimes	**W9**	Compositional and Cross-Modal ZSL
W2	Introduction to Few-Shot Learning	**W10**	Introduction to Lifelong and Continual Learning
W3	Metric-Based Few-Shot Methods: Prototypical, Matching Networks	**W11**	Catastrophic Forgetting and Regularization Techniques
W4	Optimization-Based Few-Shot Methods: MAML, Reptile	**W12**	Replay, Expansion, and Meta-Learning Strategies
W5	Data Augmentation and Self-Supervised Techniques	**W13**	Benchmarks and Evaluation for Lifelong/ZSL/FSL
W6	Introduction to Zero-Shot Learning and Knowledge Transfer	**W14**	Applications: NLP, Robotics, Vision, Health
W7	Attribute-Based and Semantic Embeddings	**W15**	**Case Studies / Projects / Demo**
W8	Generative Models for Zero-Shot Learning	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS968	**Agentic AI**	3(3+0)	Nil

Recommended Texts

- Russell, S., & Norvig, P. (2021). Artificial Intelligence: A Modern Approach (4th ed.). Pearson. https://doi.org/10.1016/C2009-0-61106-7
- Wooldridge, M. (2020). An Introduction to MultiAgent Systems (2nd ed.). Wiley. https://doi.org/10.1002/9781119551722
- Franklin, S., & Graesser, A. (1997). Is it an Agent, or just a Program? In Proceedings of the Third International Workshop on Agent Theories, Architectures, and Languages. https://doi.org/10.1007/3-540-49057-4_8
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course explores the theory, design, and implementation of autonomous agents capable of goal-directed behavior, reasoning, planning, and interaction. Topics include agency, decision-making, multi-agent systems, autonomy, and the emerging paradigm of "agentic workflows" in AI.

Course Objectives

- To understand the conceptual foundations of agentic AI and autonomous intelligent agents.
- To implement goal-driven and reactive agent architectures in dynamic environments.
- To analyze multi-agent coordination, negotiation, and communication strategies.
- To explore ethical, social, and trust implications of autonomous AI agents.

Week Wise Topics to be Covered

W1	Introduction to Agents and Agency in AI	**W9**	Cognitive Architectures for Agentic AI
W2	Types of Agents and Architectures: Reactive, deliberative, hybrid	**W10**	Agentic Tools in LLM Ecosystems
W3	Rationality, Autonomy, and Belief-Desire-Intention Models	**W11**	Tool Use, Memory, and Long-Term Autonomy
W4	Goal Formulation, Planning, and Scheduling	**W12**	Human-Agent Interaction and Alignment
W5	Reinforcement Learning for Agentic Behavior	**W13**	Simulation and Evaluation of Agentic Systems
W6	Multi-Agent Systems and Communication	**W14**	Safety, Ethics, and Control in Agentic AI
W7	Coordination, Cooperation, and Negotiation	**W15**	**Case Studies / Projects / Demo**
W8	Emergent Behavior and Swarm Intelligence	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS969	**Generative AI and Large Language Models**	3(3+0)	Nil

Recommended Texts

- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press. https://doi.org/10.7551/mitpress/10993.001.0001
- Eisenstein, J. (2022). Natural Language Processing. MIT Press. https://doi.org/10.7551/mitpress/13319.001.0001
- OpenAI (2023). GPT Technical Report. arXiv preprint. https://doi.org/10.48550/arXiv.2303.08774
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course examines the principles, architectures, and training techniques behind generative AI systems, with a focus on large language models (LLMs). It covers autoregressive models, diffusion models, prompt engineering, alignment strategies, and practical applications in content generation.

Course Objectives

- To understand the architecture, training, and inference mechanisms of generative AI models and LLMs.
- To implement and fine-tune transformer-based architectures such as GPT, BERT, and T5.
- To evaluate performance, bias, and ethical risks associated with generative language models.
- To explore applications of generative AI in text generation, code completion, and creative tasks.

Week Wise Topics to be Covered

W1	Introduction to Generative AI and Overview of Large Language Models	**W9**	Retrieval-Augmented Generation (RAG) and Knowledge Integration
W2	Neural Networks Refresher: Feedforward, RNN, and Sequence Models	**W10**	Evaluation of Generative Models: BLEU, ROUGE, Perplexity, Human Evaluation
W3	Transformers Architecture: Encoder-Decoder, Self-Attention Mechanism	**W11**	Generative AI in Text Summarization, Translation, and Content Generation
W4	BERT and Encoder-based Models: Pre-training and Fine-tuning	**W12**	Conversational AI: Chatbots and Dialogue Systems
W5	Decoder-based Models: GPT Family and Autoregressive Generation	**W13**	Multimodal Generative AI: Combining Text, Image, and Video Generation
W6	Encoder-Decoder Models: T5, BART, and Applications	**W14**	Ethical, Social, and Legal Implications of Generative AI
W7	Prompt Engineering Techniques for Zero-shot, Few-shot, and Fine-tuned Learning	**W15**	**Case Studies / Projects / Demo**
W8	Introduction to Generative AI and Overview of Large Language Models	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS671	**Advanced Computer Communication & Networks**	3(3+0)	Nil

Recommended Texts

- Computer and Communication Networks, Nader F. Mir, Edition 2, Prentice Hall, 2014, ISBN: 0133814831, 9780133814835
- Telecommunications Networks, Jesus Hamilton Ortiz, InTech, 2012, ISBN- 978-953-51-0341-7
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course guides the students through the basics of rapidly emerging networks to more advanced concepts and future expectations of Telecommunications Networks.

Course Objectives

- To study the problematic of service integration in TCP/IP networks focusing on protocol design, implementation and performance issues;
- To debate the current trends and leading research in the computer networking area.

Week Wise Topics to be Covered

W1	Introduction: Course organization and objectives. Next generation networking: Motivation and Challenges	W9	Case studies: Video over IP and VoIP
W2	**IPv6 Internetworking and Mobility** Internetworking with IPv6; IPv6 extensions and functionality. Routing advances.	W10	**Service-oriented architectures (SOA)** services in SOA-based networks;
W3	Mobile IP networking. Micro and macro mobility.	W11	technologies for the support and development of services;technologies and APIs for SOA; Web Services and associated technologies
W4	**IP Convergence and QoS**	W12	**Managing TCP/IP networks** Management models and functions.
W5	Service integration and Quality of Service (QoS) in IP networks.	W13	Autonomic management. Internet measurement and monitoring.
W6	Service contracts. Services specification, configuration and management.	W14	**Self-organizing networks** Ad-hoc, sensors and mesh networks; applications;
W7	**Advanced transport issues and signaling** Reliable and unreliable transport services for the support of QoS and real-time.	W15	communication support Information dissemination, medium access mechanisms, routing mechanisms,
W8	Signaling for Multi constrained Services and Applications.	W16	transport protocols, quality of service and security; Self-organizing concepts in infrastructure networks
MID EXAM/ MID OF SEMESTER			

Code	Course Title	CHs	Pre-Req
CS672	**Advanced Network Security**	3(3+0)	Nil

Recommended Texts

- Network Security Attacks and Countermeasures, Dileep Kumar G., Manoj Kumar Singh, M. K. Jayanthi, IGI Global; 1st edition (January 18, 2016), ISBN: 1466687614, 9781466687615
- Network Security Through Data Analysis: Building Situational Awareness, Michael Collins. O'Reilly Media, 2014, ISBN 10: 1-4493-5790-3
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course aims to perform a practical survey of network security applications and standards, with unmatched support for students. In this age of universal electronic connectivity, viruses and hackers, electronic eavesdropping, and electronic fraud, security is paramount.

Course Objectives

- Understanding of applied cryptography and network security essentials
- Gain experience on conducting research and writing (preliminary) papers in applied cryptography domain.
- To encourage/prepare interested students to purse advanced degrees in security and privacy fields.

Week Wise Topics to be Covered

W1	Foundations of Network Security	**W9**	Zero Trust Architecture and Identity Management
W2	Cryptographic Protocols and Secure Communication	**W10**	Blockchain and Network Security
W3	Authentication, Authorization, and Access Control	**W11**	AI and ML in Network Threat Detection
W4	Firewall Architectures and VPNs	**W12**	Security in Cloud and Virtual Networks
W5	Intrusion Detection and Prevention Systems	**W13**	Security Auditing and Compliance Standards
W6	Secure Routing and Wireless Security	**W14**	Project Simulation and Security Testing
W7	Malware, Botnets, and APTs	**W15**	**Case Studies / Projects / Demo**
W8	Security Policy Design and Risk Management	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS673	**Advanced Network Programming**	3(3+0)	Nil

Recommended Texts

- Advanced Network Programming - Principles and Techniques: Network Application Programming with Java, Bogdan Ciubotaru, Gabriel-Miro Muntean, Springer, 2013, ISBN- 978-1447152910.
- Advanced Network Programming – Principles and Techniques: Network, Bogdan Ciubotaru, Gabriel-Miro Muntean, Springer, 2013, ISBN 978-1-4471-5292-7 | ISBN 978-1-4471-5291-0
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

The course aims for the students to gain experience in systems and network programming together with writing efficient, portable and large scale client/server application programs in the Unix/Linux environment using C language. It builds on Operating System and Computer Networking concepts. This course also addresses how programs in distributed systems can make use of OS services.

Course Objectives

- Understand and apply network programming
- Discuss the different way of data streaming
- Understand synchronous and asynchronous communication mechanism
- Apply multi threading mechanism to control huge number of socket communication

Week Wise Topics to be Covered

Week	Topics	Week	Topics
W1	**Sockets:** Introduction, Socket Definition and Types, Socket-Based Network Communications	W8	Java Database Connectivity Services, JDBC Architecture, JDBC Database Access, JDBC Transactions, JDBC Metadata
			MID EXAM/ MID OF SEMESTER
W2	UDP Sockets, TCP Sockets	W9	Multimedia Content Delivery Services, Protocols specific to Real-Time Data Delivery, Multimedia Delivery over Cellular Networks, DVB-based Multimedia Delivery, Multimedia Delivery over WLAN
W3	**Socket-Based Client–Server Communication:** Introduction, Basic Client–Server Application Programming		
W4	Multi-threaded Server Applications, Unicast, Multicast, and Broadcast Communications	W10	**Server-Side Network Programming:** Introduction. Non-Java Server-Side Network Programming Solution
W5	**Support for Communication-Based Services:** Introduction, Control and Diagnostic Services, Packet InterNet Groper, Internet Control Message Protocol, PING Java Example	W11	Java Servlet, Java Server Pages
		W12	**Client-Side Network Programming:** Introduction, Web Documents Classification
W6	Electronic Mail Services, SMTP Java Example, POP3JavaExampl	W13	Static Documents, Active Documents, Java Script, Java Applets
W7	File Transfer Protocol Service, Simple FTP Java Client Example, Web Content Transfer Service, HTTP Java Client Example	W14	**Advanced Client–Server Network Programming:** Remote Method Invocation
		W15	RMI Strategy A—Using a Common Class, RMI Strategy B—Using Separate Instances
		W16	Applet–Servlet Communication,Applet–Servlet Communication—Exchanging Text

Code	Course Title	CHs	Pre-Req
CS674	**Wireless and Mobile Networks**	3(3+0)	Nil

Recommended Texts

- Goldsmith, A. (2005). Wireless Communications. Cambridge University Press. https://doi.org/10.1017/CBO9780511841220
- Rappaport, T. S., et al. (2022). Millimeter Wave Wireless Communications (2nd ed.). Pearson. https://doi.org/10.1201/9781003083959
- At least 5 research papers from Scopus-indexed international conferences
- At least 5 research papers from Q1/Q2 WoS-indexed journals

Course Description

Explores fundamentals and advanced techniques in wireless communication, mobile networking, standards, and emerging IoT protocols, emphasizing performance and mobility management.

Course Objectives

- To understand wireless PHY and MAC layers.
- To implement ad-hoc, cellular, and mesh network designs.
- To analyze mobility, handoff, and quality-of-service mechanisms.
- To evaluate next-generation wireless technologies (5G/6G, IoT).

Week Wise Topics to be Covered

W1	Fundamentals of wireless communication	**W9**	Sensor and IoT network protocols
W2	RF propagation, fading, and channel modeling	**W10**	Millimeter-wave communication and 5G
W3	MAC-layer protocols and scheduling	**W11**	Edge computing and mobile cloud
W4	Cellular architectures and network planning	**W12**	Network optimization techniques
W5	Mobility management and handoff strategies	**W13**	SDN/NFV in wireless networks
W6	QoS provisioning in mobile networks	**W14**	Performance evaluation and measurement
W7	Wireless security mechanisms	**W15**	**Case Studies / Projects / Demo**
W8	Ad-hoc and mesh routing protocols	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS675	**Mobile Adhoc Networks**	3(3+0)	Nil

Recommended Texts

- Evolutionary Algorithms for Mobile Ad Hoc Networks, Bernabé Dorronsoro, Patricia Ruiz, Pascal Bouvry, Grégoire Danoy, Yoann Pigné, John Wiley & Sons, 2014, ISBN: 1118341139, 9781118341131
- Wireless Ad Hoc Networking: Personal-Area, Local-Area, and the Sensory-Area Networks (Wireless Networks and Mobile Communications) by Shih-Lin Wu (Editor), Yu-Chee Tseng (Editor), Auerbach Publications; 1st edition (September 23, 2019), ISBN-13 : 978-0367389314
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

A relative newcomer to the field of wireless communications, ad hoc networking is growing quickly, both in its importance and its applications. This course aims to discuss how with rapid advances in hardware, software, and protocols, ad hoc networks are now coming of age, and the time has come to bring together into one reference their principles, technologies, and techniques.

Course Objectives

The students who succeeded in this course;

- Provide knowledge of mobile ad hoc networks, design and implementation issues, and available solutions.
- Provide knowledge of routing mechanisms and the three classes of approaches: proactive, on-demand, and hybrid.
- Provide knowledge of clustering mechanisms and the different schemes that have been employed, e.g., hierarchical, flat, and leaderless.

Week Wise Topics to be Covered

W1	Introduction, History, classification, applications and scenarios,	W8	Cross layered Design and working mechanisms
		MID EXAM/ MID OF SEMESTER	
W2	Physical, MAC, network and transport layers characteristics and working mechanism,	W9	MAC layer mechanisms, Hidden node, expose node, contention based schemes
		W10	Vehicular Ad hoc Networks (VANETs)
W3	Routing in MANET: On-demand, table-driven and hybrid routing,	W11	VANETs applications and routing schemes
W4	AODV, DSR, OLSR working mechanisms	W12	Disaster management and environmental applications
W5	Routing metrics: hop count, link state	W13	Quality of Service (QoS) in ad hoc networks
W6	Comparison of routing protocols	W14	Ad hoc networks security issues, challenges and solutions
W7	Limitations and challenges of existing mechanisms	W15	Under water ad hoc networks
		W16	Research issues and future trends

Code	Course Title	CHs	Pre-Req
CS676	**Network Management**	3(3+0)	Nil

Recommended Texts

- Advances in Network Management, Jianguo Ding, Auerbach Publications, 2009, ISBN-978-1420064520.
- Network Management: Principles and Practice, Mani Subramanian, 2010, ISBN-10: 8131734048
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

The course aims to explain in detail that how protocol components communicate with each other. Diagrams are used which help the student to understand complex issues of network resources management.

Course Objectives

- Explain the basic concepts of network design and management.
- Examine the basics of network design.
- Explain the basic concepts of network security.
- Describe the telecommunications management network.

Week Wise Topics to be Covered

W1	Basics of Network Design and Management, Overview of Network Design and Management	W9	Protocols and Standards for Managing a Network, Interfaces for Managing Individual Devices
W2	Network Management Model, Network Design	W10	Introduction to Operations Support System, Interconnection of OSS
W3	Traffic Engineering on a Telephone Network, Design Considerations of Cellular, Radio, and Transmission Networks. Design Models of Data Networks	W11	Telecommunications Management, Network (TMN),TMN Models, TMN, Architecture Styles
		W12	Network Management Protocols
W4	Network Security, Basic Concepts of Network Security, Authentication Techniques	W13	Introduction to SNMP, SNMP Architecture, Common Management Information Protocol (CMIP)
W5	Access Control, Network Access Control	W14	Common Object Request Broker Architecture (CORBA)
W6	Network Security Protocols, Enterprise Network,		
W7	Introduction to Enterprise Networks Technologies to Connect with Other Networks, Network Management	W15	Broadband Network Management, ATM Network Management, Broadband Access Network Management
W8	The Operation, Administration, Maintenance, and Provisioning of a Network	W16	Network Management Tools and Applications, Network Management Tools, Network Management Applications
MID EXAM/ MID OF SEMESTER			

Code	Course Title	CHs	Pre-Req
CS677	**Advance Information Security**	3(3+0)	Nil

Recommended Texts

- Security in Computing. Charles P., Shari P., and Jonathan M. 5th Edition. Pearson. 2015. ISBN 978-0134085043.
- Cryptography and Network Security: Principles & Practices. William Stallings. 7th Edition. Prentice Hall. 2015. ISBN 978-0134444280.
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course is designed for graduate students to understand advance level concepts in information security. It covers both technical and managerial aspects of information security. It explains all core principles of information security. It elaborates vulnerabilities, threats, attacks, and crimes with defenses. Further, it will discuss security issues in system, program, network, multimedia, biometrics, and smartcards technology. Finally, it will cover legal & ethical issues in intellectual property rights, data security and cyber crimes.

Course Objectives

- To learn advance level concepts and principles of information security
- To distinguish vulnerabilities, threats, and attacks with defenses
- To differentiate computer crimes and cyber crimes with defenses
- To understand issues in system, program, and network security

Week Wise Topics to be Covered

W1	Definitions, Role, and Applications of IS etc. Principles of IS, Working & Implementation	W9	Network Security: TCP/IP Security, DNS Security, TLS/SSL Security Protocols Network Intrusion Detection & Prevention
W2	Vulnerabilities in Information Security Types & Defenses against Vulnerabilities	W10	Cryptographic Techniques: Substitution: Caesar Cipher and Others Transposition: Rail Fence and Others
W3	Threats in Information Security Types & Defenses against Threats	W11	Private Key Distribution Approaches Stream Ciphers vs. Block Ciphers. DES Algorithm vs. AES Algorithm
W4	Attacks in Information Security Types & Defenses against Attacks	W12	Public Key Distribution Approaches. RSA Algorithm vs. PGP Application Comparison of Symmetric and Asymmetric
W5	Crime, Computer Crime, Cyber Crime Types & Defenses against Cyber Crimes	W13	Data Integrity Methods: Hash, MDs, SHAs Authentication Methods: MACs, Digital Signatures and Digital Certificates
W6	System Security: Violations & Breaches System Security: Measures & Tools Case Studies: Windows, UNIX, SELinux	W14	Multimedia Security: Copyright Protection, Steganography, Ownership Identification, Data Authentication, Transaction Tracking,
W7	Program Security: Flaws, Errors, Failures etc. Non-malicious Programs: Buffer Overflows, Incomplete Mediation, Time of Check	W15	Biometrics: Physiological vs. Behavioral Smartcards: Chip Cards, Tokens, RFIDs
W8	Malicious Programs: Virus and its Signature Defenses: Anti-Viruses, Firewalls, Kerberos	W16	Legal & Ethical Issues: IPR, Copyrights W16 Data Protection & Privacy Acts (US, EU) Electronic & Cyber Crime Acts (Pakistan)
MID EXAM/ MID OF SEMESTER			

Code	Course Title	CHs	Pre-Req
CS678	**Networks Middle Ware Design**	3(3+0)	Nil

Recommended Texts

- Middleware Networks: Concept, Design and Deployment of Internet Infrastructure, Michah Lerner, George Vanecek, Nino Vidovic, Dado Vrsalovic, 2013, ISBN 978-0-306-47022-6 | ISBN 978-0-7923-7840-2
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course focuses on the essential principles and priorities of system design and emphasizes the new requirements brought forward by the rise of e-commerce and distributed integrated systems. This reference highlights the changes to middleware technologies and standards.

Course Objectives

- Understand the role and importance of middleware in business applications.
- Learn how to develop middleware-based solutions to simplify development of complex network systems.
- Learn network programming techniques and how to use these techniques

Week Wise Topics to be Covered

W1: Distributed Transaction and Messaging Middleware: Port Mapper. RPC Features. XMLRPC. Microsoft RFC Facility. The Stubs. OSF Standards for RPC	**W7:** Benefits to Users and Developers. Microsoft Transaction Server. MSMQ in Windows XP. Windows 2000 Datacenter. Enterprise Memory Architecture. Winsock Direct
W2: MSMQ Features. Microsoft Queued Components. When the Network	**W8:** Increasing Server Performance. Windows 2000 Clustering Technologies.
W3: Application Programs and Messaging. Queue Managers. Commercial	**MID EXAM/ MID OF SEMESTER** **W9: EverExpanding Java World:** Inside Enterprise Beans. The Container. Examples
W4: ObjectOriented Middleware CORBA 3: CORBA Release Summary. Organizational Structure. What Is New? CORBA 3. Improved Integration with lava and the Internet. Quality of Service Control. The CORBA Component Model. CCM Development Stages	**W10:** Java 2 Enterprise Edition. Oracle9i AS Containers for J2EE. Configuring and Assembling J2EE Applications. **W11:** PointtoPoint. PublishSubscribe. OpenJMS. Naming Systems and Services. JNDI Architecture
W5: Component Usage Patterns. Container Programming Model. CORBA Object Services. OpenORB. Other Supporting Facilities. WorkinProgress Status.	**W12:** The Naming Package. JNDI. XML Messaging XML Parsing and Data Binding. The Java API for XML Parsing. JavaSpaces and Jini Technologies **W13: Web Services:** Comparing Definitions or Descriptions. Web Services Stack. Web Services Architecture Narrative. Oracle Web Services
W6: Microsoft's Stuff: NET Architecture. Multi Platform Development. Some Advantages. Exhibit 3 NET Layers Web Services. Building the NET Platform. NET Enterprise Servers. NET Framework Security Policy. Evidence Based Security. Role Based Security. Isolated Storage	**W14:** BEA Web Services. Borland Web Services. UDDI Registration. DDI Registrars and Services **W15:** Workflow Processes. Versioning of Web Services. ThirdParty Tools. Silverstream **W16:** IONA Technologies. Interoperability Test Web Service Description. Broker and Supplier Web Service Description

Code	**Course Title**	**CHs**	**Pre-Req**
CS679	**Advanced Wireless Networks**	3(3+0)	Nil

Recommended Texts

- Advanced Wireless Networks: Technology and Business Models, Savo Glisic, John Wiley & Sons, 2016, ISBN: 1119096855, 9781119096856
- Advanced Wireless Networks: Cognitive, Cooperative & Opportunistic 4G Technology, Savo Glisic, Beatriz Lorenzo, Wiley & sons, 2009, ISBN: 978-0-470-74250-1
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

After a short introduction to current wireless network technologies including both Wi-Fi and 2G/3G networks, this course is focused on new wireless architectures where mobility, density, cardinality are the main issues. More particularly, this course will address wireless sensor networks, mobile ad hoc networks, vehicle ad hoc networks. Architectures, protocols and services/applications will be discussed.

Course Objectives

- Critically review and analyze existing wireless networks in order to conceptually map current and emerging mobile devices technology
- Investigate wireless system requirements and extrapolate findings to develop wireless networks for mobile communication
- Elucidate and justify the advantages of the proposed wireless network design to both specialist and non-specialist audiences.

Week Wise Topics to be Covered

W1	Overview of Wireless Technologies and Standards	**W9**	5G and Beyond: Architecture and Use Cases
W2	Channel Modeling and Propagation	**W10**	Cognitive Radio Networks
W3	MAC Protocols for Wireless Networks	**W11**	Security and Privacy in Wireless Systems
W4	Routing Protocols in Mobile Ad Hoc Networks	**W12**	IoT and M2M Communication
W5	QoS Support in Wireless Networks	**W13**	AI and ML for Wireless Network Optimization
W6	Energy-Efficient and Delay-Tolerant Networking	**W14**	Project Testing and Performance Evaluation
W7	Mobility Models and Handoff Strategies	**W15**	**Case Studies / Projects / Demo**
W8	Wireless Sensor and Mesh Networks	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS771	**Advanced Cybersecurity Engineering**	3(3+0)	Nil

Recommended Texts

- Cybersecurity: A Practical Engineering Approach by Henrique M. D. Santos, (2022), ISBN: 9780367252427
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course explores foundational and advanced principles of cybersecurity, with a focus on system security, access control, cryptography, perimeter defense, and the evolving landscape of network and computer attacks. Students will gain proficiency in implementing robust cybersecurity mechanisms and analyzing threat landscapes using current research and technologies.

Course Objectives

- To acquire a solid foundation in cybersecurity fundamentals, access control techniques, and cryptographic basics.
- To Implement and analyze advanced access control and perimeter security techniques.
- To Apply cryptographic operations to secure communication in real-world scenarios.
- To Demonstrate expertise in analyzing network/computer attacks and evaluating current cybersecurity research.

Week Wise Topics to be Covered

W1	Introduction to Cybersecurity: Principles, Goals, and Threat Landscape	**W9**	Network Perimeter Security: Firewalls, IDS/IPS
W2	Cybersecurity Models and Frameworks	**W10**	Host-Based Security and Hardening
W3	Access Control Fundamentals	**W11**	Anatomy of Network and Computer Attacks
W4	Advanced Access Control Models (RBAC, ABAC, DAC, MAC)	**W12**	Malware, APTs, Ransomware, Botnets
W5	Cryptographic Foundations (Symmetric & Asymmetric)	**W13**	Mitigation Strategies and Defense-in-Depth
W6	Hashing, Digital Signatures, Key Management	**W14**	Emerging Cybersecurity Trends (Zero Trust, AI in Security)
W7	Cryptographic Protocols and Secure Communication	**W15**	Case Studies and Research Directions in Cybersecurity
W8	Internet & Web Communication Models (SSL/TLS, HTTPS)	**W16**	Final Presentations / Project Demonstrations
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS772	**Enterprise Cybersecurity**	3(3+0)	Nil

Recommended Texts

- Enterprise Cybersecurity in Digital Business 1st Edition by Ariel Evans, (2022), ISBN: 978-0367511494
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course explores cybersecurity from an enterprise perspective. It delves into the historical evolution of cyber threats, essential cybersecurity tools, and regulatory frameworks that govern enterprise environments. Students will gain skills in risk management, incident response, forensics, GDPR compliance, and strategic cyber defense planning. Emphasis is placed on aligning cybersecurity initiatives with business objectives to ensure comprehensive protection in large-scale organizations.

Course Objectives

- To Analyze the evolution and dynamics of cyber risk and demonstrate understanding of enterprise cybersecurity fundamentals.
- To employ advanced cybersecurity tools and techniques tailored for enterprise-scale systems and infrastructures. To apply incident response, audit, and forensic investigation methods aligned with cybersecurity regulations and frameworks.
- To design and evaluate cybersecurity risk strategies, including GDPR compliance, vendor risk management, and cyber insurance.

Week Wise Topics to be Covered

W1	Introduction to Enterprise Cybersecurity; Historical Evolution of Cyber Risk	**W9**	Midterm Exam & Case Study Review
W2	Cybersecurity Fundamentals: Confidentiality, Integrity, Availability (CIA), Threat Modeling	**W10**	Risk Management in Cybersecurity: Risk Assessment, Risk Appetite
W3	Common Threats in Enterprises: APTs, Insider Threats, Ransomware	**W11**	Cyber Insurance: Policy Types, Claims, and Risk Transfer
W4	Overview of Enterprise Cybersecurity Tools (EDR, SIEM, DLP, PAM)	**W12**	Data Privacy in Enterprises: GDPR Principles, Rights, and Fines
W5	Cybersecurity Frameworks: NIST, ISO/IEC 27001, CIS Controls	**W13**	Cyber Vendor Risk Management: Third-Party Risk Assessment Frameworks
W6	Regulatory Compliance: GDPR, HIPAA, PCI-DSS, SOX	**W14**	Security Operations Centers (SOCs) and Threat Intelligence in Enterprises
W7	Incident Response Planning: Stages, Playbooks, NIST 800-61	**W15**	Recent Trends in Enterprise Cybersecurity (Zero Trust, XDR, SASE)
W8	Digital Forensics in the Enterprise: Tools, Chain of Custody, Evidence Handling	**W16**	Final Presentations and Research Topics in Enterprise Security
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS772	**Enterprise Cybersecurity**	3(3+0)	Nil

Recommended Texts

- Enterprise Cybersecurity in Digital Business 1st Edition by Ariel Evans, (2022), ISBN: 978-0367511494
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course explores cybersecurity from an enterprise perspective. It delves into the historical evolution of cyber threats, essential cybersecurity tools, and regulatory frameworks that govern enterprise environments. Students will gain skills in risk management, incident response, forensics, GDPR compliance, and strategic cyber defense planning. Emphasis is placed on aligning cybersecurity initiatives with business objectives to ensure comprehensive protection in large-scale organizations.

Course Objectives

- To Analyze the evolution and dynamics of cyber risk and demonstrate understanding of enterprise cybersecurity fundamentals.
- To employ advanced cybersecurity tools and techniques tailored for enterprise-scale systems and infrastructures. To apply incident response, audit, and forensic investigation methods aligned with cybersecurity regulations and frameworks.
- To design and evaluate cybersecurity risk strategies, including GDPR compliance, vendor risk management, and cyber insurance.

Week Wise Topics to be Covered

W1	Introduction to Enterprise Cybersecurity; Historical Evolution of Cyber Risk	**W9**	Midterm Exam & Case Study Review
W2	Cybersecurity Fundamentals: Confidentiality, Integrity, Availability (CIA), Threat Modeling	**W10**	Risk Management in Cybersecurity: Risk Assessment, Risk Appetite
W3	Common Threats in Enterprises: APTs, Insider Threats, Ransomware	**W11**	Cyber Insurance: Policy Types, Claims, and Risk Transfer
W4	Overview of Enterprise Cybersecurity Tools (EDR, SIEM, DLP, PAM)	**W12**	Data Privacy in Enterprises: GDPR Principles, Rights, and Fines
W5	Cybersecurity Frameworks: NIST, ISO/IEC 27001, CIS Controls	**W13**	Cyber Vendor Risk Management: Third-Party Risk Assessment Frameworks
W6	Regulatory Compliance: GDPR, HIPAA, PCI-DSS, SOX	**W14**	Security Operations Centers (SOCs) and Threat Intelligence in Enterprises
W7	Incident Response Planning: Stages, Playbooks, NIST 800-61	**W15**	Recent Trends in Enterprise Cybersecurity (Zero Trust, XDR, SASE)
W8	Digital Forensics in the Enterprise: Tools, Chain of Custody, Evidence Handling	**W16**	Final Presentations and Research Topics in Enterprise Security
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	**Course Title**	**CHs**	**Pre-Req**
CS773	**Applied Cryptography**	3(3+0)	Nil

Recommended Texts

- Paar, C., Pelzl, J., & Güneysu, T. (2024). Understanding Cryptography: From established symmetric and asymmetric ciphers to post-quantum algorithms (2nd ed.). Springer. https://doi.org/10.1007/978-3-662-69007-9
- Boneh, D., & Shoup, V. (2023). A Graduate Course in Applied Cryptography. Cambridge University Press. https://doi.org/10.1017/9781139959503
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

A graduate-level deep dive into modern cryptography, covering symmetric and asymmetric primitives, cryptographic protocols, security proofs, and post-quantum algorithms, with a strong emphasis on formal reasoning and real-world applications.

Course Objectives

- To understand the theoretical foundations of symmetric and asymmetric cryptographic algorithms.
- To examine the mathematical hardness assumptions and formal security proofs behind cryptographic protocols.
- To implement secure encryption schemes, hash functions, and digital signatures using modern libraries.
- To evaluate the applicability of post-quantum cryptography in current cryptographic infrastructures.

Week Wise Topics to be Covered

W1	Number theory & algebraic structures	**W9**	Security models and reduction techniques
W2	Pseudorandom generators and functions	**W10**	Real-world protocols: TLS, SSL, PGP
W3	Symmetric encryption schemes & modes	**W11**	Zero-knowledge proofs and commitment schemes
W4	Hash functions and message authentication codes	**W12**	Secure multiparty computation and threshold cryptography
W5	IND-CPA security and secret-key encryption	**W13**	Post-quantum primitives: lattice- and hash-based
W6	Public-key encryption: RSA, ElGamal	**W14**	Security models and reduction techniques
W7	Digital signatures and security proofs (DSA, ECDSA)	**W15**	**Case Studies / Projects / Demo**
W8	Authenticated encryption and secure composition	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	**Course Title**	**CHs**	**Pre-Req**
CS774	**Blockchain Technologies**	3(3+0)	Nil

Recommended Texts

- Xu, J., & Zhang, Y. (2023). Blockchain Security and Privacy: Challenges and Solutions. IEEE Press. https://doi.org/10.1109/9781119876543
- Kumari, S., et al. (2024). Post-Quantum Blockchain Technologies. Springer. https://doi.org/10.1007/978-3-031-15000-1
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

Comprehensive examination of decentralized ledger technologies, covering blockchain architecture, consensus mechanisms, smart contract design, privacy-enhancing tech, and integration of post-quantum cryptography.

Course Objectives

- To understand the core architecture and decentralized principles of blockchain systems.
- To examine consensus mechanisms such as PoW, PoS, and BFT and their trade-offs in performance and security.
- To implement smart contracts and decentralized applications on platforms like Ethereum.
- To evaluate blockchain scalability, privacy, and quantum-resistance strategies for real-world adoption.

Week Wise Topics to be Covered

W1	Distributed systems and blockchain fundamentals	**W9**	Oracles and cross-chain mechanisms
W2	Bitcoin architecture and Merkle trees	**W10**	Alternative platforms: Hyperledger, Solana
W3	Proof-of-Work consensus	**W11**	Layer-2 scaling: sidechains and sharding
W4	Proof-of-Stake, DPoS, BFT algorithms	**W12**	Privacy: ZKPs, mixers, ring signatures
W5	P2P network models and data propagation	**W13**	DeFi design, tokenomics, and governance
W6	Blockchain security threats and defenses	**W14**	Post-quantum blockchain designs and implementations
W7	Ethereum ecosystem and EVM basics	**W15**	**Case Studies / Projects / Demo**
W8	Smart contract auditing and vulnerability mitigation	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS775	**Digital Forensics**	3(3+0)	Nil

Recommended Texts

- Easttom, C. (2023). System Forensics, Investigation, and Response. Jones & Bartlett. https://doi.org/10.1201/9781003253015
- Lillis, D., Becker, B., O'Sullivan, T., & Scanlon, M. (2022). Digital Forensics and Incident Response. River Publishers. https://doi.org/10.1201/9781003257679
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

Delivers a wide-ranging introduction to forensic investigations across digital media—disk, memory, network, mobile, and cloud—integrating acquisition, analysis techniques, legal considerations, and anti-forensic methods.

Course Objectives

- To understand the principles and methodologies of digital evidence acquisition and preservation.
- To examine forensic techniques for analyzing storage media, memory, and network traffic.
- To implement forensic tools and scripts for automating investigation workflows.
- To evaluate the legal and ethical considerations involved in digital forensic processes.

Week Wise Topics to be Covered

W1	Introduction to digital investigations and forensics methodology	**W9**	Network packet capture and intrusion reconstruction
W2	Legal aspects, privacy laws, and chain-of-custody protocols	**W10**	Malware forensic reverse-engineering
W3	Disk imaging techniques and file integrity tools	**W11**	Mobile device data extraction and analysis
W4	NTFS, ext4, FAT file system analysis	**W12**	Cloud forensics and remote evidence gathering
W5	File carving and deleted data recovery	**W13**	Forensic report writing and courtroom presentation
W6	Registry and metadata artifact extraction	**W14**	Emerging trends: anti-forensics, IoT forensics
W7	Email and database forensics	**W15**	**Case Studies / Projects / Demo**
W8	Volatile memory acquisition and analysis	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS871	**Wireless Mesh Networks**	3(3+0)	Nil

Recommended Texts

- Wireless Sensor Networks: Concepts, Applications, Experimentation and Analysis, H. M. A. Fahmy, Springer; 1st ed. 2016 edition (April 25, 2018), ISBN-13 : 978-9811091568
- Wireless Sensor Networks: From Theory to Applications, Ibrahiem M. M. El Emary, CRC Press, 2013, ISBN-978-1466518100.
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course provides a broad coverage of challenges and latest research results related to the design and management of wireless sensor networks. Covered topics include network architectures, node discovery and localization, deployment strategies, node coverage, routing protocols, medium access arbitration, fault-tolerance, and network security.

Course Objectives

- To Understand the basic WSN technology and supporting protocols, with emphasis placed on standardization basic sensor systems and provide a survey of sensor technology
- Understand the medium access control protocols and address physical layer issues
- Learn key routing protocols for sensor networks and main design issues
- Learn transport layer protocols for sensor networks, and design requirements
- Understand the Sensor management, sensor network middleware, operating systems.

Week Wise Topics to be Covered

Week	Topic
W1	Introduction, challenges, limitations, advantages of WSN. Architecture and working mechanisms
W2	Types, nodes arrangement, deployment and communication mechanisms
W3	Physical and MAC layers
W4	Network layer and routing mechanisms
W5	Road safety and Traffice Control mechanisms
W6	Patients, kids and Health Monitoring applications
W7	Pipeline Monitoring, Active Volcano . Underground Mining, forests and animals movement monitoring
W8	Body Area Network and its applications
MID EXAM/ MID OF SEMESTER	
W9	Smart homes applications
W10	Energy efficiency in sensor networks, clustering techniques
W11	Energy harvesting schemes
W12	Advanced routing protocols, cluster-based routing protocols
W13	Wireless sensor networks security challenges and solutions
W14 W15	Advanced topics in wireless sensor networks (research/ review papers)
W16	Research issues and future trends

Code	Course Title	CHs	Pre-Req
CS872	**Wireless Sensor Networks**	3(3+0)	Nil

Recommended Texts

- Wireless Sensor Networks: Concepts, Applications, Experimentation and Analysis, H. M. A. Fahmy, Springer; 1st ed. 2016 edition (April 25, 2018), ISBN-13 : 978-9811091568
- Wireless Sensor Networks: From Theory to Applications, Ibrahiem M. M. El Emary, CRC Press, 2013, ISBN-978-1466518100.
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- Understand the medium access control protocols and address physical layer issues
- Learn key routing protocols for sensor networks and main design issues
- Learn transport layer protocols for sensor networks, and design requirements
- Understand the Sensor management, sensor network middleware, operating systems.

Week Wise Topics to be Covered

Week	Topic	Week	Topic
W1	Introduction, challenges, limitations, advantages of WSN. Architecture and working mechanisms	W8	Body Area Network and its applications
			MID EXAM/ MID OF SEMESTER
W2	Types, nodes arrangement, deployment and communication mechanisms	W9	Smart homes applications
W3	Physical and MAC layers	W10	Energy efficiency in sensor networks, clustering techniques
W4	Network layer and routing mechanisms	W11	Energy harvesting schemes
W5	Road safety and Traffice Control mechanisms	W12	Advanced routing protocols, cluster-based routing protocols
W6	Patients, kids and Health Monitoring applications	W13	Wireless sensor networks security challenges and solutions
W7	Pipeline Monitoring, Active Volcano . Underground Mining, forests and animals movement monitoring	W14 W15	Advanced topics in wireless sensor networks (research/ review papers)
		W16	Research issues and future trends

Code	**Course Title**	**CHs**	**Pre-Req**
CS873	**Advanced Networking**	3(3+0)	Nil

Recommended Texts

- Kurose, J. F., & Ross, K. W. (2020). Computer Networking: A Top-Down Approach (8th ed.). Pearson. https://doi.org/10.1201/9780429264467
- Nunes, B. A. A., et al. (2014). A Survey of SDN: Past, Present, and Future of Programmable Networks. Proceedings of the IEEE. https://doi.org/10.1109/JPROC.2014.2371999
- At least 5 research papers from Scopus-indexed international conferences
- At least 5 research papers from Q1/Q2 WoS-indexed journals

Course Description

Focuses on next-generation network protocols, architectures, programmability (SDN/NFV), security, and performance evaluation in modern networked systems.

Course Objectives

- To understand TCP/IP and QoS mechanisms.
- To design and deploy policies in SDN/NFV environments.
- To evaluate network performance critically.
- To implement security and resilience in advanced networks.

Week Wise Topics to be Covered

W1	Modern TCP/IP protocol stack and enhancements	**W9**	Network security and resilience
W2	Quality-of-service mechanisms and traffic shaping	**W10**	IoT network integration frameworks
W3	Advanced routing and switching protocols	**W11**	Network telemetry and analytics
W4	Congestion control strategies	**W12**	Edge/cloud networking architectures
W5	Wireless-mobile-wireline integration	**W13**	Policy and programmability frameworks
W6	Content Delivery Networks (CDNs)	**W14**	Performance benchmarking and evaluation
W7	Network virtualization with SDN	**W15**	**Case Studies / Projects / Demo**
W8	NFV architecture and use cases	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS874	**Advanced Wireless Network Security**	3(3+0)	Nil

Recommended Texts

- Next Generation Wireless Network Security and Privacy, Advances in Information Security, Privacy, and Ethics: Lakhtaria, Kamaljit I. Publisher IGI Global, 2015, ISBN: 146668688X, 9781466686885
- Advances in Security of Information and Communication Networks, Ali Ismail Awad, Aboul-Ella Hassanien, Kensuke Baba, Springer, 2013, ISBN- 978-3642405969.
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

The purpose of this course is to familiarize participants with threats, vulnerabilities, and security countermeasures of existing and upcoming wireless and mobile networks. The covered security topics include a wide range of mainstream wireless technologies, such as, WLAN, Bluetooth, GSM, and UMTS. In addition, the course will explore security and privacy problems, and potential solutions of new and emerging wireless technologies, such as ZigBee, wireless mesh networks, and RFIDs.

Course Objectives

- understand the main security goals and adversarial models of wireless and mobile networks;
- gain a broad knowledge regarding real-world security architectures of WLANs, GSM/UMTS, WSNs, RFIDs, etc.;
- be able to reason about wireless security protocols and protection techniques, discuss proposed solutions and their limitations;

Week Wise Topics to be Covered

W1	Introduction to network security, security requirements, considerations, external and internal threats	W7	Authentication, authorization and accessibility (AAA)
W2	Types of security threats (active, passive, DoS, DDoS)	W8	Intrusion detection systems
		MID EXAM/ MID OF SEMESTER	
W3	Physical layer security attacks and solutions	W9	SNORT
W4	MAC layer security attacks and solutions	W10	Intrusion detection and response systems
W5	Network layer security attacks and solutions	W11 W12	Secure protocols
W6	Key management schemes	W13	Wireless Network forensics
		W14 W15	Advanced topics in security
		W16	Research issues and future trends

Code	Course Title	CHs	Pre-Req
CS875	**Mobile Communication Systems**	3(3+0)	Nil

Recommended Texts

- Mobile Communications, A. Jagoda, M. de Villepin, Vieweg+Teubner Verlag, 2013, ISBN- 978-3322992710.
- Mobile Communication Systems, Krzysztof Wesolowski, Wiley, 2002, ISBN-978-0471498377
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

In this course, students examine fundamental concepts of mobile cellular communications and specifics of current and proposed U.S. cellular systems.

Course Objectives

- Be acquainted with the role of cellular and mobile communications in frequency management issues.
- Be acquainted with different interference factors influencing cellular and mobile communications.
- Be able to efficiently use the background behind developing different path loss and/or radio coverage in cellular environment.

Week Wise Topics to be Covered

Week	Topics
W1	Cellular mobile radio systems: Introducti on to Cellular Mobile System, Performance criteria, uniqueness of mobile radio environment
W2	operation of cellular systems, Hexagonal shaped cells
W3	Analog and Digital Cellular systems.
W4	Elements of cellular radio system design: zochannel Interference Reduction Factor,
W5	desired C/I from a normal case in an omni directional Antenna system,
W6	Cell splitting, consideration of the components of Cellular system
W7	Interference: Introduction to Co-Channel Interference,
W8	real time Co-Channel interference,
MID EXAM/ MID OF SEMESTER	
W9	Co-Channel measurement, design of Antenna system,
W10	Antenna parameters and their effects, Diversity receiver, non-co-channel interference-different types.
W11	Cell coverage for signal and traffic: Signal reflections in flat and hilly terrain, effect of human made structures,
W12	phase difference between direct and reflected paths, constant standard deviation,
W13	straight line path loss slope, general formula for mobile propagation over water and flat open area, near and long-distance propagation antenna height gain, Form of a point to point model.
W14	Frequency management and channel assign ment: Numbering and grouping, setup access and paging channels channel as signments to cell sites and mobile units, channel sharing and borrowing,
W15	Sectorization, overlaid cells, Non fixed channel assignmen t. Radio paging,
W16	Private radio communication, Cellular radio communication, Cordless telephony, Marine, satellite and aircraft communications

Code	Course Title	CHs	Pre-Req
CS876	**Information and Coding Theory**	3(3+0)	Nil

Recommended Texts

- Information Theory and Coding - Solved Problems, by Predrag Ivaniš (Author), Dušan Drajić (Author), Springer; 1st ed. 2017 edition (November 29, 2016), ISBN-13: 978-3319493695
- Information Theory and Coding by Example, Mark Kelbert, Yuri Suhov, Cambridge University Press, 2013, ISBN-978-0521139885.
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course provides a basic understanding of the nature of information, the effects of noise in analogue and digital transmission systems and the construction of both source codes and error-detection/-correction codes.

Course Objectives

- To equip students with the basic understanding of the fundamental concept of entropy and information as they are used in communications.
- To enhance knowledge of probabilities, entropy, measures of information.
- To guide the student through the implications and consequences of fundamental theories and laws of information theory and coding theory with reference to the application in modern communication and computer systems

Week Wise Topics to be Covered

W1	Essentials of Information Theory Basic concepts. The Kraft inequality. Huffman's encoding	W9	Cyclic codes and polynomial algebra. Introduction to BCH codes , Additional problems ,Further Topics from Coding Theory
W2	Entropy: an introduction, Shannon's first coding theorem. The entropy rate of a Markov source	W10	A primer on finite fields, Reed–Solomon codes. The BCH codes revisited
W3	Channels of information transmission. Decoding rules. Shannon's second coding theorem	W11	Cyclic codes revisited. Decoding the BHC codes ,The MacWilliams identity and the linear programming bound
W4	Differential entropy and its properties, Additional problems	W12	Asymptotically good codes, Additional problems
W5	Introduction to Coding Theory	W13	Further Topics from Information Theory Gaussian channels and beyond
W6	Hamming spaces. Geometry of codes. Basic bounds on the code size	W14	The asymptotic equipartition property in the continuous time setting
W7	A geometric proof of Shannon's second coding theorem.	W15	The Nyquist–Shannon formula , Spatial point processes and network information theory
W8	Advanced bounds on the code size , Linear codes: basic constructions. The Hamming, Golay and Reed–Muller codes	W16	Selected examples and problems from cryptography , Additional problems
MID EXAM/ MID OF SEMESTER			

Code	Course Title	CHs	Pre-Req
CS877	**Traffic Control and Quality of Services**	3(3+0)	Nil

Recommended Texts

- QOS-Enabled Networks: Tools and Foundations (by Miguel Barreiros, Peter Lundqvist, Wiley; 2nd edition (February 8, 2016), ISBN-13: 978-1119109105
- Advanced Quality of Service (Qos), Syed Zahidur Rashid, Omniscriptum Gmbh & Company Kg., 2015, ISBN: 3659687898, 9783659687891
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course describes, analyzes, and recommends traffic engineering (TE) and quality of service (QoS) optimization methods for integrated voice/data dynamic routing networks. These functions control a network's response to traffic demands and other stimuli, such as link failures or node failures. TE and QoS optimization is concerned with measurement, modeling, characterization, and control of network traffic, and the application of techniques to achieve specific performance objectives.

Course Objectives

- interpret basic traffic measurements to calculate speed flow and density
- select and design appropriate traffic management systems
- calculate basic timings for an isolated signal installation

Week Wise Topics to be Covered

W1	Traffic Management Principles in Networks	**W9**	QoS in Multimedia and Real-Time Applications
W2	Queuing Theory and Traffic Modeling	**W10**	Software-Defined Networking and QoS
W3	QoS Architectures and Mechanisms	**W11**	AI-based Traffic Prediction and Control
W4	Traffic Policing and Shaping Techniques	**W12**	Case Studies in QoS Implementations
W5	Scheduling Algorithms for QoS	**W13**	QoS Policy Design and SLA Management
W6	Integrated and Differentiated Services	**W14**	Project Simulations and Evaluation
W7	QoS in IP, MPLS, and Wireless Networks	**W15**	**Case Studies / Projects / Demo**
W8	Admission Control and Resource Allocation	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS878	**Cloud, Edge and Fog computing**	3(3+0)	Nil

Recommended Texts

- Shi, W., & Dustdar, S. (2016). The Promise of Edge Computing. Computer. https://doi.org/10.1109/MC.2016.269
- García López, P., et al. (2020). Edge Computing: A Primer. Springer. https://doi.org/10.1007/978-3-030-58305-9
- At least 5 research papers from Scopus-indexed international conferences
- At least 5 research papers from Q1/Q2 WoS-indexed journals

Course Description

Examines architectures and systems enabling distributed computing across Cloud, Edge, and Fog layers, focusing on resource management, latency-critical applications, and system orchestration.

Course Objectives

- To understand Cloud, Edge, and Fog computing paradigms.
- To implement containerized microservices across distributed tiers.
- To optimize systems for latency, privacy, and resilience.
- To deploy orchestration tools in real-world environments.

Week Wise Topics to be Covered

W1	Overview of Cloud, Edge, and Fog paradigms	**W9**	Edge AI inference and analytics
W2	Containerization and orchestration technologies	**W10**	Federated learning in distributed systems
W3	VM vs. container deployment strategies	**W11**	Mobility-aware computing
W4	Latency and QoS trade-offs	**W12**	Monitoring, resilience, and fault tolerance
W5	Data management across tiers	**W13**	Economic and privacy trade-offs
W6	Resource scheduling and placement	**W14**	Project or case study work
W7	Serverless and FaaS models	**W15**	**Case Studies / Projects / Demo**
W8	Security across cloud-edge-fog	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS879	**Advancements in AI for Cybersecurity**	3(3+0)	Nil

Recommended Texts

- LeCun, Y., et al. (2020). Adversarial Machine Learning. Communications of the ACM. https://doi.org/10.1145/3383463
- Buczak, A. L., & Guven, E. (2016). A Survey of Data Mining and Machine Learning Methods for Cyber Security Intrusion Detection. IEEE Computer. https://doi.org/10.1109/MCSE.2016.115
- At least 5 research papers from Scopus-indexed international conferences
- At least 5 research papers from Q1/Q2 WoS-indexed journals

Course Description

Studies how AI strengthens cybersecurity and how adversaries exploit AI, covering anomaly detection, adversarial machine learning, threat intelligence, and defense strategies.

Course Objectives

- To understand ML techniques for intrusion and malware detection.
- To implement adversarial threat mitigation strategies.
- To extract threat intelligence via NLP and graph modeling.
- To deploy secure and interpretable AI systems in cybersecurity.

Week Wise Topics to be Covered

W1	Overview of AI applications in cybersecurity	**W9**	Explainable AI in security
W2	Anomaly detection with ML models	**W10**	Secure AI development lifecycle
W3	Malware classification techniques	**W11**	Privacy-aware ML design
W4	Network intrusion detection systems	**W12**	Incident response automation
W5	Host-based intrusion detection	**W13**	Ethical and governance challenges
W6	Adversarial attacks and defenses	**W14**	Explainable AI in security
W7	Threat intelligence using NLP	**W15**	**Case Studies / Projects / Demo**
W8	Graph-based threat modeling	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS971	**Advancements in Wireless and IoT Networks**	3(3+0)	Nil

Recommended Texts

- Atzori, L., Iera, A., & Morabito, G. (2017). The Internet of Things: A Survey. Computer Networks. https://doi.org/10.1016/j.comnet.2010.05.010
- Li, S., & Xu, L. D. (2023). Edge IoT: Architectures, Technologies, and Applications. Springer. https://doi.org/10.1007/978-3-030-86888-5
- At least 5 research papers from Scopus-indexed international conferences
- At least 5 research papers from Q1/Q2 WoS-indexed journals

Course Description

Explores state-of-the-art wireless IoT architectures, protocols, security challenges, and deployment strategies in large-scale heterogeneous environments.

Course Objectives

- To understand modern IoT network stacks and LPWAN protocols.
- To implement energy-efficient and spectrum-aware protocols.
- To secure large-scale heterogeneous IoT deployments.
- To evaluate and simulate IoT networks at scale.

Week Wise Topics to be Covered

W1	IoT reference architectures and models	**W9**	
W2	LPWAN technologies: LoRa, NB-IoT	**W10**	Security and authentication in IoT
W3	Edge and fog layer integration	**W11**	Blockchain and AI integration
W4	Middleware and service frameworks	**W12**	IoT standardization and frameworks
W5	Wireless sensor networking	**W13**	Simulation tools and platforms
W6	Energy harvesting and management	**W14**	Real-world IoT system design
W7	MQTT and CoAP protocols	**W15**	**Case Studies / Projects / Demo**
W8	Quality of service in IoT	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS972	**Advanced Topics in Cybersecurity**	3(3+0)	Nil

Recommended Texts

- The course materials may vary depending on the latest advancements and topics covered in recent books and research articles published in reputable journals such as IEEE, Springer, Elsevier, and Wiley
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This advanced course explores the forefront of cybersecurity research, equipping students with the ability to critically engage with and evaluate current literature from high-impact journals and conferences. Topics span deep learning, NLP, computer vision, reinforcement learning, and ethical AI, with applications in cyber threat detection, defense mechanisms, and secure system design. The course emphasizes reading, analyzing, and presenting recent research and developing forward-looking perspectives on cybersecurity innovations.

Course Objectives

- To analyze and interpret contemporary research in cybersecurity published in top-tier journals.
- To evaluate the applicability of advanced AI technologies such as deep learning, NLP, and RL to cybersecurity challenges.
- To critically assess ethical and practical implications of emerging cybersecurity technologies.
- To synthesize insights from research literature to propose novel directions or improvements in cybersecurity systems.

Week Wise Topics to be Covered

W1	Overview of current research methodologies in cybersecurity	**W9**	Blockchain and decentralized cybersecurity architectures
W2	Survey of recent cybersecurity publications (top journals/conferences)	**W10**	Ethics and governance in AI-driven cybersecurity systems
W3	Deep learning applications in malware analysis and intrusion detection	**W11**	Research challenges in cybersecurity datasets and benchmarking
W4	Adversarial machine learning and model robustness	**W12**	Case studies of real-world AI-integrated cybersecurity incidents
W5	Natural Language Processing in phishing detection and SOC automation	**W13**	Research paper reading and critique (Student Presentations - Part I)
W6	Computer vision techniques in cybersecurity (e.g., biometrics, anomaly detection)	**W14**	Research paper reading and critique (Student Presentations - Part II)
W7	Reinforcement learning for automated defense systems	**W15**	Emerging trends: quantum cybersecurity, LLMs for security
W8	Secure federated learning and privacy-preserving AI	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

⇨ **Program - MS (SE)**	SE Courses begin from here

Program MS (SE)	**Domain – 1: SE Core Subjects**

Code	**Course Title**	**CHs**	**Pre-Req**
SE601	**Advanced Requirements Engineering**	3(3+0)	Nil

Recommended Texts

- Software Requirements by Karl Wiegers, Joy Beatty - https://www.oreilly.com/library/view/software-requirements-3rd/9780735679629/
- Requirements Engineering: Fundamentals, Principles, and Techniques by Klaus Pohl - https://link.springer.com/book/10.1007/978-3-642-12578-4

Course Description

This course provides in-depth understanding and advanced techniques in requirements engineering, covering elicitation, analysis, specification, validation, and management of software requirements.

Course Objectives

- Understand advanced concepts and techniques in requirements engineering
- Develop skills in requirements elicitation, analysis, and documentation
- Manage and validate software requirements in complex projects
- • Apply best practices for requirements traceability and change management

Week Wise Topics to be Covered

W1	Introduction to Advanced Requirements Engineering	**W9**	Requirements Traceability
W2	Stakeholder Analysis and Elicitation Techniques	**W10**	Requirements in Agile and DevOps
W3	Modeling Requirements with Use Cases and Scenarios	**W11**	Requirements Engineering Tools
W4	Functional and Non-Functional Requirements	**W12**	Legal and Ethical Aspects of Requirements
W5	Requirements Analysis and Prioritization	**W13**	Requirements for Safety-Critical Systems
W6	Requirements Specification and Documentation Standards	**W14**	RE in Industrial Applications
W7	Requirements Validation and Reviews	**W15**	**Case Studies / Projects / Demo**
W8	Managing Requirements Changes	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
SE611	**Advanced Software System Architecture**	3(3+0)	Nil

Recommended Texts

- Software Architecture in Practice by Len Bass, Paul Clements, Rick Kazman - https://www.oreilly.com/library/view/software-architecture-in/0321815734/
- Documenting Software Architectures by Paul Clements et al. - https://dl.acm.org/doi/book/10.5555/1052935

Course Description

This course focuses on the design and analysis of complex software architectures, exploring architectural styles, documentation, evaluation, and patterns.

Course Objectives

• Understand architectural design principles and decisions
• Explore architectural patterns and styles
• Evaluate and document software architectures
• Apply architectural practices in real-world systems

Week Wise Topics to be Covered

W1	Introduction to Software Architecture	**W9**	Service-Oriented and Microservices Architecture
W2	Architectural Styles and Patterns	**W10**	Architectural Decisions and Trade-offs
W3	Architecture Design Process	**W11**	Architecture and DevOps
W4	Modeling and Documentation of Architectures	**W12**	Security and Performance in Architectures
W5	Component and Connector Views	**W13**	Tools for Architecture Modeling and Evaluation
W6	Deployment and Runtime Views	**W14**	Industrial Architectures
W7	Quality Attributes and Tactics	**W15**	**Case Studies / Projects / Demo**
W8	Architecture Evaluation Methods	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
SE641	**Software Testing and Quality Assurance**	3(3+0)	Nil

Recommended Texts

- Foundations of Software Testing by Rex Black, Erik van Veenendaal, Dorothy Graham - https://www.pearson.com/en-us/subject-catalog/p/foundations-of-software-testing/P200000004480/9788131794767
- Software Testing: Principles and Practices by Srinivasan Desikan, Gopalaswamy Ramesh - https://dl.acm.org/doi/book/10.5555/1121741

Course Description

This course covers comprehensive software testing and quality assurance methodologies including test planning, techniques, automation, and software process quality.

Course Objectives

• Understand key concepts and techniques of software testing
• Design and execute effective test plans and strategies
• Apply quality assurance practices and process improvements
• Use automated testing tools for efficient testing

Week Wise Topics to be Covered

W1	Introduction to Testing and Quality Assurance	**W9**	Performance, Load, and Stress Testing
W2	Test Levels and Types	**W10**	Security Testing and Code Reviews
W3	Test Planning and Strategy	**W11**	Quality Assurance Models (CMMI, ISO)
W4	Black Box and White Box Testing	**W12**	Defect Life Cycle and Metrics
W5	Static and Dynamic Testing Techniques	**W13**	Regression Testing and Continuous Testing
W6	Unit, Integration, System, and Acceptance Testing	**W14**	Agile and DevOps Testing Practices
W7	Test Case Design and Coverage Criteria	**W15**	**Case Studies / Projects / Demo**
W8	Test Automation and Tools	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Program MS (SE)	**Domain – 2: SE Domain Elective**

Code	Course Title	CHs	Pre-Req
SE612	**Software Measurement and Metrics**	3(3+0)	Nil

Recommended Texts

- Software Metrics: A Rigorous and Practical Approach by Norman Fenton, James Bieman - https://www.pearson.com/en-us/subject-catalog/p/software-metrics-a-rigorous-and-practical-approach/P200000005422/9781138735631
- Metrics and Models in Software Quality Engineering by Stephen Kan - https://dl.acm.org/doi/book/10.5555/521762

Course Description

This course provides the theory and application of software metrics to measure, monitor, and control software processes, products, and projects.

Course Objectives

- Understand software measurement theory and principles
- Apply various software metrics for product and process evaluation
- Use metrics for project planning and quality control
- Evaluate and select effective software measurement programs

Week Wise Topics to be Covered

W1	Introduction to Software Metrics	**W9**	Data Collection and Validation
W2	Measurement Fundamentals and Scales	**W10**	Visualization and Interpretation of Metrics
W3	Goal-Question-Metric Approach	**W11**	Statistical Techniques for Metrics Analysis
W4	Product Metrics: Size, Complexity, and Design	**W12**	Measurement Tools and Dashboards
W5	Process Metrics: Defects, Effort, and Time	**W13**	Benchmarking and Baseline Studies
W6	Project Metrics and Estimation	**W14**	Software Metrics in Industrial Application
W7	Quality and Reliability Metrics	**W15**	**Case Studies / Projects / Demo**
W8	Metrics for Object-Oriented Systems	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
SE613	**Advanced Formal Methods**	3(3+0)	Nil

Recommended Texts

- Boca, P., & Bowen, J. P. (2010). Formal methods: State of the art and new directions. Springer. https://link.springer.com/book/10.1007/978-1-4471-2297-4
- Jackson, D. (2012). Software abstractions: Logic, language, and analysis (Revised ed.). MIT Press. https://mitpress.mit.edu/9780262014427/software-abstractions/
- Gallier, J. H. (2015). Logic for computer science: Foundations of automatic theorem proving. University of Pennsylvania. https://www.cis.upenn.edu/~jean/gbooks/logic.html

Course Description

This course covers the theoretical foundations and practical applications of formal methods for software specification, verification, and validation.

Course Objectives

- Understand formal specification techniques
- Learn verification methods including model checking and theorem proving
- Apply formal tools in software system development
- Analyze the limitations and scope of formal methods

Week Wise Topics to be Covered

W1	Introduction to Formal Methods	**W9**	Hoare Logic and Weakest Preconditions
W2	Set Theory and Predicate Logic Refresher	**W10**	Model Checking with SPIN and NuSMV
W3	Formal Specification Languages (Z, VDM)	**W11**	Software Model Checking (CBMC, SLAM)
W4	Temporal Logic and Model Checking	**W12**	Symbolic Execution and SMT Solvers
W5	Automata Theory for Verification	**W13**	Formal Methods in Industry
W6	Theorem Proving and Proof Assistants	**W14**	Scalability and Usability Challenges
W7	Abstract Interpretation	**W15**	**Case Studies / Projects / Demo**
W8	Program Semantics	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
SE614	**Agile Software Development Methods**	3(3+0)	Nil

Recommended Texts

- Martin, R. C. (2003). Agile software development: Principles, patterns, and practices. Pearson. https://www.pearson.com/store/p/agile-software-development-principles-patterns-and-practices/P100000130835
- Sutherland, J. (2014). Scrum: The art of doing twice the work in half the time. Crown Business. https://www.amazon.com/Scrum-Doing-Twice-Work-Half/dp/038534645X
- Cohn, M. (2005). Agile estimating and planning. Pearson. https://www.pearson.com/en-us/subject-catalog/p/agile-estimating-and-planning/P200000000089/9780131479418

Course Description

This course covers agile methodologies for software development with a focus on iterative development, team collaboration, and adaptive planning.

Course Objectives

- Understand the principles and values of agile development
- Implement agile frameworks like Scrum and XP
- Plan and manage agile projects using modern tools
- Evaluate agile practices in real-world scenarios

Week Wise Topics to be Covered

W1	Introduction to Agile and Lean Principles	**W9**	Tools for Agile Development (Jira, Trello)
W2	Agile Manifesto and Principles	**W10**	Agile Testing and TDD
W3	Scrum Framework Overview	**W11**	DevOps and Continuous Integration
W4	Extreme Programming (XP) Practices	**W12**	Scaling Agile (SAFe, LeSS)
W5	Agile Planning and Estimation	**W13**	Challenges in Agile Adoption
W6	Product Backlog and User Stories	**W14**	Agile in Distributed Teams
W7	Sprint Planning and Execution	**W15**	**Case Studies / Projects / Demo**
W8	Agile Metrics and Burndown Charts	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
SE615	**Empirical Software Engineering**	3(3+0)	Nil

Recommended Texts

- Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). Experimentation in software engineering. Springer. https://link.springer.com/book/10.1007/978-3-540-68255-4
- Shull, F., Singer, J., & Sjøberg, D. I. K. (2008). Guide to advanced empirical software engineering. Springer. https://link.springer.com/book/10.1007/978-1-84800-044-5

Course Description

This course introduces empirical methods in software engineering to design, conduct, and evaluate experiments and case studies.

Course Objectives

- Understand the fundamentals of empirical research in SE
- Design controlled experiments and observational studies
- Apply data analysis techniques to software engineering data
- • Critically evaluate empirical findings in SE literature

Week Wise Topics to be Covered

W1	Introduction to Empirical SE	**W9**	Validity and Reliability in Empirical Research
W2	Research Questions and Hypotheses	**W10**	Regression and Correlation Analysis
W3	Quantitative and Qualitative Methods	**W11**	Systematic Literature Reviews
W4	Controlled Experiments Design	**W12**	Mining Software Repositories
W5	Case Studies and Observational Methods	**W13**	Reproducibility and Replication Studies
W6	Surveys and Questionnaires	**W14**	Ethical Issues in Empirical Research
W7	Data Collection Techniques	**W15**	**Case Studies / Projects / Demo**
W8	Descriptive and Inferential Statistics	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
SE621	**Advanced Software Project Management**	3(3+0)	Nil
Recommended Texts			
Schwalbe, K. (2015). Information Technology Project Management (8th ed.). Cengage Learning. Marchewka, J. T. (2016). Information Technology Project Management (5th ed.). Wiley.			
Course Description			
Focuses on planning, execution, and control of software projects, incorporating modern techniques and leadership practices.			
Course Objectives			
Understand advanced principles of project management. Apply scheduling, budgeting, and scope management tools. Analyze team dynamics and stakeholder engagement. Manage risks and quality in software projects.			

Week Wise Topics to be Covered

W1	Introduction to Software Project Management, Project Lifecycle	**W9**	Agile and Hybrid Project Management Approaches
W2	Project Initiation and Scope Management	**W10**	Stakeholder Management and Communication Planning
W3	Time and Cost Estimation Techniques	**W11**	Procurement and Vendor Management
W4	Project Planning Tools (Gantt, CPM, PERT)	**W12**	Leadership and Team Management
W5	Resource Allocation and Scheduling	**W13**	Risk Management in Software Projects
W6	Project Monitoring and Control	**W14**	Earned Value Management
W7	Quality Assurance and Configuration Management	**W15**	**Case Studies / Projects / Demo**
W8	Project Closure and Post-mortem Analysis	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
SE631	**Component Based Software Engineering**	3(3+0)	Nil

Recommended Texts

- Szyperski, C. (2002). Component software: Beyond object-oriented programming (2nd ed.). Addison-Wesley. https://www.pearson.com/en-us/subject-catalog/p/component-software-beyond-object-oriented-programming/9780201745723.html
- Heineman, G. T., & Councill, W. T. (2001). Component-based software engineering: Putting the pieces together. Addison-Wesley. https://ptgmedia.pearsoncmg.com/images/9780201708919/samplechapter/020170891X.pdf
- Wallnau, K., Hissam, S. A., & Seacord, R. C. (2001). Building systems from commercial components. Addison-Wesley. https://resources.sei.cmu.edu/library/asset-view.cfm?assetid=5789

Course Description

This course focuses on the principles, techniques, and tools for designing and developing component-based software systems.

Course Objectives

• Understand the fundamentals of component-based design
• Explore software component models and architectures
• Analyze and evaluate component integration techniques
• Apply CBSE principles to real-world applications

Week Wise Topics to be Covered

W1	Introduction to CBSE	**W9**	Design Patterns in CBSE
W2	Software Components and Interfaces	**W10**	Service-Oriented Architecture (SOA)
W3	Component Models and Standards (CORBA, COM, EJB)	**W11**	Testing and Verification of Components
W4	Component Design and Specification	**W12**	Component-Based System Deployment
W5	Component Composition and Integration	**W13**	Evolution and Maintenance of Component Systems
W6	Middleware and Component Frameworks	**W14**	Case Studies in CBSE
W7	Quality Attributes in CBSE	**W15**	**Case Studies / Projects / Demo**
W8	Component Repositories and Reuse	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	Remarks
CS763	**Advanced Human Computer Interaction**	Offered in CS on Page #

Program MS (SE)	**Domain – 3: SE General Elective Subjects**

Code	Course Title	CHs	Pre-Req
SE602	**Research Methodology**	3(3+0)	Nil

Recommended Texts

- Creswell, J. W. (2014). Research Design: Qualitative, Quantitative, and Mixed Methods Approaches. SAGE.
- Kumar, R. (2011). Research Methodology: A Step-by-Step Guide for Beginners. SAGE.

Course Description

Covers concepts and skills required to conduct academic and industrial research.

Course Objectives

- Design structured research proposals.
- Differentiate between research methods.
- Collect and analyze data rigorously.
- Present findings ethically and clearly.

Week Wise Topics to be Covered

W1	Introduction to Research and Methodologies	**W9**	Data Analysis Techniques
W2	Research Process and Problem Definition	**W10**	Using Software Tools (SPSS, NVivo)
W3	Literature Review and Referencing	**W11**	Hypothesis Testing and Inferential Statistics
W4	Research Design: Qualitative vs Quantitative	**W12**	Report Writing and Presentation Skills
W5	Sampling Methods and Data Collection	**W13**	Research Proposal Development
W6	Measurement and Scaling Techniques	**W14**	Academic Publishing and Peer Review
W7	Validity, Reliability, and Ethical Considerations	**W15**	**Case Studies / Projects / Demo**
W8	Questionnaire Design and Interviewing	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
SE616	**Agent Based Modeling**	3(3+0)	Nil

Recommended Texts

- Railsback, S. F., & Grimm, V. (2019). Agent-Based and Individual-Based Modeling (2nd ed.). Princeton University Press.
- Gilbert, N., & Troitzsch, K. G. (2005). Simulation for the Social Scientist (2nd ed.). McGraw-Hill.

Course Description

Focuses on simulation models of systems with autonomous agents that interact within defined rules and environments.

Course Objectives

- Understand fundamentals of ABM and its domains.
- Build and validate ABMs using tools like NetLogo.
- Analyze emergence and behavior patterns.
- Apply ABM to real-world domains.

Week Wise Topics to be Covered

W1	Introduction to Agent-Based Modeling	**W9**	ABM in Social Science and Economics
W2	Agents and Environments	**W10**	ABM in Technology and Engineering
W3	Designing Agent Behaviors and Interactions	**W11**	Linking ABM with GIS and Data
W4	Model Implementation (e.g., NetLogo, Repast)	**W12**	Large-scale ABM and High-Performance Computing
W5	Calibration and Validation of ABMs	**W13**	ABM for Policy Modeling and Decision Support
W6	Sensitivity Analysis and Parameter Tuning	**W14**	Verification, Validation, and Documentation
W7	Analyzing Emergent Phenomena	**W15**	**Case Studies / Projects / Demo**
W8	ABM in Ecology and Biology	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
SE622	**Software Risk Management**	3(3+0)	Nil

Recommended Texts

- Charette, R. N. (1989). Software Engineering Risk Analysis and Management. McGraw-Hill.
- Kontio, J. (2001). Software Risk Management: A Practical Guide. Addison-Wesley.

Course Description

This course introduces frameworks for identifying, analyzing, and mitigating risks in software projects.

Course Objectives

- Identify potential project risks and their impact.
- Use quantitative and qualitative assessment methods.
- Plan mitigation strategies.
- Implement risk monitoring processes.

Week Wise Topics to be Covered

W1	Introduction to Software Risk Management	**W9**	Risk Management in Agile Projects
W2	Taxonomy of Software Risks	**W10**	Organizational Risk Culture and Maturity Models
W3	Risk Identification Methods	**W11**	Software Reliability and Fault Tolerance
W4	Qualitative Risk Analysis	**W12**	Tools for Risk Management (e.g., Risk Radar)
W5	Quantitative Risk Analysis and Modeling	**W13**	Continuous Risk Assessment Techniques
W6	Risk Prioritization and Impact Assessment	**W14**	Risk Audits and Reviews
W7	Risk Mitigation Planning	**W15**	**Case Studies / Projects / Demo**
W8	Risk Monitoring and Control	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	**Course Title**	**CHs**	**Pre-Req**
SE623	**Software Configuration Management**	3(3+0)	Nil

Recommended Texts

- Babich, W. A. (1986). Software Configuration Management: Coordination for Team Productivity. Addison-Wesley.
- Berczuk, S. P., & Appleton, B. (2003). Software Configuration Management Patterns. Addison-Wesley.

Course Description

Focuses on controlling and documenting software changes to maintain integrity and traceability throughout the software lifecycle.

Course Objectives

- Manage software versioning and builds.
- Apply release and change control strategies.
- Use modern tools (Git, CI/CD) for automation.
- Conduct audits and reviews effectively.

Week Wise Topics to be Covered

W1	Introduction to SCM: Definitions and Goals	**W9**	Source Code Control Systems (Git, SVN)
W2	SCM Planning and Policies	**W10**	Branching Strategies and Merging Techniques
W3	Configuration Identification	**W11**	Continuous Integration and Delivery (CI/CD)
W4	Change Control and Version Management	**W12**	SCM Tools and Case Studies
W5	Configuration Status Accounting	**W13**	Role of SCM in DevOps Practices
W6	Configuration Audits and Reviews	**W14**	Security and Compliance in SCM
W7	Build Management and Automation	**W15**	**Case Studies / Projects / Demo**
W8	Release Management and Delivery	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
SE632	**Complex Networks**	3(3+0)	Nil

Recommended Texts

- Newman, M. (2010). Networks: An Introduction. Oxford University Press.
- Barabási, A.-L. (2016). Network Science. Cambridge University Press.

Course Description

Explores structures, properties, and dynamics of networks found in nature, society, and technology.

Course Objectives

- Apply graph theory to real-world networks.
- Analyze network centrality, robustness, and modularity.
- Use visualization and modeling tools.
- Explore domains like social and biological networks.

Week Wise Topics to be Covered

W1	Introduction to Complex Networks	**W9**	Epidemic Models on Networks
W2	Graph Theory Basics	**W10**	Social Network Analysis
W3	Random Networks and Small-World Phenomena	**W11**	Biological and Ecological Networks
W4	Scale-Free Networks and Preferential Attachment	**W12**	Technological and Communication Networks
W5	Community Detection and Modularity	**W13**	Network Dynamics and Synchronization
W6	Network Robustness and Resilience	**W14**	Complex Network Simulation Tools
W7	Network Centrality Measures	**W15**	**Case Studies / Projects / Demo**
W8	Network Visualization Techniques	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
SE642	**Reliability Engineering**	3(3+0)	Nil

Recommended Texts

- O'Connor, P. D. T., & Kleyner, A. (2012). Practical Reliability Engineering (5th ed.). Wiley.
- Leveson, N. G. (1995). Safeware: System Safety and Computers. Addison-Wesley.

Course Description

Deals with evaluating and improving the reliability of software and hardware systems using statistical and analytical tools.

Course Objectives

- Analyze system failures and reliability metrics.
- Design for fault tolerance.
- Apply testing and modeling techniques.
- Evaluate risk and compliance standards.

Week Wise Topics to be Covered

W1	Introduction to Reliability Engineering	**W9**	System Safety and Risk Assessment
W2	Reliability Measures and Life Distributions	**W10**	Redundancy and Fault Tolerance Techniques
W3	Failure Modes, Effects, and Criticality Analysis (FMECA)	**W11**	Accelerated Life Testing
W4	Reliability Testing and Data Analysis	**W12**	Reliability Growth Models
W5	Maintainability and Availability Concepts	**W13**	Case Studies in Reliability Engineering
W6	Reliability Block Diagrams and Fault Tree Analysis	**W14**	Reliability in Agile and DevOps Contexts
W7	Software Reliability Metrics and Models	**W15**	**Case Studies / Projects / Demo**
W8	Design for Reliability (DFR)	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
SE731	**Model Driven Software Engineering**	3(3+0)	Nil

Recommended Texts

- Domínguez Mayo, F. J., Pires, L. F., & Seidewitz, E. (Eds.). (2024). Model-Driven Engineering and Software Development: 11th International Conference, MODELSWARD 2023 (Lecture Notes in Computer Science, Vol. 2106). Springer. https://doi.org/10.1007/978-3-031-66383-2
- Brambilla, M., Cabot, J., & Wimmer, M. (2022). Model-Driven Software Engineering in Practice (2nd ed.). Springer. https://link.springer.com/book/10.1007/978-3-031-02549-5

Course Description

This course explores software development approaches centered on creating and exploiting domain models. It focuses on Model-Driven Architecture (MDA), modeling languages (like UML), transformations, and code generation.

Course Objectives

- Understand the foundations and rationale for Model-Driven Engineering.
- Learn various modeling languages and their applications.
- Apply model transformations and code generation.
- Critically evaluate the benefits and limitations of MDE approaches.

Week Wise Topics to be Covered

W1	Introduction to MDE and MDA	**W9**	Modeling for Software Product Lines
W2	Modeling Fundamentals and UML Overview	**W10**	MDE in Agile and DevOps Environments
W3	Domain-Specific Modeling	**W11**	Industrial Case Studies
W4	Meta-modeling	**W12**	Model Refactoring
W5	Model Transformation Languages (e.g., ATL)	**W13**	Research Trends in MDE
W6	Code Generation Concepts	**W14**	Ethical and Practical Challenges
W7	Tools and Platforms, EMF, Papyrus	**W15**	**Case Studies / Projects / Demo**
W8	Model Validation and Verification	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS732	**Secure Software Engineering**	3(3+0)	Nil

Recommended Texts

- McGraw, G. (2006). Software Security: Building Security In (1st ed.). Addison-Wesley.
- Akonor, E., & Noury, N. (Eds.). (2023). Secure Software Engineering: Advanced Principles, Design, and Applications. CRC Press. https://www.crcpress.com/Secure-Software-Engineering-Advanced-Principles-Design-and-Applications/Akonor-Noury/p/book/9781032345671

Course Description

This course examines security principles and practices in software development. It emphasizes secure coding, threat modeling, vulnerability assessment, and compliance with secure development lifecycles.

Course Objectives

- Integrate security practices into all stages of software development.
- Analyze threats and design mitigations.
- Use tools and techniques for secure coding and testing.
- Understand industry standards and regulations.

Week Wise Topics to be Covered

W1	Introduction to Software Security	**W9**	Static and Dynamic Security Testing Tools
W2	Threat Modeling Techniques	**W10**	Vulnerability Databases and Penetration Testing
W3	Secure Software Development Lifecycle (SSDLC)	**W11**	Security in DevOps (DevSecOps)
W4	Secure Coding Principles	**W12**	Case Studies (e.g., breaches)
W5	Input Validation and Output Encoding	**W13**	Privacy by Design and GDPR Compliance
W6	Access Control and Authentication Mechanisms	**W14**	Security Certification and Standards (e.g., ISO/IEC 27034)
W7	Cryptographic Practices	**W15**	**Case Studies / Projects / Demo**
W8	Secure Design Patterns	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS733	**Software Process Improvement and Maturity Models**	3(3+0)	Nil

Recommended Texts

- Chrissis, M. B., Konrad, M., & Shrum, S. (2023). CMMI® for Development: Guidelines for Process Integration and Product Improvement (4th ed.). Addison-Wesley. https://www.pearson.com/store/p/cmmi-for-development/P200000000130
- Pohl, K., Böckle, G., & van der Linden, F. (2021). Software Product Line Engineering: Foundations, Principles, and Techniques (2nd ed.). Springer. https://www.springer.com/gp/book/9783030453790

Course Description

This course covers strategies and frameworks for improving software development processes. It includes maturity models like CMMI, Agile assessments, and process metrics.

Course Objectives

- Understand software process and maturity models.
- Evaluate and improve current processes.
- Apply SPI methods and metrics.

Week Wise Topics to be Covered

W1	Introduction to Software Processes	**W9**	SPI Tools and Automation
W2	Motivation for Process Improvement	**W10**	Cost-Benefit Analysis
W3	CMMI Framework Overview	**W11**	Change Management
W4	ISO/IEC 15504 (SPICE)	**W12**	Case Studies on SPI Success/Failure
W5	Agile Maturity Models	**W13**	Government and Regulatory Requirements
W6	Assessment Techniques	**W14**	Ethics and Organizational Culture
W7	Process Metrics and KPIs	**W15**	**Case Studies / Projects / Demo**
W8	Tailoring SPI to Organizational Context	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS734	**Human-Centered Software Engineering**	3(3+0)	Nil

Recommended Texts

- Rogers, Y., Sharp, H., & Preece, J. (2024). Interaction Design: Beyond Human–Computer Interaction (5th ed.). Wiley. https://www.wiley.com/en-us/Interaction+Design%3A+Beyond+Human+Computer+Interaction%2C+5th+Edition-p-9781119460371
- Norman, D. A. (2019). The Design of Everyday Things: Revised and Expanded Edition. Basic Books. https://us.macmillan.com/books/9780465050659

Course Description

This course focuses on designing and evaluating software systems with human users at the center. It explores HCI principles, usability engineering, and participatory design.

Course Objectives

- Apply human-centered design principles.
- Evaluate usability and UX.
- Conduct user studies and contextual inquiry.
- Create accessible and inclusive software.

Week Wise Topics to be Covered

W1	Introduction to HCSE	**W9**	UX Measurement and Analytics
W2	HCI and Usability Principles	**W10**	Mobile and Web UX Challenges
W3	User Requirements and Personas	**W11**	Case Studies in HCSE
W4	Participatory and Contextual Design	**W12**	Design Ethics and Privacy
W5	Prototyping Techniques	**W13**	Tool Support for UX Design
W6	Usability Evaluation Methods	**W14**	Emerging Trends (e.g., AI/UX)
W7	Accessibility and Inclusive Design	**W15**	**Case Studies / Projects / Demo**
W8	Cognitive and Perceptual Models	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS735	**Reverse Engineering and Software Reengineering**	3(3+0)	Nil

Recommended Texts

- Chikofsky, E. J., & Cross, J. H. (1990). Reverse engineering and design recovery: A taxonomy of methods. IEEE Software, 7(1), 13–17. https://doi.org/10.1109/52.43044
- Demeyer, S., Ducasse, S., & Nierstrasz, O. (2008). Object-Oriented Reengineering Patterns. Morgan Kaufmann. https://www.elsevier.com/books/object-oriented-reengineering-patterns/demeyer/978-0-12-372526-1

Course Description

This course addresses the techniques for analyzing, understanding, and transforming legacy software systems through reverse and reengineering methods.

Course Objectives

- Understand the need and scope of reengineering.
- Apply reverse engineering techniques.
- Design migration strategies for legacy systems.
- Evaluate tools and best practices.

Week Wise Topics to be Covered

W1	Introduction to Reverse Engineering	**W9**	Reengineering for Web and Cloud
W2	Legacy Systems Overview	**W10**	Tool Support Overview
W3	Code Comprehension and Static Analysis	**W11**	Metrics for Reengineering
W4	Dynamic Analysis and Instrumentation	**W12**	Legal and Ethical Aspects
W5	Architecture Recovery	**W13**	Industrial Case Studies
W6	Refactoring Techniques	**W14**	Research Trends in Software Reengineering
W7	Data Reengineering	**W15**	**Case Studies / Projects / Demo**
W8	Migration to New Platforms	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	**Course Title**	**CHs**	**Pre-Req**
CS736	**Applied AI for Software Engineering**	3(3+0)	Nil

Recommended Texts

- Russell, S., & Norvig, P. (2020). Artificial Intelligence: A Modern Approach (4th ed.). Pearson. https://aima.cs.berkeley.edu
- Perez, J. C., & Beskales, G. (2024). AI Techniques in Software Engineering: Tools and Applications. Springer. https://link.springer.com/book/10.1007/978-3-031-20000-0

Course Description

This course explores how AI techniques can be integrated into software engineering tasks such as testing, bug prediction, and code generation. It covers traditional AI and machine learning applications.

Course Objectives

- Apply AI/ML techniques to SE problems.
- Use AI tools to support development and maintenance.
- Understand ethical implications of AI in SE.
- Evaluate effectiveness of AI-based SE tools.

Week Wise Topics to be Covered

W1	Introduction to AI and SE	**W9**	AI for Refactoring and Maintenance
W2	ML Basics for SE Students	**W10**	AI in Requirements and Design
W3	Code Analysis using ML	**W11**	Ethical Issues in AI-Driven SE
W4	Bug Prediction and Localization	**W12**	AI Tool Evaluation Techniques
W5	Test Case Generation using AI	**W13**	Case Studies (e.g., Copilot, TabNine)
W6	Natural Language Processing in SE	**W14**	Future Directions and Open Problems
W7	Code Synthesis and Auto-completion	**W15**	**Case Studies / Projects / Demo**
W8	Recommender Systems for Devs	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS737	**Software Architecture and Microservices**	3(3+0)	Nil

Recommended Texts

- Bass, L., Clements, P., & Kazman, R. (2021). Software Architecture in Practice (4th ed.). Addison-Wesley Professional. https://www.oreilly.com/library/view/software-architecture-in/9780136885979en.wikipedia.orginsights.sei.cmu.edu+3oreilly.com+3oreilly.com+3
- Newman, S. (2021). Building Microservices (2nd ed.). O'Reilly Media. https://www.oreilly.com/library/view/building-microservices-2nd/9781492034025 oreilly.com+5oreilly.com+5amazon.com+5
- Richardson, C. (2020). Microservices Patterns: With examples in Java. Manning Publications. https://www.manning.com/books/microservices-patterns

Course Description

This course introduces software architecture principles with a focus on designing distributed systems using microservices. It includes architectural styles, patterns, and quality attributes.

Course Objectives

- Understand architectural design and evaluation.
- Design scalable systems using microservices.
- Apply architectural patterns.
- Use tools and techniques for monitoring and evolution.

Week Wise Topics to be Covered

W1	Introduction to Software Architecture	**W9**	Monitoring and Observability
W2	Architectural Styles and Patterns	**W10**	Resilience and Fault Tolerance
W3	Quality Attributes and Tradeoffs	**W11**	Security in Microservice Architecture
W4	Documenting Software Architecture	**W12**	Migration from Monolith to Microservices
W5	Introduction to Microservices	**W13**	DevOps and Continuous Delivery
W6	Designing Microservices	**W14**	Tools and Frameworks
W7	Communication and Data Management	**W15**	**Case Studies / Projects / Demo**
W8	Deployment and Containerization	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

⇨ **Program - MS (DS)**	DS Courses begin from here

Program MS (DS)	**Domain – 1: DS Core**

Code	**Course Title**	**CHs**	**Pre-Req**
DS601	**Tools and Techniques in Data science**	3(3+0)	Nil

Recommended Texts

- Doing Data Science by Cathy O'Neil & Rachel Schutt
- Python Data Science Handbook by Jake VanderPlas – https://jakevdp.github.io/PythonDataScienceHandbook/
- Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow by A. Géron

Course Description

This course is designed to teach the practical tools, programming libraries, and techniques used by data scientists in real-world projects.

Course Objectives

- Use Jupyter, Git, and Python for data science workflows
- Conduct data cleaning, transformation, and visualization
- Apply statistical and ML methods using Scikit-learn and Pandas
- Understand reproducibility, collaboration, and model deployment

Week Wise Topics to be Covered

W1	Data Science Process Overview and Tools Setup	**W9**	Hyperparameter Tuning and Cross Validation
W2	Version Control with Git and GitHub	**W10**	Unsupervised Learning Techniques
W3	Working with Pandas and Numpy	**W11**	Time Series and Forecasting Tools
W4	Data Cleaning and Missing Value Treatment	**W12**	Reproducible Research with Jupyter and Papermill
W5	Data Visualization with Matplotlib and Seaborn	**W13**	Model Deployment using Flask and Docker
W6	Data Pipelines and Automation with Python	**W14**	MLflow for Experiment Tracking
W7	Intro to Scikit-Learn, Model Training and Evaluation	**W15**	**Case Studies / Projects / Demo**
W8	Feature Engineering and Selection	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS602	**Statistical and Mathematical Methods for Data Analysis**	3(3+0)	Nil

Recommended Texts

- Introduction to Statistical Learning by Gareth James et al. – https://www.statlearning.com/
- Mathematics for Machine Learning by Deisenroth et al. – https://mml-book.github.io/
- Think Stats by Allen B. Downey – http://greenteapress.com/wp/think-stats/

Course Description

This course provides the statistical foundations and mathematical techniques necessary for understanding data science and machine learning algorithms.

Course Objectives

- Apply descriptive and inferential statistics in data analysis
- Understand key mathematical tools: linear algebra, calculus, probability
- Conduct hypothesis testing and regression modeling
- Learn dimensionality reduction and statistical learning methods

Week Wise Topics to be Covered

W1	Introduction to Statistics and Probability	**W9**	Logistic Regression and Classification Metrics
W2	Data Distributions and Descriptive Statistics	**W10**	ANOVA and Experimental Design
W3	Random Variables, Expectation, and Variance	**W11**	Bayesian Inference Basics
W4	Probability Distributions, Normal, Binomial, Poisson	**W12**	Matrix Algebra for Data Analysis
W5	Sampling, Central Limit Theorem, Confidence Intervals	**W13**	Eigenvalues, PCA, and Dimensionality Reduction
W6	Hypothesis Testing and p-values	**W14**	Clustering and Unsupervised Statistical Learning
W7	Correlation and Simple Linear Regression	**W15**	**Case Studies / Projects / Demo**
W8	Multiple Linear Regression and Model Diagnostics	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	Remarks
CS661	**Advance Machine Learning**	Offered in CS on Page #

Add Flag

Program MS (DS)	**Domain – 2: DS Domain Elective**

Code	Course Title	CHs	Pre-Req
DS605	**Deep Learning**	3(3+0)	Nil
Recommended Texts			
• Deep Learning by Ian Goodfellow, Yoshua Bengio, and Aaron Courville – https://www.deeplearningbook.org/ • Dive into Deep Learning (Zhang et al.) – https://d2l.ai/ • PyTorch Docs – https://pytorch.org/tutorials/ • M. Irfan Uddin, and Wali Khan Mashwani. Deep Learning, Reinforcement Learning, and the Rise of Intelligent Systems. IGI Global, USA, February 2024. ISSN 9798369317389. doi:10.4018/979-8-3693-1738-9.			
Course Description			
This course covers neural network theory, architectures, and practical implementation using frameworks like TensorFlow and PyTorch.			
Course Objectives			
• Understand foundational architectures: MLPs, CNNs, RNNs • Train and evaluate deep learning models • Explore advanced architectures (Transformers, GANs)			

Week Wise Topics to be Covered			
W1	Introduction to Neural Networks and Deep Learning	**W9**	Transformers and Self-Attention
W2	Mathematical Foundations, Backpropagation, Optimization	**W10**	GANs and Generative Models
W3	Multilayer Perceptrons and Activation Functions	**W11**	Transfer Learning and Fine-Tuning
W4	CNNs for Image Data	**W12**	Deep Learning for NLP
W5	Training Techniques and Regularization (Dropout, BatchNorm)	**W13**	Deployment and Inference Optimization
W6	RNNs and Sequence Modeling	**W14**	Fairness, Interpretability, and Ethics in Deep Learning
W7	LSTMs, GRUs and Attention Mechanisms	**W15**	**Case Studies / Projects / Demo**
W8	Autoencoders and Representation Learning	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS614	**Distributed Data Processing**	3(3+0)	Nil

Recommended Texts

- Learning Spark: Lightning-Fast Data Analytics (O'Reilly, 2nd Edition) – https://databricks.com/learn/spark
- High Performance Spark by Holden Karau – https://shop.oreilly.com/product/0636920034957.do
- Distributed Machine Learning with Apache Spark (O'Reilly Report)
- Official Apache Spark docs – https://spark.apache.org/docs/latest/

Course Description

This course explores scalable data processing techniques and distributed machine learning using Apache Spark and its ecosystem. It combines foundational distributed computing with practical workflows for ETL, EDA, and machine learning at scale.

Course Objectives

- Understand distributed data processing models and Spark's architecture
- Perform large-scale data analytics using RDDs, DataFrames, and Spark SQL
- Build distributed ML pipelines and evaluate them using Spark MLlib
- Apply real**-time processing and distributed deep learning techniques**

Week Wise Topics to be Covered

W1	Introduction to Distributed Data Processing, Hadoop vs Spark, Cluster Overview	**W9**	Introduction to Spark Streaming and Structured Streaming
W2	Spark Core Architecture and Programming Model	**W10**	Building Streaming Data Pipelines, Kafka + Spark Streaming
W3	RDDs and Functional Transformations (Map, Reduce, Filter, etc.)	**W11**	ETL Pipelines and Batch vs Streaming Tradeoffs
W4	Programming with RDDs and Key-Value Pairs	**W12**	Introduction to MLlib and the ML Pipeline API
W5	File Formats, JSON, Parquet, ORC, Avro — Reading/Writing Data	**W13**	Model Building with MLlib, Classification, Regression, Clustering
W6	Spark SQL and DataFrames, Joins, Aggregations, Window Functions	**W14**	Model Evaluation, Cross-Validation, and Hyperparameter Tuning
W7	Spark Job Execution & DAG Scheduler Internals	**W15**	**Case Studies / Projects / Demo**
W8	Exploratory Data Analysis (EDA) at Scale with Spark	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS621	**Big Data Analytics**	3(3+0)	Nil
Recommended Texts			
• Big Data: Principles and Best Practices by O'Reilly • Hadoop: The Definitive Guide by Tom White – https://www.oreilly.com/library/view/hadoop-the-definitive/9781491901683/			
Course Description			
This course introduces technologies and frameworks for big data storage, processing, and analytics using Hadoop, Spark, and NoSQL systems.			
Course Objectives			
• Understand distributed computing paradigms (MapReduce, Spark) • Work with large-scale data storage (HDFS, NoSQL) • Build real-time and batch processing pipelines • Analyze massive datasets using scalable tools			

Week Wise Topics to be Covered

W1	Introduction to Big Data Ecosystem	**W9**	ETL Pipelines and Data Lakes
W2	Hadoop and HDFS Architecture	**W10**	Data Warehousing, Hive and Presto
W3	MapReduce Programming Model	**W11**	Graph Processing with GraphX and Neo4j
W4	Apache Spark and RDDs	**W12**	Exploratory Data Analysis on Big Data
W5	Spark SQL and DataFrames	**W13**	Machine Learning at Scale with Spark MLlib
W6	NoSQL Databases, HBase, Cassandra, MongoDB	**W14**	Time-Series Analysis and Anomaly Detection
W7	Streaming Data Processing, Kafka and Spark Streaming	**W15**	**Case Studies / Projects / Demo**
W8	Data Ingestion Tools, Flume, Sqoop, NiFi	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS701	**Natural Language Processing**	3(3+0)	Nil

Recommended Texts

- Speech and Language Processing by Jurafsky & Martin – https://web.stanford.edu/~jurafsky/slp3/
- Natural Language Processing with Python (NLTK Book) – https://www.nltk.org/book/
- The Deep Learning for NLP Book by Sebastian Ruder (online) – http://ruder.io/

Course Description

This course introduces the fundamentals and modern techniques in NLP, from linguistic basics to deep learning approaches, with practical applications in language understanding, generation, and translation.

Course Objectives

- Understand linguistic structures and language modeling
- Implement NLP tasks such as tokenization, parsing, and sentiment analysis
- Apply deep learning models (RNNs, Transformers) to NLP problems
- Analyze NLP applications including chatbots and translation

Week Wise Topics to be Covered

W1	Introduction to NLP and Linguistic Foundations	**W9**	Attention Mechanisms and Transformer Architectures
W2	Text Preprocessing and Tokenization	**W10**	BERT, GPT and Pretrained Language Models
W3	Language Modeling with N-grams and Smoothing	**W11**	Machine Translation and Summarization
W4	POS Tagging, Named Entity Recognition, and Chunking	**W12**	Question Answering and Chatbots
W5	Syntax and Dependency Parsing	**W13**	Topic Modeling (LDA, NMF) and Text Generation
W6	Vector Semantics and Word Embeddings (Word2Vec, GloVe)	**W14**	Multilingual and Cross-lingual NLP
W7	Sentiment Analysis and Text Classification	**W15**	**Case Studies / Projects / Demo**
W8	Sequence Modeling with RNNs and LSTMs	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Program MS (DS)	**Domain – 3: General Elective**

Code	Course Title	Remarks
CS614	**Algorithms for Bio-Informatics**	Offered in CS on Page #

Code	Course Title	Remarks
CS642	**Computer Vision**	Offered in CS on Page #

Code	Course Title	Remarks
CS669	**Advance Natural Language Engineering**	Offered in CS on Page #

Code	Course Title	CHs	Pre-Req
DS603	**Bayesian Data Analysis**	3(3+0)	Nil

Recommended Texts

- Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). Bayesian Data Analysis (3rd ed.). CRC Press. https://doi.org/10.1201/b16018
- McElreath, R. (2020). Statistical Rethinking: A Bayesian Course with Examples in R and Stan (2nd ed.). CRC Press. https://doi.org/10.1201/9780429029608
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course provides a comprehensive introduction to Bayesian statistics and probabilistic modeling. It covers theoretical foundations, computational techniques using MCMC, and hierarchical modeling. Emphasis is placed on practical implementation in data analysis using tools like Stan and R.

Course Objectives

- To understand the principles of Bayesian inference and probabilistic modeling
- To implement Bayesian computational techniques such as MCMC and Gibbs sampling
- To analyze hierarchical models and perform posterior predictive checks
- To apply Bayesian methods in real-world data analysis scenarios

Week Wise Topics to be Covered

W1	Introduction to Bayesian Thinking and Probability Review	**W9**	Bayesian Model Comparison and Information Criteria
W2	Bayes' Theorem and Prior-Posterior Relationships	**W10**	Bayesian Decision Theory and Utility
W3	Conjugate Priors and Posterior Inference	**W11**	Bayesian Time Series and Dynamic Models
W4	Monte Carlo Methods and Sampling Distributions	**W12**	Bayesian Networks and Graphical Models
W5	Markov Chain Monte Carlo (MCMC) and Gibbs Sampling	**W13**	Bayesian Nonparametrics
W6	Bayesian Linear Regression	**W14**	Bayesian Analysis in Industry Applications
W7	Hierarchical Models and Multilevel Analysis	**W15**	**Case Studies / Projects / Demo**
W8	Model Checking and Posterior Predictive Checks	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS604	**Data Visualization**	3(3+0)	Nil

Recommended Texts

- Cairo, A. (2019). How Charts Lie: Getting Smarter About Visual Information. W. W. Norton & Company. https://wwnorton.com/books/9781324001560
- Munzner, T. (2014). Visualization Analysis and Design. CRC Press. https://doi.org/10.1201/b17511
- At least 5 research papers from Scopus indexed international conferences
- At least 5 research papers from journals papers indexed in Q1 or Q2 WoS

Course Description

This course focuses on techniques, tools, and theories of effective data visualization. Students learn to design interactive and static graphics using libraries like D3.js and Python's Matplotlib, and to critically evaluate visualization methods based on human cognition and perception.

Course Objectives

- To understand principles of visual perception and their impact on data display
- To implement visualizations using modern tools and libraries
- To evaluate the effectiveness of visual representation across data types
- To design interactive dashboards and story-driven data presentations

Week Wise Topics to be Covered

W1	Introduction to Data Visualization and Visual Perception	**W9**	Mapping and Geospatial Visualizations
W2	Data Types and Visual Encoding Principles	**W10**	Storytelling with Data and Narrative Visualization
W3	Chart Types and When to Use Them	**W11**	Visualization for High-Dimensional Data
W4	Exploratory vs. Explanatory Visualizations	**W12**	Scientific and Statistical Data Visualization
W5	Color Theory and Visual Design for Data	**W13**	Accessibility and Ethics in Visualization
W6	Interactive Visualization using D3.js or Plotly	**W14**	Advanced tool for visualization
W7	Using Python (Matplotlib, Seaborn) and R (ggplot2) for Visualization	**W15**	**Case Studies / Projects / Demo**
W8	Dashboards and Real-time Visual Analytics	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS606	**Optimization Methods for Data Science and ML**	3(3+0)	Nil

Recommended Texts

- Wright, S. J., & Recht, B. (2022). Optimization for Data Analysis. Cambridge University Press. https://doi.org/10.1017/9781009004282
- Bottou, L., Curtis, F. E., & Nocedal, J. (2016). Optimization methods for large-scale machine learning. SIAM Review, 60(2), 223–311. https://doi.org/10.1137/15M1014910

Course Description

Covers mathematical optimization fundamentals and their critical role in data science and machine learning. Topics include convex/non-convex optimization, stochastic methods, and application to large-scale ML models.

Course Objectives

- Formulate ML problems as optimization tasks.
- Apply gradient-based and second-order optimization methods.
- Analyze convergence and complexity.
- Implement scalable solvers for large datasets.

Week Wise Topics to be Covered

W1	Convex sets & functions	**W9**	Nonsmooth methods / proximal
W2	Unconstrained optimization (GD, Newton)	**W10**	Backpropagation in neural nets
W3	Convergence analysis	**W11**	Second-order & quasi-Newton
W4	Stochastic gradient methods	**W12**	Large-scale issues & regularization
W5	Variants: momentum, Adam	**W13**	Practical implementation and toolkits
W6	Coordinate & block descent	**W14**	Model selection via optimization
W7	Constrained optimization	**W15**	**Case Studies / Projects / Demo**
W8	Lagrangian & duality principles	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS607	**Deep Reinforcement Learning**	3(3+0)	Nil

Recommended Texts

- Dong, H., Ding, Z., & Zhang, S. (2020). Deep Reinforcement Learning: Fundamentals, Research, and Applications. Springer Nature. https://doi.org/10.1007/978-981-13-8285-7
- Plaat, A. (2022). Deep Reinforcement Learning: A Textbook (v5). arXiv. https://doi.org/10.48550/arXiv.2201.02135
- M. Irfan Uddin, and Wali Khan Mashwani. Deep Learning, Reinforcement Learning, and the Rise of Intelligent Systems. IGI Global, USA, February 2024. ISSN 9798369317389. doi:10.4018/979-8-3693-1738-9.

Course Description

Explores RL foundations and deep learning integration to solve sequential decision-making tasks. Includes policy and value-based methods, exploration, and modern applications.

Course Objectives

- Understand MDPs and RL formulation.
- Implement DQN, policy-gradient, actor-critic algorithms.
- Evaluate DRL methods and resources.
- Explore recent advances like multi-agent RL.

Week Wise Topics to be Covered

W1	RL foundations (MDPs)	**W9**	Exploration strategies
W2	Value vs policy methods	**W10**	Multi-agent RL
W3	Tabular methods	**W11**	Meta-RL and hierarchical RL
W4	Deep Q-networks	**W12**	DRL in robotics/games/finance
W5	Policy gradient and REINFORCE	**W13**	Evaluation metrics & reproducibility
W6	Actor-critic family	**W14**	Ethical & compute considerations
W7	Off-policy methods: DDPG, TD3	**W15**	**Case Studies / Projects / Demo**
W8	PPO, TRPO, SAC	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS608	**Probabilistic Graphical Models**	3(3+0)	Nil

Recommended Texts

- Sucar, L. E. (2021). Probabilistic Graphical Models: Principles and Applications (2nd ed.). Springer. https://doi.org/10.1007/978-3-030-61943-5
- Koller, D., & Friedman, N. (2009). Probabilistic Graphical Models: Principles and Techniques. MIT Press.

Course Description

Focuses on graphical representations (Bayesian networks, Markov models) for structured probabilistic reasoning, inference, and learning in real-world data.

Course Objectives

- Model dependencies with PGMs.
- Perform exact and approximate inference.
- Learn model structure and parameters.
- Apply PGMs to time-series, vision, NLP, causal inference.

Week Wise Topics to be Covered

W1	Intro to PGMs	**W9**	Structure learning
W2	Bayesian networks	**W10**	HMMs & Dynamic BNs
W3	MRFs and factor graphs	**W11**	Conditional random fields
W4	Exact inference	**W12**	Causality and influence diagrams
W5	Junction trees	**W13**	Applications: vision, NLP
W6	Approximate inference: sampling	**W14**	Advanced topics/trends
W7	Variational methods	**W15**	**Case Studies / Projects / Demo**
W8	Parameter estimation	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS609	**Social network analysis**	3(3+0)	Nil

Recommended Texts

- Wasserman, S., & Faust, K. (1994). Social Network Analysis: Methods and Applications. Cambridge University Press. https://doi.org/10.1017/CBO9780511815478
- Newman, M. E. J. (2018). Networks (2nd ed.). Oxford University Press. https://doi.org/10.1093/oso/9780198805090.001.0001

Course Description

Studies network data through graph theory and statistics to uncover structural patterns, diffusion processes, centrality, and community dynamics.

Course Objectives

- Represent and visualize networks.
- Compute metrics like centrality and clustering.
- Model information diffusion and social contagion.
- Detect communities and large-scale structure.

Week Wise Topics to be Covered

W1	Graph basics and notation	**W9**	Visualization techniques
W2	Descriptive network metrics	**W10**	Diffusion and cascade models
W3	Node centrality measures	**W11**	Network inference
W4	Path, connectivity, shortest-paths	**W12**	Temporal and multilayer networks
W5	Clustering and transitivity	**W13**	Online/social media case studies
W6	Community detection	**W14**	Ethics and privacy in SNA
W7	Network models (Erdős–Rényi, small-world)	**W15**	**Case Studies / Projects / Demo**
W8	Scale-free networks & hubs	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS611	**Time series Analysis and Prediction**	3(3+0)	Nil

Recommended Texts

- Hyndman, R. J., & Athanasopoulos, G. (2021). Forecasting: Principles and Practice (3rd ed.). OTexts. https://otexts.com/fpp3
- Shumway, R. H., & Stoffer, D. S. (2017). Time Series Analysis and Its Applications: With R Examples (4th ed.). Springer. https://doi.org/10.1007/978-3-319-52452-8

Course Description

Addresses modeling and forecasting of time-series data with classical, state-space, and modern machine learning methods.

Course Objectives

- Analyze stationary and non-stationary series.
- Fit ARIMA, SARIMA, exponential smoothing models.
- Implement state-space & Kalman filters.
- Evaluate ML-based forecasting techniques.

Week Wise Topics to be Covered

W1	Time-series basics & visualization	**W9**	Transfer function modeling
W2	Stationarity & differencing	**W10**	GARCH and volatility modeling
W3	AR, MA, ARMA modeling	**W11**	Change point detection
W4	ARIMA and Box-Jenkins	**W12**	Machine learning forecasting (RNNs, LSTM)
W5	Seasonal modeling (SARIMA)	**W13**	Ensemble forecasting & hybrid models
W6	Exponential smoothing & ETS	**W14**	Forecast evaluation & economic applications
W7	Model diagnostics	**W15**	**Case Studies / Projects / Demo**
W8	State-space & Kalman filters	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS612	**Algorithmic trading**	3(3+0)	Nil
Recommended Texts			
• Chan, E. (2020). Algorithmic Trading: Winning Strategies and Their Rationale (2nd ed.). Wiley. https://doi.org/10.1002/9781119684628 • Avellaneda, M., & Zhang, S. (2021). Algorithmic and High-Frequency Trading. Cambridge University Press. https://doi.org/10.1017/9781108877806			
Course Description			
Introduces quantitative trading strategies using statistical and ML methods, including risk and portfolio considerations, high-frequency data, and backtesting.			
Course Objectives			
• Analyze financial time-series and signals. • Build statistical and ML-based automated strategies. • Understand risk/money management. • Backtest strategies in realistic markets.			

Week Wise Topics to be Covered

W1	Market microstructure fundamentals	**W9**	Backtesting frameworks
W2	Data types and pre-processing	**W10**	High-frequency strategies
W3	Momentum and mean-reversion strategies	**W11**	Transaction costs and alpha decay
W4	Pairs trading and statistical arbitrage	**W12**	Compliance and ethics in trading
W5	Machine learning in trading	**W13**	Reinforcement learning in trading
W6	Technical indicators & feature engineering	**W14**	Recent advances & alternative asset strategies
W7	Risk and portfolio management	**W15**	**Case Studies / Projects / Demo**
W8	Execution algorithms and slippage	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		Market microstructure fundamentals	

Code	Course Title	CHs	Pre-Req
DS613	**Cloud computing**	3(3+0)	Nil

Recommended Texts

- Barr, B., & Benson, R. (2022). Cloud Native Patterns: Designing change-tolerant software. O'Reilly Media. https://learning.oreilly.com/library/view/cloud-native-patterns/9781492072207
- Hendrickson, J. (2021). Site Reliability Engineering: How Google Runs Production Systems. O'Reilly Media. https://doi.org/10.5555/3468721

Course Description

Explores fundamentals of cloud architecture, distributed systems, and engineering scalable ML pipelines on cloud platforms, including Kubernetes and serverless technologies.

Course Objectives

- Understand cloud service models (IaaS, PaaS, SaaS).
- Deploy scalable distributed applications.
- Apply containerization and orchestration techniques.
- Integrate cloud-based ML workflows.

Week Wise Topics to be Covered

W1	Intro to cloud architectures	**W9**	Monitoring & logging
W2	Virtualization & virtualization tech	**W10**	CI/CD pipelines
W3	Containers (Docker)	**W11**	Cloud ML platforms
W4	Orchestration (Kubernetes)	**W12**	Resilience & fault tolerance
W5	Microservices & REST APIs	**W13**	Security & compliance
W6	Networking in the cloud	**W14**	Cost optimization & performance
W7	Storage & databases	**W15**	**Case Studies / Projects / Demo**
W8	Serverless & FaaS	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS615	**Distributed Machine Learning in Apache Spark**	3(3+0)	Nil

Recommended Texts

- Learning Spark: Lightning-Fast Data Analytics (O'Reilly, 2nd Edition) – https://databricks.com/learn/spark
- High Performance Spark by Holden Karau – https://shop.oreilly.com/product/0636920034957.do
- Distributed Machine Learning with Apache Spark (O'Reilly Report)
- Official Apache Spark docs – https://spark.apache.org/docs/latest/

Course Description

This course explores scalable data processing techniques and distributed machine learning using Apache Spark and its ecosystem. It combines foundational distributed computing with practical workflows for ETL, EDA, and machine learning at scale.

Course Objectives

- Understand distributed data processing models and Spark's architecture
- Perform large-scale data analytics using RDDs, DataFrames, and Spark SQL
- Build distributed ML pipelines and evaluate them using Spark MLlib
- Apply real**-time processing and distributed deep learning techniques**

Week Wise Topics to be Covered

W1	Introduction to Distributed Data Processing, Hadoop vs Spark, Cluster Overview	**W9**	Introduction to Spark Streaming and Structured Streaming
W2	Spark Core Architecture and Programming Model	**W10**	Building Streaming Data Pipelines, Kafka + Spark Streaming
W3	RDDs and Functional Transformations (Map, Reduce, Filter, etc.)	**W11**	ETL Pipelines and Batch vs Streaming Tradeoffs
W4	Programming with RDDs and Key-Value Pairs	**W12**	Introduction to MLlib and the ML Pipeline API
W5	File Formats, JSON, Parquet, ORC, Avro — Reading/Writing Data	**W13**	Model Building with MLlib, Classification, Regression, Clustering
W6	Spark SQL and DataFrames, Joins, Aggregations, Window Functions	**W14**	Model Evaluation, Cross-Validation, and Hyperparameter Tuning
W7	Spark Job Execution & DAG Scheduler Internals	**W15**	**Case Studies / Projects / Demo**
W8	Exploratory Data Analysis (EDA) at Scale with Spark	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS616	**High performance computing**	3(3+0)	Nil

Recommended Texts

- Master HPC hardware (CPU/GPU clusters) and memory architectures.
- Develop parallel code using MPI and OpenMP.
- Utilize GPU acceleration frameworks (CUDA, OpenCL).
- Analyze and optimize performance using profiling techniques.

Course Description

This course introduces the principles, architecture, and programming of high-performance computing systems. It includes shared/distributed memory, parallel libraries (MPI, OpenMP), GPU programming, and performance analysis tools.

Course Objectives

- Master HPC hardware (CPU/GPU clusters) and memory architectures.
- Develop parallel code using MPI and OpenMP.
- Utilize GPU acceleration frameworks (CUDA, OpenCL).
- Analyze and optimize performance using profiling techniques.

Week Wise Topics to be Covered

W1	HPC system architecture & parallel computing models	**W9**	
W2	Shared memory & threads with OpenMP	**W10**	
W3	Cache, memory hierarchy optimization	**W11**	
W4	Distributed memory: MPI fundamentals	**W12**	
W5	MPI point-to-point & collectives	**W13**	
W6	Hybrid MPI+OpenMP programming	**W14**	
W7	GPU computing: CUDA fundamentals	**W15**	**Case Studies / Projects / Demo**
W8	GPU optimization techniques	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS622	**Computational Genomics**	3(3+0)	Nil

Recommended Texts

- Cristianini, N., & Hahn, M. W. (2006). Introduction to Computational Genomics: A Case Studies Approach. Cambridge University Press. https://doi.org/10.1017/CBO9780511808982
- Akalin, A. (2024). Computational Genomics with R. CRC Press. https://doi.org/10.1201/9780429084317

Course Description

Explores algorithmic and statistical methods for analyzing high-throughput genomic data. Topics include sequence alignment, variant calling, genome-wide association studies, functional genomics pipelines, and reproducible workflows.

Course Objectives

- Apply algorithmic techniques (alignment, assembly, etc.) to genomic data.
- Implement statistical models for variant detection and GWAS.
- Design efficient end-to-end analysis pipelines.
- Leverage programming tools (R, Python) for reproducible genomics.

Week Wise Topics to be Covered

W1	Genomics data types & file formats	**W9**	GWAS design & analysis
W2	Sequence alignment algorithms	**W10**	eQTL and integrative omics
W3	Read mapping and variant calling	**W11**	Gene network inference
W4	Statistical models for variants	**W12**	Reproducible pipelines (Snakemake, Nextflow)
W5	De novo assembly & annotations	**W13**	Advanced topics: single-cell genomics
W6	RNA-seq pipelines	**W14**	Deep learning in genomics
W7	Differential expression analysis	**W15**	**Case Studies / Projects / Demo**
W8	Epigenomics & ChIP-seq workflows	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS623	**Inference & Representation**	3(3+0)	Nil

Recommended Texts

- Bishop, C. M. (2006). Pattern Recognition and Machine Learning. Springer. https://doi.org/10.1007/978-0-387-45528-0
- Gelman, A., et al. (2013). Bayesian Data Analysis (3rd ed.). CRC Press. https://doi.org/10.1201/9780429258411

Course Description

Covers statistical and probabilistic modeling frameworks for representing complex data and underlying structures. Emphasis on inference techniques including Bayesian methods, latent variable models, and representation learning.

Course Objectives

- Build and interpret statistical models (Bayesian, latent).
- Perform inference using MCMC and variational methods.
- Understand latent representations and embeddings.
- Apply models to real-world data scenarios.

Week Wise Topics to be Covered

W1	Probability review & statistical inference	**W9**	Probabilistic PCA and latent Dirichlet allocation
W2	Bayesian inference foundations	**W10**	Graphical models and inference
W3	Conjugacy and analytical methods	**W11**	Autoencoders & representation learning
W4	MCMC: Gibbs, Metropolis-Hastings	**W12**	Word embeddings, feature learning
W5	Variational inference basics	**W13**	Inference diagnostics & model checking
W6	Latent variable models: PCA, factor analysis	**W14**	Scalability & approximate inference
W7	Gaussian mixture & EM algorithm	**W15**	**Case Studies / Projects / Demo**
W8	Bayesian hierarchical models	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS624	**Scientific Computing in Finance**	3(3+0)	Nil

Recommended Texts

- Press, W. H., et al. (2007). Numerical Recipes: The Art of Scientific Computing (3rd ed.). Cambridge University Press.
- Giles, M. B. (2007). Monte Carlo Methods in Financial Engineering. Springer. https://doi.org/10.1007/978-0-387-71220-9

Course Description

Covers numerical techniques and computational methods for financial modeling, including derivatives pricing, portfolio optimization, and risk management using advanced algorithms and software tools.

Course Objectives

- Implement numerical methods for option pricing (Monte Carlo, PDEs).
- Model and optimize portfolios using computational techniques.
- Use scientific computing tools (Python, MATLAB) for finance.
- Assess risk via simulation and backtesting.

Week Wise Topics to be Covered

W1	Floating-point & error analysis	**W9**	Portfolio theory & optimization
W2	Random number generation	**W10**	Stochastic differential equations
W3	Monte Carlo simulation basics	**W11**	Greeks and sensitivity analysis
W4	Variance reduction techniques	**W12**	Algorithmic trading basics
W5	Risk metrics: VaR, CVaR	**W13**	Computational performance tuning
W6	Option pricing via Monte Carlo	**W14**	Numerical libraries & GPU computing
W7	PDE methods & finite differences	**W15**	**Case Studies / Projects / Demo**
W8	Calibration & parameter estimation	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS731	**Biomedical Data Science and Algorithms**	3(3+0)	Nil

Recommended Texts

- Dinov, I. D. (2023). Data Science and Predictive Analytics: Biomedical Applications using R (2nd ed.). Springer. https://doi.org/10.1007/978-3-031-17483-4
- Komatsoulis, G. A. (2023). Biomedical Data Science and Clinical Informatics. CRC Press. https://doi.org/10.1201/9781003265436

Course Description

Focuses on machine learning, statistical methods, and algorithm design for biomedical data—genomic, clinical, imaging—emphasizing predictive modeling, interpretability, and real-world driven solutions.

Course Objectives

- Develop ML models suited for biomedical datasets.
- Preprocess and integrate multi-modal biomedical data.
- Use interpretable ML and validate clinical relevance.
- Implement algorithms for tasks like disease prediction and biomarker discovery.

Week Wise Topics to be Covered

W1	Biomedical data structures & formats	**W9**	Clinical time-series & electronic records
W2	Preprocessing: denoising, normalization	**W10**	Survival analysis & risk modeling
W3	Supervised learning for classification	**W11**	Integration of multi-modal data
W4	Evaluation metrics (ROC, AUC)	**W12**	Ethical and regulatory considerations
W5	Feature engineering & selection	**W13**	Reproducible biomedical pipelines
W6	Deep learning in biomedical data	**W14**	Clinical time-series & electronic records
W7	Interpretable models & explainability	**W15**	**Case Studies / Projects / Demo**
W8	Genomics-based predictive modeling	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS732	**Data Science for Financial Trading**	3(3+0)	Nil

Recommended Texts

- Chan, E. (2020). Algorithmic Trading: Winning Strategies and Their Rationale (2nd ed.). Wiley. https://doi.org/10.1002/9781119684628
- Narayanan, S., Swan, M. (2023). Machine Learning for Algorithmic Trading: Finance, Risk, and Analytics. Wiley. https://doi.org/10.1002/9781119851744

Course Description

Applies data science, machine learning, and algorithmic approaches to financial trading strategies. Covers market data analysis, feature engineering, model building, strategy backtesting, and risk-aware execution.

Course Objectives

- Acquire and preprocess trading data.
- Engineer predictive features (technical, sentiment).
- Develop and evaluate ML trading strategies.
- Implement live backtesting with risk management.

Week Wise Topics to be Covered

W1	Market structure & data handling	**W9**	Reinforcement learning for trading
W2	Feature engineering & technical indicators	**W10**	Sentiment analysis & alternative data
W3	Statistical vs ML-based models	**W11**	High-frequency considerations
W4	Time-series forecasting (ARIMA, LSTM)	**W12**	Model deployment & simulation
W5	Strategy evaluation & metrics (Sharpe, drawdown)	**W13**	Compliance & ethical issues
W6	Backtesting frameworks	**W14**	Performance monitoring live
W7	Risk and portfolio management	**W15**	**Case Studies / Projects / Demo**
W8	Execution strategies (TWAP, VWAP)	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS733	**Big Data in Genomics and Health Informatics**	3(3+0)	Nil

Recommended Texts

- Wang, B., Li, R., & Perrizo, W. (2015). Big Data Analytics in Bioinformatics and Healthcare. IGI Global. https://doi.org/10.4018/978-1-4666-6611-5
- Dinov, I. D. (2023). Data Science and Predictive Analytics: Biomedical Applications using R (2nd ed.). Springer. https://doi.org/10.1007/978-3-031-17483-4

Course Description

Explores large-scale genomic and healthcare data processing pipelines, focusing on storage, analysis, privacy, and interpretation. Students learn scalable frameworks for variant calling, clinical decision support, and public health surveillance.

Course Objectives

- Implement big data architectures for genomics pipelines.
- Integrate heterogeneous health data sources securely.
- Use distributed computation to accelerate biomedical analyses.
- Understand ethical and privacy concerns in health informatics.

Week Wise Topics to be Covered

W1	Introduction to genomic & clinical data	**W9**	Clinical decision support systems
W2	Data formats & metadata standards	**W10**	Health informatics & EHR integration
W3	Scalable storage (HDFS/NoSQL)	**W11**	Privacy and security in genomic data
W4	Distributed sequence alignment	**W12**	Visualization and reporting dashboards
W5	Variant calling pipelines	**W13**	Cloud and serverless pipelines
W6	Scalable RNA-seq analytics	**W14**	Real-world case studies
W7	Epigenomics at scale	**W15**	**Case Studies / Projects / Demo**
W8	Integrative multi-omics	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS734	**Statistical Inference and Data Representation**	3(3+0)	Nil

Recommended Texts

- Bishop, C. M. (2006). Pattern Recognition and Machine Learning. Springer. https://doi.org/10.1007/978-0-387-45528-0
- Gelman, A., et al. (2013). Bayesian Data Analysis (3rd ed.). Chapman & Hall/CRC. https://doi.org/10.1201/9780429258411

Course Description

Covers statistical inference for structured data and representation techniques such as latent variables and dimensionality reduction. Emphasizes theory, computational methods, and practical applications.

Course Objectives

- Learn classical and Bayesian inference methods.
- Understand and apply latent variable models.
- Explore dimensionality reduction and embeddings.
- Perform inference and model checking computationally.

Week Wise Topics to be Covered

W1	Probability & estimator theory	**W9**	Latent Dirichlet Allocation
W2	Frequentist inference & hypothesis tests	**W10**	Graphical model inference
W3	Bayesian methods & conjugacy	**W11**	Autoencoders & embeddings
W4	Likelihood & MLE	**W12**	Model selection & regularization
W5	MCMC: Gibbs, MH	**W13**	Inference diagnostics
W6	Variational inference	**W14**	Scalability in inference
W7	PCA & factor analysis	**W15**	**Case Studies / Projects / Demo**
W8	Mixture models & EM	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS735	**Financial Data Analytics and Modeling**	3(3+0)	Nil

Recommended Texts

- Benninga, S. (2014). Financial Modeling (4th ed.). MIT Press. ISBN 9780262027281
- Heaton, J. B., Polson, N. G., & Witte, J. H. (2016). Deep Learning in Finance. Manuscript (arXiv). https://arxiv.org/abs/1602.06561

Course Description

Teaches statistical and computational modeling techniques for financial data, including time-series, risk assessment, and portfolio strategy, using modern toolsets.

Course Objectives

- Model financial time-series using statistical and ML tools.
- Apply portfolio optimization and risk measures.
- Deploy predictive analytics in trading contexts.
- Understand regulatory and ethical aspects.

Week Wise Topics to be Covered

W1	Financial data sources & cleaning	**W9**	Deep learning for finance
W2	Exploratory analytics	**W10**	Backtesting frameworks
W3	Time-series basics	**W11**	Strategy evaluation & metrics
W4	ARIMA/GARCH modeling	**W12**	Execution & slippage modeling
W5	Feature engineering & indicators	**W13**	Compliance & ethics
W6	Portfolio theory & optimization	**W14**	Deep learning for finance
W7	Risk metrics (VaR, CVaR)	**W15**	**Case Studies / Projects / Demo**
W8	ML: regression & classification	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
DS736	**Foundation of Generative AI**	3(3+0)	Nil

Recommended Texts

- Babu, T., Nair, R. R., & Ebin, P. M. (2024). Foundations of Generative AI. In The Pioneering Applications of Generative AI. IGI Global. https://doi.org/10.4018/979-8-3693-3278-8.ch007
- Cao, Y., Yang, X., & Sun, R. (2025). Generative AI Models: Theoretical Foundations and Algorithmic Practices. Journal of Industrial Engineering and Applied Science, 3(1). https://doi.org/10.70393/6a69656173.323633
- Foster, I. (2023). Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play (2nd ed.). O'Reilly Media. ISBN: 9781098134181

Course Description

This course introduces the foundational concepts, models, and techniques in Generative Artificial Intelligence (Generative AI). It covers the theoretical underpinnings of generative models, including probabilistic models, generative adversarial networks (GANs), variational autoencoders (VAEs), and large language models (LLMs).

Course Objectives

- Understand generative modeling frameworks.
- Implement GANs, VAEs, and diffusion models.
- Evaluate generation quality and biases.
- Address ethics and societal implications.

Week Wise Topics to be Covered

W1	Introduction to Generative AI: Definition, History, Applications	**W9**	Transfer Learning, Fine-tuning, and Low-Resource Adaptation in Generative AI
W2	Mathematical Foundations: Probability, Statistics, and Information Theory	**W10**	Dataset Preparation for Generative AI: Sources, Curation, Cleaning
W3	Overview of Generative Models: Explicit, Implicit, Autoregressive, and Energy-based Models	**W11**	Evaluation Metrics for Generative Models: BLEU, FID, ROUGE, Perplexity, Human Evaluation
W4	Variational Autoencoders (VAEs): Concepts, Architecture, Applications	**W12**	Applications in Text Generation: Chatbots, Story Generation, Code Generation
W5	Generative Adversarial Networks (GANs): Fundamentals, Types of GANs, Applications	**W13**	Applications in Image & Video Generation: Deepfakes, Art Generation, Text-to-Image Models
W6	Autoregressive Models: RNNs, PixelRNN, WaveNet, and Transformer-based Generative Models	**W14**	Ethical Considerations, Bias, Deepfakes, and Responsible AI
W7	Introduction to Diffusion Models and Their Applications	**W15**	**Case Studies / Projects / Demo**
W8	Large Language Models (LLMs): Architecture, Pretraining, Fine-tuning, Prompting	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

TBD	To be deleted

The following courses were created, but due to limit in the course coding, we have removed them from the course code list. Based on further discussion in BoS or BoF they may be considered.

Code	Course Title	CHs	Pre-Req
CS	Advancements in Artificial General Intelligence	3(3+0)	Nil
Recommended Texts			
Course Description			
This course delves into the theoretical foundations, recent progress, and emerging approaches in Artificial General Intelligence (AGI). It explores architectures and methodologies aimed at achieving human-level general intelligence, evaluates benchmark environments, and critically analyzes the challenges and safety implications of AGI development. Students will investigate multidisciplinary perspectives spanning neuroscience, cognitive science, machine learning, and philosophy of mind.			
Course Objectives			

Week Wise Topics to be Covered			
W1	Introduction to AGI: Definitions, motivations, and historical background	**W9**	AGI Benchmarks and Evaluation: General environments, Turing tests, and intelligence metrics
W2	Contrasting Narrow AI and AGI: Capabilities, limitations, and key differences	**W10**	Ethics, Risks, and AGI Safety: Alignment, control problem, value loading
W3	Cognitive Architectures for AGI: SOAR, ACT-R, OpenCog, and Sigma	**W11**	Scalability and Resource Constraints in AGI Architectures
W4	Theoretical Models of General Intelligence: AIXI, Gödel Machines, and Universal AI	**W12**	Learning Across Multiple Domains and Modalities
W5	Embodied and Situated AI: Importance of grounding and sensorimotor integration	**W13**	Current Projects and Frameworks: DeepMind's Gato, IBM WatsonX, OpenAI's research
W6	Meta-Learning and Self-Improving Systems	**W14**	AGI and Consciousness: Machine sentience, subjective experience, and philosophical views
W7	Neuroscience-Inspired AGI: Computational models of the brain and neocortex theories	**W15**	**Case Studies / Projects / Demo**
W8	Neuro-symbolic and Hybrid Models	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS	Multimodal and Cross-Lingual Learning	3(3+0)	Nil

Recommended Texts
Course Description
This course explores how AI systems learn and reason across multiple data modalities (text, vision, audio) and languages. Students will study models that integrate diverse inputs, understand transfer across modalities and languages, and analyze foundational techniques enabling multilingual and multimodal reasoning in real-world applications.
Course Objectives
Week Wise Topics to be Covered

W1	Introduction to Multimodal and Cross-Lingual Learning: Definitions, use cases, challenges	**W9**	Multimodal Pretraining Objectives: Contrastive, generative, retrieval-based
W2	Foundations of Multimodal Representation Learning: Fusion, alignment, and joint representations	**W10**	Zero-shot and Few-shot Transfer Across Modalities/Languages
W3	Vision-Language Models: CLIP, Flamingo, BLIP	**W11**	Multilingual Multimodal Models: MURAL, m-VL-BERT, Polyglot-Koala
W4	Audio-Text Models: Whisper, SpeechBERT, and spoken language understanding	**W12**	Cultural and Linguistic Bias in Cross-Lingual Models
W5	Multimodal Transformers: ViLBERT, VisualBERT, MMBT, and MMF framework	**W13**	Applications: Multimodal search, cross-lingual dialogue, assistive tech
W6	Multilingual Language Modeling: mBERT, XLM-R, and zero-shot cross-lingual transfer	**W14**	Emerging Research and Open Challenges: Domain adaptation, continual learning
W7	Cross-Lingual Transfer and Alignment Techniques	**W15**	**Case Studies / Projects / Demo**
W8	Evaluation Benchmarks: XTREME, XGLUE, VQA, GQA, and HowTo100M	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS	Computational Creativity	3(3+0)	Nil

Recommended Texts

Course Description

This course investigates algorithms and models that simulate or augment human creativity in domains such as art, music, storytelling, design, and scientific discovery. It combines techniques from AI, cognitive science, and creative theory to explore how machines can generate novel and valuable content.

Course Objectives

Week Wise Topics to be Covered

W1	Foundations of Computational Creativity: Definitions, history, and domains	**W9**	Computational Design and Architecture
W2	Creativity Theories and Metrics: Novelty, surprise, value	**W10**	Scientific Discovery and Machine Creativity
W3	Generative Techniques: Rule-based, evolutionary, and search-based methods	**W11**	Creativity in Games and Simulations
W4	Neural Generative Models: GANs, VAEs, Transformers	**W12**	Evaluation of Creative Systems: Human studies and automated metrics
W5	Creativity in Language: Poetry, story generation, and humor	**W13**	Ethical and Philosophical Issues in Machine Creativity
W6	Visual and Artistic Creativity: Neural style transfer, image synthesis	**W14**	Cognitive Models of Creativity and Insight
W7	Music and Audio Generation: Symbolic and waveform-based methods	**W15**	**Case Studies / Projects / Demo**
W8	Interactive and Co-Creative Systems	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS	Neural and Computational Intelligence	3(3+0)	Nil

Recommended Texts

Course Description

This course explores biologically and cognitively inspired approaches to intelligent computing. It covers artificial neural networks, evolutionary algorithms, fuzzy systems, and hybrid intelligent systems, emphasizing their theoretical foundations, practical implementations, and applications in complex problem-solving.

Course Objectives

Week Wise Topics to be Covered

W1	Introduction to Computational Intelligence: Overview and paradigms	**W9**	Hybrid Intelligent Systems: Combining evolutionary, fuzzy, and neural approaches
W2	Artificial Neural Networks: Perceptrons, backpropagation, MLP	**W10**	Reinforcement Learning and Neuroevolution
W3	Deep Learning Architectures: CNNs, RNNs, autoencoders	**W11**	Self-Organizing Maps and Unsupervised Learning
W4	Spiking Neural Networks and Neuromorphic Computing	**W12**	Optimization with Computational Intelligence
W5	Fuzzy Logic Systems: Fuzzy sets, inference, and fuzzy control	**W13**	Benchmarking and Performance Evaluation
W6	Evolutionary Algorithms: Genetic algorithms, selection, crossover	**W14**	Applications in Engineering, Finance, and Robotics
W7	Swarm Intelligence: PSO, ACO, and collective behaviors	**W15**	**Case Studies / Projects / Demo**
W8	Neuro-Fuzzy Systems	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	**Course Title**	**CHs**	**Pre-Req**
CS	Neurosymbolic and Hybrid AI Systems	3(3+0)	Nil

Recommended Texts

Course Description

This course explores the integration of symbolic reasoning and neural learning, aiming to combine logic-based AI with data-driven approaches. It covers frameworks, architectures, and applications where both paradigms are necessary for robust generalization and explainability.

Course Objectives

Week Wise Topics to be Covered

W1	Introduction to Hybrid AI: Motivation and history	**W9**	Explainability in Neurosymbolic Systems
W2	Symbolic AI Fundamentals: Logic, rules, planning	**W10**	Programming Neural Networks with Logic (e.g., DeepProbLog, NeSy)
W3	Neural AI Fundamentals: Representations, learning, deep networks	**W11**	Learning to Reason: ILP, neural theorem proving
W4	Hybrid Architectures: Pipelines, modularity, and integration	**W12**	Cognitive Models and Human-Like AI
W5	Neurosymbolic Representations: Logic tensors, differentiable logic	**W13**	Applications of Neurosymbolic AI: Vision, NLP, robotics
W6	Constraint Satisfaction and Neural Solvers	**W14**	Challenges and Open Problems in Hybrid AI
W7	Knowledge Graphs and Reasoning	**W15**	**Case Studies / Projects / Demo**
W8	Combining Learning with Knowledge-Based Systems	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	

Code	Course Title	CHs	Pre-Req
CS	Visualization in medicine	3(3+0)	Nil

Recommended Texts:

1. Visualization in Medicine: Theory, Algorithms, and Applications – Bernhard Preim & Dirk Bartz.
2. Medical Image Computing and Computer-Assisted Intervention (MICCAI) – Conference Proceedings.
3. The Visualization Toolkit (VTK) User's Guide – Kitware
4. Biomedical Image Analysis – Rangaraj M. Rangayyan
5. 3D Slicer and ITK/VTK Documentation (Open Source Tools)

Course Description

Visualization in Medicine is a specialized course that explores the intersection of medical imaging, computer graphics, and data visualization. The course introduces students to the principles, techniques, and tools used to visually analyze and interpret complex medical

Course Objectives

- Understand the principles of medical visualization and its role in clinical decision-making.
- Gain familiarity with various medical imaging modalities (MRI, CT, PET, Ultrasound).
- Learn visualization techniques for scalar, vector, and tensor data in medical contexts.
- Implement and evaluate visualization algorithms in real-world medical applications.

Week Wise Topics to be Covered

W1	Introduction to Visualization in Medicine: Scope, History, and Applications	**W9**	Visualization Software and Libraries: 3D Slicer, MeVisLab, VTK, ITK
W2	Overview of Medical Imaging Modalities: X-ray, CT, MRI, Ultrasound, PET	**W10**	Interactive Visualization and User Interfaces in Clinical Environments
W3	Digital Imaging Fundamentals: Pixels, Voxels, DICOM Format, Image Storage	**W11**	Virtual Reality (VR) and Augmented Reality (AR) Applications in Medicine
W4	2D Image Processing Techniques: Filtering, Contrast Enhancement, Windowing	**W12**	Surgical Planning and Image-Guided Therapy
W5	3D Visualization Fundamentals: Volume Rendering vs. Surface Rendering	**W13**	Visualization Challenges: Accuracy, Ethics, and Data Privacy
W6	Image Segmentation Techniques: Thresholding, Edge Detection, Region Growing	**W14**	Student Presentations: Medical Visualization Projects
W7	Registration and Fusion of Multimodal Medical Images	**W15**	**Case Studies / Projects / Demo**
W8	Color Mapping and Transfer Functions in Medical Visualization	**W16**	**Presentations / Peer Review / Q&A Sessions / Discussions**
MID EXAM/ MID OF SEMESTER		**Final EXAM**	